
Core Mathematics

A Practical Guide on Mathematics for Physics Learners

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Lecture notes for first-year university students in physics or physical science, covering mathematical tools you the students need most. Calculus, logic, complex numbers, and vectors, with introductions to vector calculus, differential equations, and statistical analysis. Exercises are central to these notes. Practice each topic repeatedly until you can work through problems quickly and accurately.

Preface

Physics is an activity to describe the world, but how? At present, mathematics (+ *broken* English) is the only language we can use. So, *unfortunately*, we need to learn mathematics! (and English... 🙄) But it means we are *fortunate*: we have a language to describe the world. In fact, physics have developed together with mathematics. We understand physics better if we study mathematics well!

When you do DIY, you need to use the tools properly, smoothly, and accurately. It is the same here; when we use math as a tool, it must be **logical**, **quick**, and **accurate**. To be logical, you must think each problem carefully; otherwise your discussion might be incomplete and you can't convince others. To be quick and accurate, you must "*drill*" repeatedly, as if top NBA players do shooting practice almost everyday. You should reach the point where basic calculations feel automatic.

This is not a math textbook. **This is rather** a math "drill" book. Rigorous proofs are mostly omitted. Drill problems are the core of this document, through which students are expected to achieve better conceptual understanding.

Target of this document

This document is primarily for first-year undergraduate students in their second semester, who

- completed basic calculus (differentiation and integration),
- are going to major in physics or physical science, and
- do not hesitate to do repetitive practice to achieve better conceptual understanding.

How to use this document

1. Buy a A4-sized paper notebook.

Digital notebooks on tablets are not suitable (less effective).

2. Read the text and solve **Quizzes**.

Read all the text carefully. The texts are kept short!! Quizzes should be solved while reading, but you don't have to write down the solution process for Quizzes.

3. Solve **Drill Problems**, until you feel "I can solve them *fluently and accurately*".

This is to improve your fluency ($\neq$ speed) and accuracy. Write down solution process clearly.

4. Solve other **Problems**.

This is to have better conceptual understanding, but also to develop your writing skills. Write down the solution process clearly *as an English text*, taking care of *logical completeness*.

Optionally, you are encouraged to prepare a reference book, so that you can consult it when you want to know more. Shown below is the best option.



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Visit <https://github.com/misho104/LecturePublic> for further information, updates, and to report issues.

Problems

-  **Drill problems**, marked with , are simple and easy. They help you practice the basic calculations until you can do them fluently and accurately. If they feel too easy for you, feel free to skip some of them.
- ★ These are **minimal basic** problems. To pass this course, you should be able to solve all of them. If you struggle with them, please come talk to Sho—don't wait until the exam!
 - *** These **basic** problems are the same difficulty as the ★-problems. Solve these for extra practice, to check that you really understand the ★-problems and were not just lucky.
 - ** These are **intermediate** problems. Solve these if you are aiming for an A+ in this course and also outperform others in your future courses. Note also that, at this level, you are asked to write answers carefully with clear explanations.
 - * These are **advanced** problems, beyond the scope of the lecture, for motivated students who are interested in research activities, a top-level engineering career, or demanding analytical-thinking work.

Bibliography

This document is written under the influence of the following references.

- [The Math Book by Hal Tasaki](#) (数学：物理を学び楽しむために, in Japanese)
A free book with full description, but unfortunately in Japanese. Topic selections and the depth of descriptions are based on this reference.
- [First-Year Math Practice](#) (in Japanese)
This document is grounded in a “learn steadily” approach (“じっくり” in Japanese), which originates from Prof. Gocho and Prof. Kiyono, who were instructors of Sho in his freshman. Problems and rigorous math descriptions are taken from their works.
- **[Boas]** Mary L. Boas, *Mathematical Methods in the Physical Sciences*, 3rd ed., Wiley, 2006.
- **[AWH]** George B. Arfken, Hans J. Weber, and Frank E. Harris, *Mathematical Methods for Physicists*, 7th ed., Academic Press, 2023.

These two books are widely used in physics departments around the world. [Boas] seems more friendly to beginners, while [AWH] seems more detailed and complete. Sho recommends you to study [Boas] *in parallel with* this document, using it as a reference, and to consult [AWH] when you want more details or more exercises.

Other references are:

- [1] Bureau International des Poids et Mesures, “The International System of Units (SI),” BIPM. [Online]. Available: <https://doi.org/10.59161/AUEZ1291>
- [2] Herbert Goldstein, Charles P. Poole, and John L. Safko, *Classical Mechanics*, 3rd ed. Pearson.
- [3] Sadri Hassani, *Mathematical Physics*, 2nd ed. Springer.

- [4] John Hubbard and Barbara Hubbard, *Vector Calculus, Linear Algebra and Differential Forms*, 5th ed. Matrix Editions.

Symbols

SI Units

Historically, different civilizations used different units. Today, science and engineering worldwide use the **SI units** (Système International d’Unités)—the single international standard. It is built from seven **SI base units**:

Table 1: SI base units and their symbols for dimension.

quantity	symbol and name	symbol for dimension
time	s (second)	T
length	m (meter)	L
mass	kg (kilogram)	M
electric current	A (ampere)	I
temperature	K (kelvin)	Θ
amount of substance	mol (mole)	N
luminous intensity	cd (candela)	J

See [1] for details. Recall that units are case-sensitive.

Mathematical Notations

Chapter 1		Chapter 3	
$\mathbb{C}$	complex numbers	$\wedge$	logical “and”
$\mathbb{R}$	real numbers	$\vee$	logical “or”
$\mathbb{Q}$	rational numbers	$\neg$	logical “not”
$\mathbb{Z}$	integers	$\implies$	implication
$\mathbb{N}^0$	natural numbers (0, 1, 2, ...)	$\iff$	logical equivalence
$\mathbb{N}^+$	natural numbers (1, 2, ...)	$\forall$	for all
$x \in A$	a member of	$\exists$	there exists
$A \cup B$	union	$:=$	is defined as
$A \cap B$	intersection		
$A \setminus B$	set difference		
$f'(x)$	derivative $\left(\text{also } \frac{df(x)}{dx} \right)$		
$f^{(n)}(x)$	n -th derivative $\left(\text{also } \frac{d^n f(x)}{dx^n} \right)$		

Greek symbols

A	α	alpha	H	η	eta	N	ν	nu	T	τ	tau
B	β	beta	Θ	θ	theta	Ξ	ξ	xi	Y	υ	upsilon
Γ	γ	gamma	I	ι	iota	O	$\omicron$	omicron	Φ	$\varphi(\phi)$	phi
Δ	δ	delta	K	κ	kappa	Π	π	pi	X	χ	chi
E	ε	epsilon	Λ	λ	lambda	P	ρ	rho	Ψ	ψ	psi
Z	ζ	zeta	M	μ	mu	Σ	σ	sigma	Ω	ω	omega

Gray-outed ones are almost never used, but others are almost always used. You need to write them so that **others can distinguish each from others**, but Sho thinks Japanese- or Chinese-speakers are OK with it, as we everyday read and write 日 vs 日 vs 白 or 土 vs 土 vs 工 properly!

Quiz

Q1. Read out the following words.

- | | | |
|---|----------------------------------|---|
| (1) α -particle | (2) β -decay | (3) Γ -function |
| (4) ε - δ definition | (5) angle θ and φ | (6) μ - and τ -leptons |
| (7) $\psi(x)$ and $\varphi(x)$ | (8) ζ and ξ | (9) metric $\eta_{\mu\nu}$ |
| (10) X-ray and γ -ray | (11) Ω -baryon | (12) $\varepsilon_{\mu\nu\rho\sigma}$ -tensor |

Problems

 0.1 Write the following letters so that people can distinguish each from others.

- | | | | |
|--------------------|------------------------------------|---------------------|-------------------------|
| (1) a, α | (2) b, β | (3) c, C | (4) e, E, ε |
| (5) ζ, ξ | (6) $\Theta, \theta, Q, \sigma, 6$ | (7) $g, q, 9$ | (8) $i, l, 1, I$ |
| (9) k, K, κ | (10) m, μ | (11) p, ρ | (12) s, S |
| (13) t, τ, T | (14) u, v, ν | (15) w, W, ω | (16) x, X, χ |

Lecture Plan

The content is designed for 150-minute $\times$ 14-week lectures, as it is originally for a lecture *Mathematics for Fundamental Physics* (基礎物理數學) in National Sun Yat-sen University (國立中山大學).

1. Derivative
2. Units. Significant Digits.
3. Logic. Theorems. a^x and $\log_a x$.
4. Logic. Theorems. a^x and $\log_a x$.
5. Vectors are arrows.
6. $a \cdot b$ and $a \times b$
7. Matrices (rotation, reflection, scaling)
8. Complex numbers (polar form, Euler's formula, multi-valued-ness)
9. Complex vectors, Hermitian inner product, bra-ket notation
10. Vector calculus (grad, div, rot; example of point charge)
11. ODE basics
12. ODE basics
13. Probability and error analysis
14. Probability and error analysis

♣TODO♣ a bit on (abstract) vector space to prep for QM?

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Chapter 1

Derivative (Review)

1.1 The first step

Birds sing, fish swim, flowers bloom, stars twinkle, and university students calculate derivatives. Let us begin with basic **derivatives**. Try the next quiz—and note how long it takes.

Quiz

Q1. Calculate the first derivatives of the following functions, **measuring how many minutes it takes**.

(1) $(x + 1)^3$

(2) $\tan x$

(3) $2 \cos^2 x$

(4) $2 \cos x^2$

(5) $x^3 \cos x$

(6) $\frac{3 \sin x}{x}$

(7) $x^{3/2}$

(8) $\sqrt{\sin x}$

- No mistake, less than two minutes? Amazing! It is as fast as Sho!
- No mistake, less than four minutes? Great, it is exactly as Sho anticipates. Please continue your effort!
- No mistake but takes more than four? That's okay, do not worry. You are just not *fluent*; a little more practice will help you. Try more problems in [the Derivative Boot Camp](#).^{#1}
- If you make any mistakes in these eight calculations, then it is a serious issue—just as serious as forgetting how to do the calculations at all. In university, you will perform similar calculations more than 100 times. You need to be both fast and accurate.



At university, students often underestimate the importance of basic calculations. In physics, simple calculations appear constantly, so your speed and accuracy directly affect how well you follow lectures, how efficiently you study, and ultimately your grade. Both can be improved with practice.

This course will help you strengthen these basic calculations—some of which you already know from high school—while also introducing new topics that are essential for physics. You will solve many problems and drills, just as an athlete repeats the same move hundreds of times to master it.

This document (this lecture course) is designed for first-year students in their second semester, **who have already studied derivatives well** in their first-semester calculus. As we do not cover the details of derivatives here, if you are not confident with derivatives, please first study [the Derivative Boot Camp](#).

^{#1}Visit <https://misho104.github.io/LecturePublic> and find gp1_boot1_deriv_true.pdf.

1.2 Typical pitfalls

1.2.1 Notations

Consider a function $f(x)$. As we vary the value of x , x is called a **variable**. For a function $f(x)$, there are several equivalent ways to write its derivative:

$$f'(x) = \frac{df}{dx}(x) = \frac{df(x)}{dx} = \frac{d}{dx}f(x) \quad [\text{All means the same thing}]. \quad (1.1)$$

Similarly, the value of $f'(x)$ at a specific point $x = 3$ can be written as

$$f'(3) = f'(x)\Big|_{x=3} = \frac{df}{dx}\Big|_{x=3} = \frac{df}{dx}(3) = \frac{d}{dx}f(3), \quad (1.2)$$

and they are all equivalent. For example, if $f(x) = 3x^2 + 6$, then $f'(x) = 6x$ and $f'(3) = 18$.

Most of students get confused when a **constant** a appears. If a is *declared as a constant*, we can define a function such as $g(x) = ax^2 + 2a$, for which

$$g'(x) = 2ax. \quad (1.3)$$

We can evaluate $g'(x)$ at $x = a$. The result is $g'(a) = 2a^2$, written as

$$g'(a) = g'(x)\Big|_{x=a} = \frac{dg}{dx}(a) = \frac{d}{dx}g(a) = 2a^2. \quad (1.4)$$

Be careful: The notation $\frac{d}{dx}g(a)$ does **not** mean $\frac{d}{dx}[g(a)] = \frac{d}{dx}(a^3 + 2a) = 0$.

Fail safe note: If you can't see Eq. (1.3), try setting $a = 3$. Then, you notice $g(x)$ is the same function as $f(x)$, and $g'(x)$ should equal $f'(x)$.

Higher-order derivatives is written as

$$\frac{d}{dx}\left(\frac{df}{dx}\right) = \frac{d^2f}{dx^2} = f''(x) = f^{(2)}(x), \quad \frac{d}{dx}\left(\frac{d}{dx}\left(\frac{df}{dx}\right)\right) = \frac{d^3f}{dx^3} = f'''(x) = f^{(3)}(x), \quad (1.5)$$

etc. Please be careful on the position of “2” and “3”.

Quiz

Q2. Let a be a constant and $g(x) = ax^2 + 2a$. Calculate

- | | | | |
|-------------|-------------|--------------|------------------------|
| (1) $g(1)$ | (2) $g(a)$ | (3) $g(0)$ | (4) $g(7)$ |
| (5) $g'(1)$ | (6) $g'(a)$ | (7) $g''(2)$ | (8) $\frac{d}{dx}g(7)$ |

Q3. Let $f(x) = x^3 + 2x^2 + 5x + 1$. Calculate

- | | | | |
|------------------|---------------|----------------------------|---|
| (1) $f^{(2)}(1)$ | (2) $f'''(1)$ | (3) $\frac{d^2f}{dx^2}(2)$ | (4) $\frac{d}{dx}\left(\frac{d}{dx}\left(\frac{df}{dx}\right)\right)$ |
|------------------|---------------|----------------------------|---|

One more bad news. *Physicists are often lazy* and write $f(x)$ as f . If $f(x) = 2x^2 + 1$, we may write $f' = 4x$ and $f'' = 4$. For $g(t) = 2t^2 - 1$, you may write $g' = 4t$ and $g'' = 4$. Therefore, when you see a function f , you must **identify its variable from the context**.

In physics, a variable can be a function of another variable. The kinetic energy $K(v) = mv^2/2$ is a good example. Its variable v is a function of time t —we write this fact by $v = v(t)$ —and thus

$$K(v) = \frac{1}{2}mv^2 \quad \text{but also} \quad K(t) = K(v(t)) = \frac{1}{2}mv(t)^2. \quad (1.6)$$

Now, what does K' mean? —It is ambiguous and we must avoid such notation. Even though, if you see it in a textbook, you need to guess the author's intention from the context.

Advanced note: This rewrite, $K(t)$ and $K(v)$, is common in physics. However, mathematicians do not like this practice because they consider K a unique object: a function should always mean the same rule applied to its input. If they see $K(v) = mv^2/2$, they would say [K is an object that converts its input into $m/2 \times (\text{input})^2$, and thus if you feed t into K , you must get $K(t) = mt^2/2$]. This is of course not what we mean. For example, imagine $v(t) = \alpha t$ with α a constant. Then

$$\text{physicists: } K(v) = \frac{1}{2}mv^2, \quad K(t) = \frac{1}{2}m\alpha^2 t^2; \quad (1.7)$$

but mathematicians would say, [since it returns $m\alpha^2/2 \times (\text{input})^2$, it is a different object] and hence use a different name for it, such as $\tilde{K}$:

$$\text{mathematicians: } K(v(t)) = \frac{1}{2}m \times v(t)^2 = \tilde{K}(t) = \frac{1}{2}m\alpha^2 t^2. \quad (1.8)$$

Both conventions are reasonable, and eventually you will get used to both.

1.2.2 Radian and Trigonometric functions

At university, angles are almost always measured in **radians**:

$$360 \text{ degree (or: } 360^\circ) = 2\pi \text{ radian (or: } 2\pi \text{ rad)} \quad (1.9)$$

Furthermore, we usually omit “radian” (because we are lazy!). So,

$$\text{a right angle is } \pi/2. \quad \text{The sum of the interior angles of a triangle is } \pi. \quad (1.10)$$

You may write $180^\circ = \pi$, $\pi/180 = 1^\circ$, and so on, but the small circle $^\circ$ must be written *clearly*.

Quiz

Q4. Express the following angles in radians, and radians in angles.

- | | | | | |
|-----------------|----------------|----------------|---------------|-----------------|
| (1) 180° | (2) 45° | (3) 60° | (4) 90 | (5) -30° |
| (6) 1° | (7) 1 | (8) -0.3 | (9) x° | (10) x |

Why do we insist on radians? The answer comes from the derivative formula

$$\frac{d}{dx} \sin x = \cos x. \quad (1.11)$$

Because we have chosen radians as the standard, $(\sin x)'$ becomes this simple.

Advanced note: The simpleness of Eq. (1.11) originates in the fact $\sin(0.01 \text{ rad}) \approx 0.01$. If we used degrees, we would have $\sin(0.01^\circ) \approx 0.01 \times 0.017453$ and everything is messed up with this number 0.017453.

There are a few remarks in the notation of **trigonometric functions**:

$$\begin{aligned} \sin^2 x &\neq \sin x^2. \quad \text{Namely, } (\sin x)^2 = \sin^2 x \quad \neq \quad \sin x^2 = \sin(x^2). \\ \tan^{-1} x &\neq \frac{1}{\tan x}. \quad \text{Namely, } \tan^{-1} x = \arctan x \quad \neq \quad (\tan x)^{-1} = \frac{1}{\tan x} = \cot x. \end{aligned} \quad (1.12)$$

The following expressions are not incorrect but confusing;

$$\sin^{-2} x, \quad \sin^{1/2} x, \quad \sin(x)^2, \quad \sin(x+1)^2, \quad \dots$$

We should avoid ambiguity, so please *never* use confusing these notations. Sho thinks we should use $\sin^k x$ only for $k = 2, 3, 4, \dots$, and use $\arcsin x$ instead of $\sin^{-1} x$.

1.2.3 Several interpretations of derivatives

When you, physics learners, discuss $f'(t)$, you should have the following three interpretations:

- Regarding t as the time, $f'(t)$ is the **rate of change** of $f(t)$ per unit time. If $f'(t) > 0$, the function f is increasing at the time t , while $f'(t) < 0$ means it is decreasing.
- Consider a graph of $f(t)$. Then, $f'(t)$ is the slope of the **tangent line** to the graph at the point t .
- Mathematically, $f'(t)$ is defined by (Check that they are equivalent.)

$$f'(t) := \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t} = \lim_{h \rightarrow 0} \frac{f(t + h) - f(t - h)}{2h} = \lim_{s \rightarrow t} \frac{f(s) - f(t)}{s - t}. \quad (1.13)$$

If you are unsure of them, please consult your first-year Calculus textbooks for further information.

Quiz

Q5. What is the definition of $f'(x)$? Explain.

In physics, it is important to memorize and understand the **definition** of each concept. We will talk about “definitions” in Section 3.5.

1.3 Taylor expansion

Reviewing the definition of $f'(x)$, we have

$$f'(x) := \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}, \quad f'(a) := \lim_{\delta \rightarrow 0} \frac{f(a + \delta) - f(a)}{\delta}. \quad (1.14)$$

The second equation is interpreted as follows:

$$\text{If } \delta \approx 0, \quad f'(a) \approx \frac{f(a + \delta) - f(a)}{\delta}, \quad \text{i.e., } f(a + \delta) \approx f(a) + \delta f'(a) \quad (1.15)$$

and this interpretation is useful in the following example:

Example 1.1

Find the approximate value of the following expressions without using calculators. Then, check your answer with a calculator.

(1) $\sqrt{1.002}$

(2) $(1.002)^{10}$

(3) $\sqrt{4.004}$

Solution

(1) Apply Eq. (1.15) for $f(x) = \sqrt{x}$, $a = 1$, and $\delta = 0.002$. Then,

$$f'(x) = \frac{1}{2\sqrt{x}}, \quad f'(a) = f'(1) = \frac{1}{2}, \quad f(a + \delta) \approx f(a) + \delta f'(a) = 1 + \frac{\delta}{2} = \underline{1.001}.$$

(2) Doing the same thing for $g(x) = x^{10}$ with $a = 1$ and $\delta = 0.002$,

$$g'(x) = 10x^9, \quad g'(a) = g'(1) = 10, \quad g(a + \delta) \approx g(a) + \delta g'(a) = 1^{10} + 10\delta = \underline{1.02}.$$

(3) Doing the same thing for $h(x) = \sqrt{x}$, $a = 4$, and $\delta = 0.004$, we have

$$h'(4) = \frac{1}{4}, \quad h'(4.004) \approx h(4) + 0.004 \times \frac{1}{4} = \underline{2.001}.$$

Advanced note: “Solutions” contain not only the answer but also how you reached the answer. You are, of course, asked to write such explanations when you solve problems.

Namely, with this technique, you can find **the value of $f(x)$ around a point $x = a$** . This technique, Eq. (1.15), is a special case of Taylor’s theorem, which is discussed in ♣TODO♣???

Quiz

Q6. Find the approximate value of the following expressions without using calculators. Then, check your answer with a calculator.

(1) $\sqrt{1.001}$

(2) $(1.0001)^{30}$

(3) $\frac{1}{1.001}$

To summarize, we have the following statement:

Statement 1.2 Taylor expansion: basic

For a physicists-friendly function $f(x)$,

$$f(a + \delta) \approx f(a) + \delta f'(a), \quad (1.16)$$

or as an equivalent expression,

$$f(x) \approx f(x_0) + (x - x_0)f'(x_0). \quad (1.17)$$

Quiz

Q7. Check these two equations are equivalent. You will be ready to use either of them in the future.

Q8. Review how to derive Eq. (1.16) and summarize it in your own words.

Problems

 **1.1** Calculate the following expression.

- (1) $\sin 30^\circ$ (2) $\cos 45^\circ$ (3) $\cos 120^\circ$ (4) $\sin 150^\circ$ (5) $\tan 180^\circ$
 (6) $\cos 0$ (7) $\sin 2\pi$ (8) $\cos \pi$ (9) $\sin(\pi/3)$ (10) $\cos(-\pi/4)$
 (11) $\tan(2\pi/3)$ (12) $\sin(\pi/6)$ (13) $\tan(5\pi/6)$ (14) $\cos(-\pi/6)$ (15) $\tan(-\pi/2)$

 **1.2** For the following $f(x)$, calculate $f'(x)$, $f''(x)$, $f^{(3)}(x)$, $f^{(4)}(x)$, and $f^{(5)}(x)$.

- (1) x^3 (2) $\sin 2x$ (3) e^{2x} (4) $\ln 2x$

 **1.3** Find the approximate value for the following expressions without calculators. Then, check your answer with a calculator.

- (1) $(1.002)^7$ (2) $(2.002)^7$ (3) $(0.998)^4$ (4) $(10.01)^3$ (5) $1 \div 0.998$
 (6) $\sqrt{0.998}$ (7) $\sqrt{4.001}$ (8) $(0.999)^{1/2}$ (9) $(0.999)^{-1/2}$ (10) $(0.999)^{-3/2}$
 (11) $\frac{1}{(2.02)^2}$ (12) $\tan 0.001$ (13) $\cos 0.002$ (14) $\sin 0.003$ (15) $\ln 1.001$

****1.4** Find the approximate value for the following expressions without calculators. Then, check your answer with a calculator.

- (1) $\sqrt[3]{1.001}$ (2) $\sqrt[3]{8.012}$ (3) $1.001^{0.5}$ (4) $1.001^{-0.9}$ (5) $\sin(3.14)$
 (6) $\cos(1.57)$ (7) $\sin 1^\circ$ (8) $\cos 61^\circ$ (9) $\ln 0.999$ (10) $\log_{10}(10.01)$
 (11) $\sqrt{1.001} \sin 0.001$

Fail safe note: If you are not sure about $\sqrt[3]{1.001}$, $(1.001)^{-0.9}$, etc., study [Chapter 4](#) first.

****1.5** For the following $f(x)$, calculate $f^{(n)}(x)$ for all positive integers $n = 1, 2, 3, \dots$

- (1) x^{10} (2) $\sin 2x$ (3) e^{2x} (4) $\ln x^2$ (5) $\sqrt{x}$

***1.6** Write down the definition of the second derivative $f''(x)$. Repeat the discussion of Eq. (1.15) to find the expansion $f'(a + \varepsilon) \approx f'(a) + \varepsilon f''(a) + (\varepsilon^2/2)f''(a)$.

Advanced note: The two equations in Eq. (1.14) have similar but different meanings: the first one defines a new function $f'(x)$, while the second one defines a number that is eventually equal to $f'(x)|_{x=a}$. Anyway, we don't care the difference.

1.4 Mathematical notation

As *physicists are lazy*, we usually use the following notation:

$x \in \mathbb{C}$	means “ x is a complex number”.	$x \in \mathbb{Z}$	means “ x is an integer ”. ^{#2}
$x \in \mathbb{R}$	means “ x is a real number”.	$x \in \mathbb{N}^+$	means “ x is a positive integer”.

^{#2} $\mathbb{Z}$ comes from “Zahlen” (Zahl).

$x \in \mathbb{Q}$ means “ x is a **rational number**”. $x \in \mathbb{N}^0$ means “ x is a non-negative integer”.

The symbol “ $\in$ ” means “*is a member of*”, and $\mathbb{N}^+$ means a **set** of positive integers, $\{1, 2, 3, 4, \dots\}$. So,

$x \in \mathbb{N}^+$ and $x \in \{1, 2, 3, 4, 5, \dots\}$ means the same thing: “ x is a positive integer”.

Be careful: In mathematics, “natural numbers” usually mean $\mathbb{N}^0$, including 0. Meanwhile, physicists and other people tend to think 0 is not a natural number.

Quiz

Q9. What is the difference between $\mathbb{N}^+$ and $\mathbb{N}^0$?

Q10. What is the difference between $\mathbb{N}^0$ and $\mathbb{Z}$?

Q11. What is “rational numbers”? What is the difference between $\mathbb{Z}$ and $\mathbb{Q}$?

We write “ x is a positive number” by $x \in \mathbb{R}, x > 0$ but also by $x > 0$. Similarly,

$x \in \mathbb{R}, x \leq 0$ or $x \leq 0$ means “ x is a non-positive number”.

Be careful: Namely, when we say something is “positive” or “negative”, we implicitly assume it is a real number.

Advanced note: This is because we cannot compare complex numbers with 0. The operator $>$ is defined only for real numbers. When we write $a > b$, we implicitly consider that a and b are real numbers.

Problems

****1.7** The **intersection** of two sets A and B is a set made by elements both in A and B ; we write it by $A \cap B$. The **union** of A and B is a set made by elements in A , B , or both; we write it by $A \cup B$. The **set difference**, $A \setminus B$, is a set made by elements in A but not in B .

For example, let $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$, and $C = \{2, 4\}$. Then,

$$A \cap B = \{2, 3\}, \quad A \cup B = \{1, 2, 3, 4\}, \quad A \setminus C = \{1, 3\}, \quad B \setminus C = \{3\}, \quad C \setminus B = \{\},$$

where $\{\}$ (or $\emptyset$) means the **empty set**, the set without any elements.

(1) Write the following sets.

(a) $A \cap C$ (b) $B \cap C$ (c) $A \cup C$ (d) $B \cup C$ (e) $A \cup B \cup C$

(f) $B \setminus A$ (g) $C \setminus A$ (h) $C \setminus B$ (i) $(B \setminus A) \cup A$ (j) $\mathbb{N}^0 \setminus \mathbb{N}^+$

(2) Express the following statements only with $\mathbb{R}$, $\mathbb{Q}$, $\mathbb{Z}$, $\mathbb{N}^+$, $\mathbb{N}^0$, $\cap$, $\cup$, $\setminus$, and “ $x \in$ ”.

- | | |
|--|------------------------------------|
| (a) x is a rational number. | (b) x is an irrational number. |
| (c) x is a rational number but not an integer. | (d) x is a non-positive integer. |
| (e) x is a negative integer. | (f) x is a non-zero integer. |

Chapter 2

Units and Significant Figures

A **physical quantity** is usually not just a number, but also has a **unit** and an **uncertainty** (error). For example, “ $(72 \pm 2) \text{ kg}$ ” has a number “72” and a unit “kg”, which compose the **central value** “72 kg”. It also has uncertainty because every measurement has an uncertainty. In experimental physics, uncertainty is usually more important than the central value, and thus uncertainty handling is a fundamental skill of physicists.

In this chapter, we first review physical quantities and how to handle units. Then, we discuss **significant figures**, a simple and convenient method to express the uncertainty. Further discussions on uncertainty analysis are given in Chapter [♣TODO♣chapter](#).

2.1 Physical quantity

The expression “ $(72 \pm 2) \text{ kg}$ ” means the central value is 72 kg and the **absolute uncertainty** is 2 kg. Accordingly, its **relative uncertainty** is given by $(2 \text{ kg})/(72 \text{ kg}) = 0.028$. Let’s see other examples.

	unit	central value	absolute uncertainty	relative uncertainty
$10 \text{ m} \pm 1 \text{ cm}$	m (meter)	10 m	1 cm = 0.01 m	0.001 (or 0.1%)
$1.6 \text{ A} \pm 0.04 \text{ A}$	A (ampere)	1.6 A	0.04 A	0.025 (or 2.5%)
$(5 \pm 1) \times 10^{-3} \text{ C}$	C (coulomb)	0.005 C	0.001 C	0.2 (or 20%)
$(50 \pm 1) \text{ m/s}$	m/s	50 m/s	1 m/s	0.02 (or 2%)

Remark: Upper- and lowercase letters are distinguished. “M” **does not** mean “meter”. The unit “A” is “ampere”, not “Ampere”. The unit “coulomb” is “C”, not “c”.

Quiz

Q1. For each of the following values, find its unit, central value, absolute uncertainty, and relative uncertainty.

- | | |
|---|---|
| (1) $50 \text{ kg} \pm 500 \text{ g}$ | (2) $(0.05 \pm 0.001) \text{ A}$ |
| (3) $(1.6 \pm 0.001) \times 10^{-19} \text{ C}$ | (4) $72 \text{ km/h} \pm 1 \text{ m/s}$ |

Q2. The next passage has six (6) errors in the use of uppercase and lowercase letters. Find them out.

In the SI system, temperature is expressed in k (kelvin), a unit named after Lord Kelvin. The unit of force is N (newton), named after the british scientist Isaac newton. The units A (Ampere) and c (coulomb) are named after French scientists. In contrast, Kg (kilogram) is not named after a person. , but comes from Greek.

As an undergraduate student, you need to follow the following rules:

Statement 2.1 Rules for Physical Quantities: Basic

1. Numbers are always with units, even in calculations.
2. Include units in a symbol.
3. For physical quantities, use decimals (e.g., 1.6 A). Do not use fractions like (8/5) A.
4. Use significant figures to express the measurement precision.

These rules are explained below.

Be careful: We use *decimals for physical quantities* and *fractions for mathematical concepts*. Since this lecture is mainly on math and math concepts, you will use decimals **♣TODO♣only in this section**.

2.2 Units

When we educate kids, we write “my height is $\square$ cm”, or “my height is h cm”, where $h = 172$ is just a number. But this is not nice! We want to convert the units freely and write equations such as $1.72 \text{ m} = 172 \text{ cm} = 0.00172 \text{ km}$. So, **we always include units in symbols** $h = 172 \text{ cm}$. Then

we can write: $h = 172 \text{ cm} = 1.72 \text{ m} = 0.00172 \text{ km} = 1.82 \times 10^{-16} \text{ light-year}$.

Similarly, if $m = 110 \text{ g}$ and $g = 9.8 \text{ m/s}^2$,

we may write: $w = mg = 110 \text{ g} \times 9.8 \text{ m/s}^2 = 1.1 \text{ kg} \cdot \text{m/s}^2$

but not: $w = mg = 0.11 \times 9.8 = 1.1 \text{ kg} \cdot \text{m/s}^2$

This second equation is incorrect because m is not equal to 0.11; m is equal to 0.11 kg or 110 g.

Remark: Usually, physicists use upright fonts for units and *italic fonts* for quantities. For example, m , T , and C are quantities, which can be mass, temperature, capacitance, etc. Meanwhile, m, T, and C are units: meter, tesla, and coulomb, respectively.

Quiz

- Q3.** (1) If $mr\omega^2 = 10.0 \text{ kg} \cdot \text{m/s}^2$, $m = 5.0 \text{ kg}$, and $r = 1.0 \text{ m}$, what is ω ?
- (2) If $v_0 t + (1/2)at^2 = 5.0 \text{ m}$ with $a = -4.0 \text{ m/s}^2$ and $v_0 = 7.0 \text{ m/s}$, what is t ?
- (3) At time $t = 0$, a particle is at $(x, y) = (2.0 \text{ m}, 0)$. It moves with a constant velocity $(v_x, v_y) = (3.0, 4.0) \text{ m/s}$. What is its position at $t = 1.0 \text{ s}$?

Every physical concept has its own unit. For example, speed has m/s, acceleration has m/s², and energy has $\text{kg} \cdot \text{m/s}^2 = \text{N} \cdot \text{m} = \text{J}$. The next table lists the quantities you have learned. Notice that angle (rad) has no dimension. It is called a **dimensionless** quantity.

concept	dimension	typical unit
speed	$L T^{-1}$	m/s
acceleration	$L T^{-2}$	m/s^2
linear momentum	$M L T^{-1}$	$kg \cdot m/s$
force	$M L T^{-2}$	$kg \cdot m/s^2$ (= N)
energy / work	$M L^2 T^{-2}$	$kg \cdot m^2/s^2$ (= J)
frequency	T^{-1}	1/s (= Hz)
angle	1 (dimensionless)	rad

Advanced note: To understand why angle has no dimension, you may consider the definition of the radian: it is defined as the ratio of the arc length to the radius, so the units cancel (meter / meter = 1).

Here, please do not write: The force is $kg \cdot m^2/s^2$. This is also called newton (N)., because we cannot mix units and concepts. A correct (and easy) way is to use English words:

the unit of the force is $kg \cdot m/s^2$ or the force has the unit $kg \cdot m/s^2$.

Similarly, the following statement is incorrect because it mixes concepts and units:

Since $m = kg$ and $g = m/s^2$, $mg = kg \cdot m/s^2 = N$.

Instead, you need to write

Since m has a unit of kg and g has a unit of m/s^2 , mg has a unit of $kg \cdot m/s^2 = N$.

Notice the last equation: units can be equal to other units, so we may write $N = J/m = kg \cdot m/s^2$

$m = J/N$ $s = \sqrt{kg \cdot m/N}$. These are scientifically correct equations.

Remark: We sometimes use informal notations, such as $[F] = N$ or $F \sim N$ to express “the unit of F is N (newton)”. Still, using an equal symbol ($=$) is **not allowed**. Namely, $F = N$ is always incorrect.

Quiz

Q4. What are the units of the following concepts? Write English sentences to explain it.

(1) speed (2) velocity (3) force (4) energy (5) power

Advanced note: In very formal situations, we use **symbols for dimensions** (see Table 1 on page 5):

Since $\dim(m) = M$ and $\dim(g) = L T^{-2}$, $\dim(mg) = M L T^{-2}$, with the operator “dim” [1]. However, it seems too complicated for most situations.

Problems

★2.1 Express the following units only with SI base units, e.g., $\text{N} = \text{kg} \cdot \text{m}/\text{s}^2$.

- (1) W (watt) (2) Pa (pascal) (3) J (joule) (4) C (coulomb)

***2.2 Find a few more examples of dimensionless quantities in physics.

**2.3 Use “dim” notation to express the dimensions of the following quantities. For example, if v is speed, then $\text{dim}(v) = \text{L T}^{-1}$.

- | | |
|--------------------------|--------------------------------------|
| (1) speed v | (2) force F |
| (3) angular momentum L | (4) electric charge Q |
| (5) resistance R | (6) specific heat capacity c |
| (7) Planck constant h | (8) fine-structure constant α |

2.3 Significant figures

In researches, we need to treat uncertainties in the method given in [♣TODO♣chap](#). However, most of lectures use **significant figures** to express the accuracy of a value, so that students becomes familiar with the concept of uncertainties. For example,

3.14	means the last digit “4” is uncertain. It can be 3.16, 3.15, 3.11, 3.10,
3.141	means “3.14” is for sure but it can be 3.143, 3.142, 3.140, 3.139,
3.1415	might be 3.1416, 3.14153, etc., but the writer is sure <i>it is not</i> 3.145 or 3.135.

We write the “uncertain digit” in **a different style** for clarity. You should notice the difference between:

1.0000	meaning the last digit “0” is uncertain so it may be 1.0002, 1.0001, 0.9998,
1.00	meaning it can be 1.03 or 0.99, but the writer is sure <i>it is not</i> 1.2 or 0.8.

It is usually convenient to use **scientific notation**:

0.000123	is OK but we prefer	1.23×10^{-4} .	Both mean the value can be 1.22×10^{-4} etc.
0.00100	is OK but we prefer	1.00×10^{-3} .	
29979	is OK but we prefer	2.9979×10^4 .	

Fail safe note: $10^3 = 1000$ and $10^{-3} = 1/10^3 = 1 \div 1000$. Go to [♣TODO♣chap](#) for details!

We need to *avoid ambiguous notations*. Namely,

2040	is ambiguous and not nice. We are not sure the author means 2.04×10^3 or 2.040×10^3 .
120	is not nice. We are not sure the author means It may mean 1.20×10^2 or 1.2×10^2 .
42000	is not nice because it has four ways to interpret the author’s intention.

Quiz

Q5. What are the four interpretations of 42000? Write them in scientific notation.

We use **rounding**-to-the-nearest (四捨五入) when necessary. For example, if you need to convert to three significant figures, it will be

$1.234 \rightarrow 1.23$, $1.235 \rightarrow 1.24$, $31.98 \rightarrow 32.0$, $1234 \rightarrow 1.23 \times 10^3$, and so on.

Quiz

Q6. Round the following numbers to three significant figures.

- | | | | |
|------------|---------------|------------|-------------------------|
| (1) 41.11 | (2) 98.76 | (3) 100.12 | (4) 15.449 |
| (5) 2.2360 | (6) 0.0123456 | (7) 9876 | (8) 9.999×10^4 |

Remark: Programmers use $2.99\text{e}8$ (or $2.99\text{E}8$) to mean 2.99×10^8 , or $1.6\text{e}-12$ (or $1.6\text{E}-12$) to mean 1.6×10^{-12} . However, we **should not** use them in handwriting.

Problems

 2.4 Write the following numbers in scientific notation. However, some of them are ambiguous, so answer “ambiguous” if so.

152 340 1000 9999 43210 0.0300 0.00213 31.0 31.00 31

 2.5 The expressions 0.123, 1.23, and 123 are numbers with **three significant figures**. Similarly, 1234 and 1.234×10^{-3} are numbers with **four significant figures**. How about the following expressions?

- | | | | |
|------------|--------------|--------------------------|----------------------------|
| (1) 3.14 | (2) 0.11 | (3) 1.1×10^{-2} | (4) 1.1×10 |
| (5) 1.0008 | (6) 1.0020 | (7) 1.0000 | (8) 4.00×10^{-10} |
| (9) 0.0008 | (10) 0.00080 | (11) 12345 | (12) 67890 |

2.4 Calculation with Significant figures

We need to do calculations of numbers with uncertainties, such as $1.23 + 4.56$ or $1.23/4.56$, but how? Here we discuss a simple method for

- addition ($x + y$) and subtraction ($x - y$)
- multiplication ($x \times y$) and division ($x \div y$ or x/y)

Other calculations, such as $\sqrt{x}$, e^x or $\sin(x)$, need the professional method given in  **TODO**  chap.

Multiplication and Division

As a rule, if x has m significant figures and y has n significant figures, then $x \times m$ or $x \div y$ should be rounded to have $\min(m, n)$ significant figures. ...But the rule is tough to understand. Probably you should see the example, and learn through practice.

Example 2.2

Calculate the following expressions, using calculators.

- (1) 8.912×2.4 (2) $5.0 \div 3.14$ (3) 2.01×49.8 (4) 1.210^3

Solution

Here we use a shorthand notation “3-SF” to mean “three significant figures”.

- (1) Since 8.912 has 4-SF and 2.4 has 2-SF, we round the result to (the smaller) 2-SF:

$$8.912 \times 2.4 = 21.3888 \rightarrow \underline{21}.$$

- (2) Since 5.0 has 2-SF and 3.14 has 3-SF, we round the result to 2-SF:

$$5.0 \div 3.14 = 1.592... \rightarrow \underline{1.6}.$$

- (3) Both 2.01 and 49.8 have 3-SF, so we keep 3-SF, but don't forget to avoid ambiguity!

$$2.01 \times 49.8 = 100.098 \rightarrow 100 \rightarrow \underline{1.00 \times 10^2}.$$

- (4) $1.210^3 = 1.210 \times 1.210 \times 1.210$, so we round the result to 4-SF:

$$1.210^3 = 1.771561 \rightarrow \underline{1.772}.$$

If you want, we can justify this treatment by carrying out a hand calculation of the long multiplication.

$$\begin{array}{r} 1.20 \\ \times 1.31 \\ \hline 120 \\ 360 \\ 120 \\ \hline 15720 \end{array}$$

$$\begin{array}{r} 8.912 \\ \times 2.4 \\ \hline 35648 \\ 17824 \\ \hline 213888 \end{array}$$

$$\begin{array}{r} 2.01 \\ \times 49.8 \\ \hline 1608 \\ 1809 \\ 804 \\ \hline 100.098 \end{array}$$

Observe which digits are uncertain, and how they “pollute” the results in each step. In 1.20×1.31 , the last 1 pollutes the part of $1 + 6$ and we get an uncertain number 7. So, the result is 1.57.

Addition and subtraction

Different rules are applied for addition and subtraction. You can understand the rules easily if you do long addition and observe which digits are polluted.

Example 2.3

Calculate the following expressions.

- (1) $12.3 + 4.56$ (2) $123 + 4.56$ (3) $0.50 - 0.032$ (4) $1.20 \times 10^3 - 27$

Solution

$$\begin{array}{r} 12.3 \\ + 4.56 \\ \hline 16.86 \end{array}$$

$$\begin{array}{r} 123 \\ + 4.56 \\ \hline 127.56 \end{array}$$

$$\begin{array}{r} 0.50 \\ - 0.032 \\ \hline 0.468 \end{array}$$

$$\begin{array}{r} 1200 \\ - 27 \\ \hline 1173 \end{array}$$

and we round the results. So, the answers are 16.9, 128, 0.47, and 1.17×10^3 .

Problems

 **2.6** Calculate the following, taking care of significant figures. You may use calculators.

- | | | | |
|--|---|------------------------|-----------------------------|
| (1) 1.23×4.5 | (2) 1.50×2.0 | (3) 2.00×5.00 | (4) 4.0×2.5 |
| (5) $1.23 + 4.5$ | (6) $1.50 + 2.0$ | (7) $2.00 + 8.00$ | (8) $100.0 + 0.123$ |
| (9) $1.23 \div 4.5$ | (10) $6.0 \div 3.00$ | (11) $9.00 \div 4.5$ | (12) $1.50 \div 3.0$ |
| (13) $1.23 - 4.5$ | (14) $3.10 - 0.10$ | (15) $3.10 - 0.1$ | (16) $3.10 - 3.0$ |
| (17) $131 + 69$ | (18) $131 + 6.9$ | (19) $131 + 0.69$ | (20) $1.3 \times 10^2 + 69$ |
| (21) $(1.2 \times 10^5) \times 9.4$ | (22) $1.2 \div (5.2 \times 10^5)$ | | |
| (23) $(3.33 \times 10^5) \times (6.3 \times 10^4)$ | (24) $(3.33 \times 10^5) \times (6.3 \times 10^{-4})$ | | |
| (25) $(3.33 \times 10^5) \div (6.3 \times 10^4)$ | (26) $(3.33 \times 10^5) \div (6.3 \times 10^{-4})$ | | |
| (27) $1.00 \times 10^4 + 2.5 \times 10^3$ | (28) $3.00 \times 10^3 - 4.50 \times 10^2$ | | |
| (29) $2.99 \times 10^2 + 0.814$ | (30) $1.23 \times 10^{-4} + 4.5 \times 10^{-6}$ | | |

****2.7** Calculate the following, taking care of significant figures. You may use calculators.

- | | | | |
|-------------------------------------|--------------------------------|---|-----------------|
| (1) $1.50 \times 2.000 \times 3.50$ | (2) $8.00 \div 4.00 \div 0.25$ | (3) 5.0^3 | (4) 2.10^3 |
| (5) $1.0 + 2.0 + 3.0 \times 4.0$ | (6) $17 - 8.0 - 3.0$ | (7) $3.0^3 + 1.2$ | (8) $5.0^3 - 5$ |
| (9) $(1.2 + 3.45) \times 2.1$ | (10) $(8.0 - 1.25) \div 2.0$ | (11) $1.2 \times 3.4 + 5.6$ | |
| (12) $18.0 - 2.5 \times 1.2$ | (13) $1.20 \times 3.40 + 5.60$ | (14) $(5.0 \times 10^2 + 3.4) \times 1.2$ | |

 **2.8** Calculate the following with calculators, taking care of significant figures and units.

- | | |
|---|--|
| (1) $3.1 \text{ m/s} \times 1.0 \text{ hour}$ | (2) $3.1 \text{ m} \div 25 \text{ cm}$ |
| (3) $1.5 \text{ kg} \times 9.8 \text{ m/s}^2$ | (4) $1.50 \text{ km} + 195 \text{ m}$ |
| (5) $(5.2 \text{ kg/m}^3) \times (4.0 \text{ }\mu\text{m})^3$ | (6) $(5.2 \text{ kg/m}^3) \times (4.0 \text{ }\mu\text{m}^3)$ |
| (7) $1.2 \text{ }\mu\text{m} \div 2.00 \text{ nm}$ | (8) $(1.7 \times 10^{-27} \text{ kg}) \times (3.00 \times 10^8 \text{ m/s})^2$ |
| (9) $2.000 \text{ kg} \times (3.0 \text{ m/s}^2)$ | (10) $2.000 \text{ kg} \times (3.0 \text{ m/s})^2$ |
| (11) $(1.60 \times 10^{-19} \text{ C}) \div 2.0 \text{ }\mu\text{s}$ | (12) $(1.60 \times 10^{-19} \text{ C}) \times 5.0 \text{ kV}$ |
| (13) $(6.63 \times 10^{-34} \text{ J} \cdot \text{s}) \times (3.0 \times 10^{12} \text{ Hz})$ | (14) $1.50 \text{ MJ} \div 20.0 \text{ m}$ |

Advanced note: Uncertainties can often be determined subjectively, but central values are also without specific definitions; it is often the averaged value of the measurements, but one may assume some probabilistic distribution and use its median, mean, or mode as the central value.

Chapter 3

Logic

Solve [Problem 1.7](#) (on page 21) again.

- (1) If $mr\omega^2 = 10.0 \text{ kg}\cdot\text{m/s}^2$, $m = 5.0 \text{ kg}$, and $r = 1.0 \text{ m}$, what is ω ?
- (2) If $v_0t + (1/2)at^2 = 5.0 \text{ m}$ with $a = -4.0 \text{ m/s}^2$ and $v_0 = 7.0 \text{ m/s}$, what is t ?
- (3) At time $t = 0$, a particle is at $(x, y) = (2.0 \text{ m}, 0)$. It moves with a constant velocity $(v_x, v_y) = (3.0, 4.0) \text{ m/s}$. What is its position at $t = 1.0 \text{ s}$?

The answers are

- (1) $\omega = \pm 1.4 \text{ s}^{-1}$
- (2) $t = 1.0 \text{ s}, 2.5 \text{ s}$
- (3) $x = 5.0 \text{ m}, y = 4.0 \text{ m}$

but **what do they mean?**

Quiz

Q1. Guess the correct meaning of each of above expressions.

- | | | |
|--|------|---|
| (1) ω is both $+1.4 \text{ s}^{-1}$ and -1.4 s^{-1} | v.s. | ω is either $+1.4 \text{ s}^{-1}$ or -1.4 s^{-1} |
| (2) t is both 1.0 s and 2.5 s | v.s. | t is either 1.0 s or 2.5 s |
| (3) x is 5.0 m and y is 4.0 m | v.s. | Either x is 5.0 m , or y is 4.0 m . |

Elementary educations do not discuss these differences because kids do not know **logical thinking**. Now, you are a grown-up university student. You need to think about it.

3.1 Basic logics

Since elementary school, you have written many equalities and inequalities, such as

$$3 + 5 = 8, \quad 6 + 2 = 10, \quad 1 + 3 > -3, \quad 1 + 3 \neq 0, \quad 3^3 = 9, \quad 1 > 2, \quad \text{and} \quad \sin(\pi) \neq 0.$$

Quiz

Q2. The above statements are either **true** (T) or **false** (F). For each of them, state whether it is true or false.

Q3. Confirm that the following statements are all true:

- | | |
|-------------------------------|-------------------------------|
| (1) $3 + 5 = 8$ is true. | (2) $6 + 2 = 10$ is false. |
| (3) $3 \neq 3$ is false. | (4) False is false. |
| (5) "False is false" is true. | (6) "False is true" is false. |

We can only write true things, such as " $3 + 5 = 8$ ", " $3^2 = 9$ is true", or " $6 + 2 = 10$ is false". In other words, **when you write something, you must confirm that it is true**. Sometimes, you are asked to confirm a statement, or **prove** a statement. The process is called **proof** of the statement.

Numbers can be manipulated by operators such as $+$, $\div$. Similarly, true (T) and false (F) can be manipulated by **and**, **or**, and **not** operators.

and (conjunction) “A and B” ($A \wedge B$) is true if both A and B are true, and false otherwise.

or (disjunction) “A or B” ($A \vee B$) is true if at least one of A and B is true. False if both are false.

not (negation) “not A” ($\neg A$) is true if A is false, and false if A is true.

We can write these property as equations:

$$(T \wedge T) = T, \quad (T \wedge F) = (F \wedge T) = (F \wedge F) = F,$$

$$(F \vee F) = F, \quad (T \vee T) = (T \vee F) = (F \vee T) = T,$$

$$(\neg T) = F, \quad (\neg F) = T,$$

A	B	A and B	A or B	not A
T	T	T	T	F
T	F	F	T	F
F	T	F	T	T
F	F	F	F	T

but the **truth table**, shown to the right, is more useful.

Remark: Note that “ $1 + 1 = 2$ or $2 + 2 = 4$ ” is true, which says the word “or” is a bit different from everyday English. In daily conversation, “*Sho will eat ramen or sushi tonight*” usually means “but not both”. However, in mathematics, if Sho eats both ramen and sushi for a dinner^{#3}, the statement “*Sho had sushi or ramen tonight*” is true.

Quiz

Q4. State whether it is true or false for the following statements.

(1) $1 + 2 = 3$ and $3 + 4 < 5$.

(2) $10 \div 3 > 0$ or $10 \div 2 > 0$.

(3) $\neg(3 > 5)$.

(4) $(6 + 3 > 0) \wedge \neg(5 - 2 > 0)$.

Fail safe note: Don't be confused by daily English.

“ $x = 1$ and $x = -1$ are the two solutions of $x^2 = 1$.” (correct daily English)

“ $x^2 = 0$ is satisfied for $x = 1$ or $x = -1$.” (correct mathematical English)

are both correct, but if you say “ $x = 1$ and $x = -1$ ”, then it means an impossible equality $x = 1 = -1$.

3.2 Implication

We can discuss the following statements: are they true or false?

Example 3.1

- If $x > 5$, then $x > 1$ This is true. (Number larger than 5 are larger than 1.)
- If $x < 9$, then $x < 2$ This is false, because we have a counterexample $x = 5$.
- If $x^2 = 1$, then $x \leq 1$ This is true.
- If $x^2 = 1$, then $x \leq 0$ This is false, because we have a counterexample $x = 1$.
- If $x^2 = 1$, then $x = -1$ This is false, because we have a counterexample $x = 1$.

^{#3}Sho sometimes does it.

In general, it is easy to claim that a statement is false, i.e., to *refute* or *disprove* a statement. You just have to find one **counterexample**. Meanwhile, it is more difficult to claim that a statement is true: you need to write a rigorous proof.

Now, let's find some counterexamples.

Quiz

Q5. The following statements are false. Find a counter example for each.

- (1) If $x^2 > 0$, then $x > 0$. (2) If $x^2 - x = 0$, then $x = 0$.
 (3) If $x^2 = y^2$, then $x = y$. (4) If $x + y < 1$, then $x < 1$ and $y < 1$.

We use the symbol " $\implies$ " to express "if ... then ..." statement.

$A \implies B$ means "If A is true, then B is true." or equivalently, " A implies B ."

and this is called **implication**. For example, the statements in the above quiz can be written as

$$x^2 > 0 \implies x > 0, \quad x^2 - x = 0 \implies x = 0, \quad x^2 = y^2 \implies x = y, \quad x + y < 1 \implies x < 1 \wedge y < 1.$$

Quiz

Q6. Similarly rewrite the statements in Example 3.1 with using the symbol " $\implies$ ".

Advanced note: In formal logic, $A \implies B$ is defined by $(\neg A) \vee B$, which means "if A is true, B must also be true; but if A is false, we do not care about B ". So, if A is always false, $A \implies B$ is always true, no matter what B is. For example, $(x \in \mathbb{R} \wedge x^2 < -1) \implies x = 999$ is a true statement. Through this definition, it is clear that $A \implies B$ is equivalent to $\neg B \implies \neg A$, which we will see below.

3.3 Equivalence and "Solving an equation"

Since elementary school, you have solved many equations, but what does "solving an equation" mean?

Quiz

Q7. (1) Choose true statements.

- (a) $2x - 1 = 0 \implies x = 1/2$. (b) $x^2 = 4 \implies x = 2$.
 (c) $2x - 1 = 0 \implies x > -10$. (d) $x^2 = 4 \implies x = 0$.

(2) Do you think these statements can be considered as "solving an equation"?

(b) and (d) are false statements (find counterexamples!), so we cannot write them. (a) and (c) are true statements, and (a) looks "solving an equation", but we do not consider (c) is "solving an equation". The difference between (a) and (c) is that

$$2x - 1 = 0 \iff x = 1/2 \text{ is true, but } 2x - 1 = 0 \iff x > -10 \text{ is false.}$$

So, for (a), both $\implies$ and $\impliedby$ are correct: we use $\iff$ to express such cases.

Definition 3.2 Equivalent

If both $A \implies B$ and $A \impliedby B$ are true, we call “ A and B are **equivalent**”, and write $A \iff B$.

$A \implies B$ is read by B if A in English.
 $A \iff B$ is read by B if and only if A or B **iff** A (namely, **if-and-only-if**).

Now you need to do a bit of practice. It is actually tough, unfortunately, but this drill is important for logical thinking.

Problems

 **3.1** Fill in the blanks with $\implies$, $\impliedby$, or $\iff$, where x is a real number.

- | | | |
|---|--|--|
| (1) $x = 2$ <input type="text"/> $x > 0$ | (2) $x > 1$ <input type="text"/> $x > 0$ | (3) $x = 1$ <input type="text"/> $x > 0$ |
| (4) $x = -3$ <input type="text"/> $x^2 = 9$ | (5) $x = 0$ <input type="text"/> $x^2 = 0$ | (6) $x < 0$ <input type="text"/> $x^2 > 0$ |
| (7) $x = 3$ <input type="text"/> $x \neq 5$ | (8) $3x = 3$ <input type="text"/> $x = 1$ | (9) $x^3 = x^2$ <input type="text"/> $x = 1$ |

If the statement “ $A \Rightarrow B$ ” is true, then

- B is called a **necessary condition** for A , because B is necessary for A ; if not B , then not A .
- A is called a **sufficient condition** for B , because if A is true, B is “sufficiently” true.

So, if A and B are equivalent, B (resp. A) is called *necessary-and-sufficient condition* for A (resp. B). For example, because $x = 1 \implies x^2 = 1$,

- $x = 1$ is sufficient to satisfy $x^2 = 1$. ($x = 1$ is a sufficient condition for $x^2 = 1$.)
- $x^2 = 1$ is necessary to satisfy $x = 1$. ($x^2 = 1$ is a necessary condition for $x = 1$.)

Quiz

Q8. Fill in the blanks with “a sufficient condition” or “a necessary condition”.

- | | |
|--|---|
| (1) $x = -1$ is <input type="text"/> for $x^2 = 1$. | (2) $x^2 = 4$ is <input type="text"/> for $x = 2$. |
| (3) $x > 1$ is <input type="text"/> for $x > 0$. | (4) $x^2 > 0$ is <input type="text"/> for $x < 0$. |
| (5) $x < 0$ is <input type="text"/> for $x^2 > 0$. | (6) $x^2 > 0$ is <input type="text"/> for $x > 0$. |



Now we have the answer for the question: What does “solving an equation” mean?

Definition 3.3 Solving an equation

“Solving an equation” means “finding an *equivalent* equation in the form of $x = \text{[]}$ ”.

Consider the equations in [Problem 2.8](#). As $2x - 1 = 0 \Leftrightarrow x = 1/2$ and $x^2 = 4 \Leftrightarrow x = 2 \vee x = -2$ are true statements, we say $x = 1/2$ and $x = \pm 2$ are the solutions of the equations, respectively.

When you solve an equation, you have to check that the solution is **necessary and sufficient**.

- If you don't check necessity, you may have an "incomplete solution".
- If you don't check sufficiency, you may have an "extraneous solution".

Quiz

- Q9. (1) Solve $\sqrt{x+2} = x$. You might reach an *extraneous solution*, which you need to *exclude* it.
- (2) Solve $|2x| = |x+1|$. Obviously $x = 1$ satisfies this equation, but it is an *incomplete solution* and you need to find more solutions.

Before discussing advanced topics on logics, you should do some drills.

Problems

 **3.2** State whether each statement is true (T) or false (F). For example, $2 > 5$ is "F".

- | | | |
|----------------------------------|------------------------------------|--------------------------------------|
| (1) $(3 > 1) \wedge (2 < 5)$ | (2) $3 < 1 \vee 2 > 5$ | (3) $\neg(3 > 1)$ |
| (4) $\neg(2 > 5)$ | (5) $(\neg(3 < 1)) \wedge (2 < 5)$ | (6) $\neg(3 < 1 \wedge 2 < 5)$ |
| (7) $(4 = 4) \wedge \neg(2 > 5)$ | (8) $(3 < 1) \vee \neg(3 = 1)$ | (9) $(\neg(3 < 1)) \vee \neg(3 > 1)$ |

 **3.3** State whether each statement is true or false. If false, give a counter example.

- | | |
|--|--|
| (1) If $x = 2$, then $x^2 = 4$. | (2) If $x^2 = 4$, then $x = 2$. |
| (3) $x^2 = 0 \implies x = 0$ | (4) $x > 2 \implies x > 0$ |
| (5) $x > 0 \implies x > 2$ | (6) $x = 3 \implies x = 3$ |
| (7) $(x = 1) \vee (x = -1) \implies x^2 = 1$ | (8) $x^2 = 1 \implies (x = 1) \vee (x = -1)$ |

★3.4 Solve each equation, assuming x is a real number.

- | | | | |
|----------------------|----------------------|-----------------|------------------|
| (1) $\sqrt{x} = 3$ | (2) $\sqrt{x-1} = 2$ | (3) $ x = 5$ | (4) $ x = -3 $ |
| (5) $\sqrt{x^2} = 3$ | (6) $ x^2 = 5$ | (7) $ x-2 = 3$ | (8) $ 2x+1 = 7$ |

****3.5** Solve each equation, assuming x is a real number.

- | | | |
|----------------------------|----------------------------|----------------------------|
| (1) $\sqrt{x-3} = 2$ | (2) $\sqrt{x-3} = x-3$ | (3) $\sqrt{x-3} = 3-x$ |
| (4) $\sqrt{(x-3)^2} = 2$ | (5) $\sqrt{(x-3)^2} = x-1$ | (6) $\sqrt{(x-3)^2} = x-3$ |
| (7) $\sqrt{(x-3)^2} = 3-x$ | (8) $\sin x = 1/2$ | (9) $\tan x = 0$ |

★3.6 The following “solutions” have logical errors. Identify errors and find the right answer.

- (1) A student solves $x^2 = 9$ and writes $x = 3$.
- (2) He solves $x^2 - 3x = 0$ by dividing both sides by x , obtaining $x - 3 = 0$, so $x = 3$.
- (3) He solves $x^6 = x^4$ by dividing both sides by x^4 , obtaining $x^2 = 1$, so $x = \pm 1$.

Next, fill in the blanks with $\implies$, $\impliedby$, or $\iff$.

- (4) $x^2 = 9$ $x = 3$
- (5) $x^2 - 3x = 0$ $x - 3 = 0$ $x = 3$
- (6) $x^6 = x^4$ $x^2 = 1$ $x = \pm 1$

★3.7 Let x and y be real numbers. Fill in the blanks with $\implies$, $\impliedby$, or $\iff$. If your answer is $\implies$ or $\impliedby$, explain the reason by giving relevant counter examples.

- (1) $x = 2$ $x^2 = 4$
- (2) $x^2 = 4$ $x = 2 \vee x = -2$
- (3) $x > 0$ $x^2 > 0$
- (4) $|x| = 2$ $(x = 2 \vee x = -2)$
- (5) $x + 1 = 0$ $x = -1$
- (6) $x^2 = x$ $x(x - 1) = 0$ $x = 0$
- (7) $\sqrt{x} = 123$ $x = 123^2$
- (8) $\sqrt{x^2} = 0$ $x = 0$
- (9) $\sqrt{x^2} = 3$ $x = 3$
- (10) $\sqrt{x^2} = 3 \wedge x > 0$ $x = 3$
- (11) $\sqrt{x^2} = \sqrt{y^2}$ $x = y$
- (12) $x + y > 0$ $x > 0 \wedge y > 0$
- (13) $xy = 0$ $x = y = 0$
- (14) $x^2 + y^2 = 0$ $x = y = 0$

**3.8 Let a be a real constant and x be a real number. For (1)–(4), fill in the blanks with $\implies$, $\impliedby$, or $\iff$. If your answer is $\implies$ or $\impliedby$, explain the reason by counter examples.

- (1) $\sqrt{x} = 4$ $x = 16$
- (2) $\sqrt{x^2} = 4$ $x = 4$
- (3) $\sqrt{x} = a^2$ $x = a^4$
- (4) $\sqrt{x^2} = a^2$ $x = a^2$
- (5) Solve the equation $\sqrt{x^2} = a$, noting that a can be negative.

**3.9 (1) Recall that $A \Rightarrow B$ is defined by $(\neg A) \vee B$. Also, recall that $A \Leftarrow B$ is defined by $A \Rightarrow B \wedge A \Leftarrow B$. Write a truth table (see page 28) for $A \Rightarrow B$, $A \Leftarrow B$, and $A \Leftrightarrow B$.

- (2) Explain the reason we can understand $A \Leftrightarrow B$ as $A = B$.
- (3) Write a truth table for the following expressions:

$$A \wedge B, \quad \neg(A \wedge B), \quad (\neg A) \vee (\neg B), \quad \neg(A \vee B), \quad (\neg A) \wedge (\neg B)$$

Explain that this truth table is considered as a *proof* of **de Morgan's theorem**

$$\neg(A \vee B) = (\neg A) \wedge (\neg B), \quad \neg(A \wedge B) = (\neg A) \vee (\neg B). \quad (3.1)$$

- (4) Prove the following, which we will discuss in the next section.

$$(A \Rightarrow B) = ((\neg B) \Rightarrow (\neg A)), \quad (A \Rightarrow B) = \neg(A \wedge \neg B). \quad (3.2)$$

***3.10** This lecture does not cover **quantifiers** “for-all $\forall$ ” and “exists $\exists$ ”. Learn them by yourselves. In particular, prove the following:

$$(1) \exists x, P(x) \iff \neg(\forall x, \neg P(x))$$

$$(2) \neg(\exists x, P(x)) \iff \forall x, \neg P(x)$$

$$(3) \forall x, P(x) \iff \neg(\exists x, \neg P(x))$$

$$(4) \neg(\forall x, P(x)) \iff \exists x, \neg P(x)$$

***3.11** For each statement, write its negation, such as “not ($x \geq 0$ for all real x)”, in a natural form. Then, state whether the original statement and the negated one are true or false.

$$(1) x^2 \geq 0 \text{ for all real } x.$$

$$(2) \text{ There exists a real } x \text{ such that } x^2 = -1.$$

$$(3) \text{ For all real } x, x > 0 \Rightarrow x^2 > 0.$$

$$(4) |x| = x \text{ for all real } x.$$

3.4 A few more notes about Logic and Proof

There is a useful theorem for “not” operator, called **de Morgan’s theorem**.

Theorem 3.4 De Morgan's theorem

1. “not (A and B)” is equivalent to “(not A) or (not B)”, i.e., $\neg(A \wedge B) = (\neg A) \vee (\neg B)$.
2. “not (A or B)” is equivalent to “(not A) and (not B)”, i.e., $\neg(A \vee B) = (\neg A) \wedge (\neg B)$.

The proof is given as [Problem 3.9](#).

Example 3.5

For each of the following statements, write their negation in a natural form.

- (1) $x = 3$ (2) $y < 2$ (3) $x = 3$ or $y < 2$ (4) $x = 3$ and $y < 2$

Solution

The negation of (1) is $x \neq 3$ and the negation of (2) is $y \geq 2$. Since de Morgan’s theorem says $\neg(A \text{ or } B) = (\neg A \text{ and } \neg B)$, the negation of (3) is $x \neq 3$ and $y \geq 2$. Similarly, with de Morgan’s theorem, the negation of (4) is $x \neq 3$ or $y \geq 2$.

Quiz

Q10. For each of the following, write its negation.

- (1) $x > 0$ or $x < 1$. (2) $x \neq 4$ and $x \neq 5$. (3) $0 < x < 1$.
 (4) $x > 0$ or $y > 0$. (5) $x = 0$ and $y = 0$. (6) $x \neq 0$ and $y \neq 0$.

In general, $A \Rightarrow B$ and $B \Rightarrow A$ are different. However, if $A \Rightarrow B$, then $\neg B \Rightarrow \neg A$ ($\text{not } B \Rightarrow \text{not } A$) is always correct. It is called the **contrapositive** of $A \Rightarrow B$. Let’s see an example.

Example 3.6

If $0 < x < 2$, then x^2 is always less than 4. We can write it by $(0 < x < 2) \Rightarrow (x^2 < 4)$. Let A be the statement “ $0 < x < 2$ ” and B be the statement “ $x^2 < 4$ ”. “Not A ” is “ $x \leq 0$ or $x \geq 2$ ”. Meanwhile, “not B ” is “ $x^2 \geq 4$ ”.

- Its **converse** ($A \Leftarrow B$) is $(x^2 < 4) \Rightarrow (0 < x < 2)$, which is false (counterexample: $x = -1$).
- Its **inverse** ($\text{not } A \Rightarrow \text{not } B$) is $(x \leq 0 \text{ or } x \geq 2) \Rightarrow (x^2 \geq 4)$, which is false.
- Its **contrapositive** ($\text{not } A \Leftarrow \text{not } B$) is $(x^2 \geq 4) \Rightarrow (x \leq 0 \text{ or } x \geq 2)$, which is true.

Because the contrapositive of a true statement is always true, we can prove a statement by proving its contrapositive. This method is called **proof by contrapositive**.

Quiz

Q11. For each of the following statements, write its converse, inverse, and contrapositive. Then state whether each of them is true or false.

- | | |
|-----------------------------------|--------------------------------------|
| (1) If $x > 3$, then $x > 0$. | (2) If $x = 2$, then $x^2 = 4$. |
| (3) If $ x = 0$, then $x = 0$. | (4) If $x^2 = 4$, then $x \neq 3$. |

Q12. Prove the following statements by *proof by contrapositive*. Namely, first write the contrapositive, then prove it.

- | | |
|--|---|
| (1) If $x^2 > 1$, then $ x > 1$. | (2) If $x^3 + x^2 + x < 0$, then $x < 0$. |
| (3) If $xy \neq 0$, then $x \neq 0$. | (4) If n^2 is not integer, then n is not integer. |

Another powerful method is **proof by contradiction** (deductio ad absurdum). Imagine you prove “if A , then B ”. You assume “ A ” and “not B ” are both true, and derive a contradiction. If you reach a contradiction, it means “ A and not B ” is false, i.e., “not (A and not B)” is true, which means $A \Rightarrow B$.

Quiz

Q13. Prove the following statements by *proof by contradiction*.

- (1) If x^2 is an even integer, then x cannot be an odd integer.
- (2) If a , b , and c are integers and abc is even, at least one of a , b , or c is even.

Remark: The validity of “proof by contrapositive” and “proof by contradiction” was already discussed on page 32, Eq. (3.2).

Advanced note: You learned another useful method for proof, **mathematical induction**, in high school. In university, you may learn more advanced methods, such as **infinite descent** and **transfinite induction**.

3.5 Roles of statements

One of the main reasons students find university physics difficult is that they treat all statements in the same way. In physics, every statement has its own role:

- experimental fact
- definition (in mathematics: definition)
- assumption / approximation (in mathematics: axiom or assumption)
- derived statements (in mathematics: theorem, corollary, or proposition)

You should always know which role each statement plays.

Remark: Mathematics does not have experimental facts because it is not based on experiments; it is purely based on definitions.

Example 3.7 Types of statements in a calculation

A particle moves as $x(t) = kt^2 + x_0$, where k and x_0 are constants. Find the averaged velocity between $t = 0$ and $t = t_0$.

Solution

We should first clarify assumptions and definitions:

- Assumptions (explicitly written): $x(t) = kt^2 + x_0$. k and x_0 are constants.
- Assumptions (implicit): k , x_0 , and t_0 are real numbers, and t means time.
- Definition: averaged velocity is *defined* by $v_{\text{avg}} := \frac{x(t_0) - x(0)}{t_0 - 0}$.

Now, we can calculate $\frac{x(t_0) - x(0)}{t_0 - 0} = \frac{(kt_0^2 + x_0) - x_0}{t_0} = kt_0$ to give a derived statement:

- Derived statement (conclusion): the averaged velocity between $t = 0$ and t_0 is $v_{\text{avg}} = kt_0$.

Quiz

Q14. Solve the following problem, but with clarifying the nature of each of your statements (as is done in the above example).

A particle moves as $x(t) = kt^2 + x_0$, where k and x_0 are constants. Find the instantaneous velocity at $t = t_1$.

Consider an **experimental fact**, such as “gravity is always attractive”. We have to accept it. We need to remember it and understand what it says. “Protons have charge $+|e|$ and electrons have charge $-|e|$ ” is another experimental fact, but we can understand it as the definition of the elementary charge $|e|$.

Definitions often appear as mathematical equations, such as “instantaneous velocity is defined by $v(t) := dx/dt$ ”. Another example is “uniform circular motion is a motion with constant speed along a circle”; here, the word “uniform circular motion” is defined.

Meanwhile, **assumption** is what we can impose freely. For example, you can decide whether you ignore air resistance or not. However, every conclusion is valid only under its assumptions. If you impose bad assumptions, your conclusion will be meaningless. For example, if you want to analyze the free fall of a sheet of paper but you ignore air resistance, your result should be invalid and meaningless. You must be very careful when you make assumptions.

Remark: We often use “ $A := B$ ” to say “define A by B ”. For example, $v(t) := dx/dt$ is the definition of instantaneous velocity, and $T := 1/f$ is the definition of period. Meanwhile, “ $A = B$ ” just means “ A is equal to B ”.

Some physicists use $A \equiv B$ to say “ A is defined by B ”, but Sho does not recommend it because mathematicians use the symbol $\equiv$ for other purposes.

Problems

★3.12 Prove the following statements by contrapositive or contradiction.

- (1) $x^2 + y^2 = 0 \implies x = 0$.
- (2) $ab = 0 \implies a = 0$ or $b = 0$.
- (3) $ab = 0 \iff a = 0$ or $b = 0$.

★3.13 Find a logical mistake: Let $a = b = 1$. Then $a^2 = ab$, so $a^2 - b^2 = ab - b^2$. So, $(a - b)(a + b) = b(a - b)$, which gives $a + b = b$, so $2 = 1$.

★3.14 With de Morgan's theorem, rewrite the following statements in a form without "not".

- (1) not $(x > 0$ and $y > 0)$
- (2) not $(x = 0$ or $y = 0)$
- (3) not $(\text{if } x > 1, \text{ then } x^2 > 1)$
- (4) not $(x > 0 \text{ iff } x^2 > 0)$

***3.15 With de Morgan's theorem, rewrite the following statements in a natural form.

- (1) not $((\text{not } A) \text{ or } (\text{not } B))$
- (2) not $((\text{not } A) \text{ and } (\text{not } B))$
- (3) $\neg((\neg A) \wedge (\neg B))$
- (4) $\neg(A \vee \neg B)$

***3.16 For each of the following statements, write its converse, inverse, and contrapositive. State T or F for the original statement, its converse, inverse, and contrapositive.

- (1) If $x^2 - 5x + 6 = 0$, then $x = 2$ or $x = 3$.
- (2) If $x > 0$ and $y > 0$, then $x + y > 0$.
- (3) If x and y are both even integers, then xy is divisible by 4.

***3.17 A student claims: "I proved $\neg B \Rightarrow \neg A$, so I have proved $B \Rightarrow A$." Is this correct? Explain, and state what the student has actually proved.

***3.18 The following two statements look similar but are logically different. For each, state whether it is T or F and explain.

- (1) For all real x : if $x > 0$, then $x^2 > 0$.
- (2) For all real x : if $x^2 > 0$, then $x > 0$.

***3.19 True or false? Give a reason.

- (1) The contrapositive of a false statement is also false.
- (2) If $A \Rightarrow B$ is true and $B \Rightarrow A$ is false, then A and B are not equivalent.
- (3) If A is false, then $A \Rightarrow B$ is always true, for any B .

*3.20 (1) What is a **proposition**?

(2) Prove the following statements on propositions P , Q , and R .

- (a) $(P \Rightarrow Q \wedge Q \Rightarrow R) \implies P \Rightarrow R$. (**syllogism**)
- (b) $(P \Rightarrow Q) \wedge (P \Rightarrow R) \iff P \Rightarrow (Q \wedge R)$.

Chapter 4

Power, Exponential, and Logarithm

4.1 Powers

Let's begin with a review: can you recall all the rules for powers...?

Quiz

Q1. Calculate without a calculator.

(1) 2^4 , 3^5 , 4^2 , 10^3 , 1^{25} , 2^{10}

(2) 5^0 , 3^0 , 2^0 , 1^0

(3) 2^{-1} , 12^{-1} , 3^{-3} , 2^{-4} , 5^{-2} , 2^{-10}

(4) $\left(\frac{1}{2}\right)^{-1}$, $\left(\frac{1}{3}\right)^{-2}$, $\left(\frac{1}{12}\right)^{-1}$, $\left(\frac{1}{5}\right)^{-2}$, $\left(\frac{1}{2}\right)^{-10}$, $\left(\frac{1}{3}\right)^0$, $\left(\frac{1}{10}\right)^0$

(5) $\left(\frac{1}{2}\right)^4$, $\left(\frac{2}{3}\right)^3$, $\left(\frac{5}{2}\right)^2$, $\left(\frac{2}{5}\right)^{-2}$, $\left(\frac{7}{2}\right)^2$, $\left(\frac{7}{2}\right)^{-2}$, $\left(\frac{1}{3}\right)^2$, $\left(\frac{1}{3}\right)^{-2}$.

(6) $4^{1/2}$, $81^{1/2}$, $3^{1/2}$, $27^{1/3}$, $\left(\frac{1}{4}\right)^{1/2}$, $\left(\frac{1}{7}\right)^{1/2}$, $\left(\frac{8}{9}\right)^{1/2}$, $\left(\frac{15}{3}\right)^{1/2}$

(7) $4^{-1/2}$, $81^{-1/2}$, $3^{3/2}$, $27^{2/3}$, $27^{-2/3}$, $\left(\frac{1}{4}\right)^{3/2}$, $\left(\frac{1}{7}\right)^{-1/2}$, $\left(\frac{1}{2}\right)^{-5/2}$

Q2. Fill in the blanks, where $A > 0$, $B > 0$, and x, y are real numbers.

(1) $A^3 \cdot A^3 = A^{\square}$

(2) $(A^3)^3 = A^{\square}$

(3) $(A \cdot B)^4 = \square \cdot \square$

(4) $A^x \cdot A^y = A^{\square}$

(5) $(A^x)^y = A^{\square}$

(6) $(A \cdot B)^x = A^x \cdot \square$

(7) $A^{\square} = \frac{1}{A^x}$

(8) $A^x = \frac{1}{\square}$

(9) $\frac{A^x}{A^y} = A^{\square}$

(10) $A^0 = \square$

(11) $A^{\square} = A$

(12) $A^{\square} = \sqrt{A}$

(13) $A^{\square} = \sqrt[3]{A}$

(14) $\frac{A^x}{B^x} = \square$

(15) $\left(\frac{A}{B}\right)^{-1} = \square$

Q3. With $A > 0$, $B > 0$, $x \in \mathbb{R}$, and $y \in \mathbb{R}$, the following statements are all false. Find a counterexample for each.

(1) $A^x + A^y = A^{x+y}$

(2) $(A + B)^x = A^x + B^x$

(3) $(A \cdot B)^x = A \cdot B^x$

(4) $(A^x)^y = A^{(x^y)}$

(5) $(A^x)^y = A^{x+y}$

Do not worry if you cannot recall all the rules! Learn this section and come back here, and then you will be able to answer all the questions!

We should first recall $\sqrt{a}$... what is that?

Definition 4.1 n -th root of non-negative numbers

For $a \geq 0$, we define $\sqrt[n]{a}$ for $n = 2, 3, 4, \dots$ by the **positive** solution x of the equation $x^n = a$ and call it the **n -th root** of a . In particular, write we $\sqrt{a}$ instead of $\sqrt[2]{a}$.

Since we have assumed $a \geq 0$, the equation $x^n = a$ always has a unique positive solution. It guarantees that $\sqrt[n]{a}$ is **always and uniquely** determined; we usually say “ $\sqrt[n]{a}$ is **well-defined**” to describe this situation. Also, notice that $\sqrt[n]{a} \geq 0$ by definition.

Example 4.2 n -th root of a positive number

$\sqrt[3]{8}$ is the positive solution of $x^3 = 8$. So, $\sqrt[3]{8} = 2$. Similarly, $\sqrt[4]{1.4641}$ is the positive solution of $x^4 = 1.4641$, so $\sqrt[4]{1.4641} = 1.1$. Notice $\sqrt[n]{0}$ is always 0 for any $n = 2, 3, 4, \dots$

Be careful: We **do not** consider $\sqrt[n]{x}$ for $x < 0$.

Advanced note: It is actually not impossible to consider such cases, but Sho recommends not to use such notations because they involve mathematical subtleties. You will just be confused.

We are ready to define the **power**, a^x . Here, a is called the **base** and x is called the **exponent**.

Definition 4.3 Power for positive base, real exponent

For positive **base** $a > 0$ and real **exponent** $x \in \mathbb{R}$, we define the **power** a^x by the following steps.

- $a^0 := 1$, $a^1 := a$.
- For $x = 2, 3, 4, \dots$, we define $a^x := a \cdot a^{x-1}$ (*recursive definition*).

We have covered the case $x \in \mathbb{N}^0$. Next, we consider positive rational numbers, $x \in \mathbb{Q}$, $x > 0$. Recall that a positive rational number x can be expressed as $x = n/m$ with $n \in \mathbb{N}^0$, $m \in \mathbb{N}^+$.

- If $x > 0$ and $x \in \mathbb{Q}$ but $x \notin \mathbb{N}$, then we define $a^x = a^{n/m}$ ($n, m \in \mathbb{N}^+$, $m \geq 2$) by

$$a^{1/m} := \sqrt[m]{a}, \quad a^{n/m} := (a^{1/m})^n = (\sqrt[m]{a})^n.$$

- If $x > 0$ and $x \notin \mathbb{Q}$, we can consider a sequence of rational numbers $x_k \in \mathbb{Q}$ such that $x_k \rightarrow x$. Then we define $a^x := \lim_{k \rightarrow \infty} a^{x_k}$.

Now we have defined a^x for all $x \geq 0$. Finally,

- If $x < 0$, we define $a^x := 1/(a^{|x|})$.

and then we have defined a^x for all $x \in \mathbb{R}$, if $a > 0$.

Example 4.4 First two items: integer-power of a positive number

- $2^0 := 1$, because of the first item in the above definition.
- The second item says $2^5 := 2 \times 2^4$, $2^4 = 2 \times 2^3$, $2^3 = 2 \times 2^2$, and $2^2 = 2 \times 2^1 = 4$. So, *recursively*, $2^3 = 2 \times 4 = 8$, $2^4 = 2 \times 8 = 16$, and $2^5 = 32$.
- Similarly, $1.1^2 = 1.1 \times 1.1 = 1.21$ and $1.1^3 = 1.1 \times 1.21 = 1.331$.

Example 4.5 Third and fourth items: positive-power of a positive number

For rational number x , we use the third item of the above definition to consider a^x . For example, $27^{1/3} = \sqrt[3]{27} = 3$ and $1.4641^{1/4} = \sqrt[4]{1.4641} = 1.1$. Then, we can get $27^{2/3} = (27^{1/3})^2 = 3^2 = 9$, $1.4641^{0.25} = \sqrt[4]{1.4641} = 1.1$, and $1.4641^{0.75} = (1.4641^{0.25})^3 = 1.1^3 = 1.331$.

If x is not rational, we need to use the fourth item. For example, if $x = \sqrt{2} = 1.4142\dots$, we consider a sequence $1, 14/10, 141/100, 1414/1000, \dots$ which converges to $\sqrt{2}$. Then,

- $2^{\sqrt{2}}$ is the limit of $2^1, 2^{14/10}, 2^{141/100}, 2^{1414/1000}, \dots \rightarrow 2^{\sqrt{2}} = 2.665\dots$
- $3^\pi = 3^{3.14159\dots}$ is the limit of $3^3, 3^{31/10}, 3^{314/100}, 3^{3141/1000}, \dots \rightarrow 3^\pi = 31.544\dots$

Quiz

Q4. Using calculators, calculate the following numbers.

- | | |
|---|--|
| (1) 3^3 and $3^{3.1}$. | (2) $\sqrt[10]{3}$ and $(\sqrt[10]{3})^{31}$ |
| (3) $\sqrt[100]{3}$ and $(\sqrt[100]{3})^{314}$ | (4) $3^{3.14}$ and $3^{3.141592653579}$ |

Example 4.6 negative-power of a positive number

For a^x with $x < 0$, we use the last item in the above definition.

$$1.1^{-2} = \frac{1}{1.1^2} = \frac{1}{1.21} \quad 1.4641^{-0.75} = \frac{1}{1.4641^{0.75}} = \frac{1}{1.331} \quad 3^{-\pi} = \frac{1}{3^\pi} = \frac{1}{31.544\dots}$$

We have discussed a^x for positive base a . Powers a^x for $a \leq 0$ require very careful consideration and we postpone it to the later chapters; Table 2 gives a summary of a^x for various a and x . Two simplest cases are discussed here.

Definition 4.7 Power for negative base

For a negative base $a < 0$, we only consider a^n for integer n .

$$a^0 := 1, \quad a^n := a \cdot a^{n-1} \text{ for } n > 0, \quad a^n := 1/a^{|n|} \text{ for } n < 0. \quad (4.1)$$

This is equivalent to $a^n := (-1)^n |a|^n$, where we define $(-1)^n$ is $+1$ for even n and -1 for odd n .

Definition 4.8 Power for zero base

We define $0^x := 0$ for $x > 0$, and $\sqrt[n]{0} := 0$ for $n \in \mathbb{N}^+$. (We do not consider 0^x for $x \leq 0$.)

Quiz

Q5. Some of the following expressions are not defined. Find all the undefined ones.

π^{-3}	$\pi^{-1/\pi}$	$(-2)^0$	$(-2)^{-3}$	$(-2)^{3/8}$	$-2^{-\pi}$	$\sqrt{-4}$
$\sqrt[3]{3.5}$	$\sqrt[3.5]{3}$	$\sqrt[4]{3}$	$\sqrt[3]{0}$	$0^{2.5}$	$0^{-2.5}$	$0.1^{0.1}$

Table 2: Summary of the definition of a^x for various a and x .

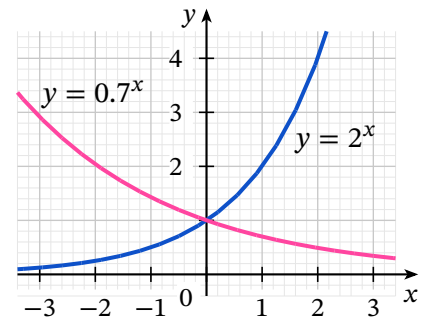
$a > 0$	$x \in \mathbb{R}$	(positive) ^(real)	Definition 4.3
	$x \in \mathbb{C} \setminus \mathbb{R}$	(positive) ^(non-real)	exp-complex-properties
$a < 0$	$x \in \mathbb{Z}$	(negative) ^(integer)	Definition 4.7: $a^x = a ^x \times (-1)^x$
	$x \notin \mathbb{Z}$	(negative) ^(others)	♣TODO♣chapter
$a = 0$	$x > 0$	0 ^(positive)	Definition 4.8: $0^x = 0$
	$x = 0$	0^0	not considered (but sometimes $0^0 := 1$)
	not $x \geq 0$	0 ^(other)	not considered

In the rest of this chapter, we will only consider powers of positive base, i.e., a^x for $a > 0$.

4.2 Exponential and Logarithmic functions

Powers with positive bases, such as 2^x and 0.7^x , are considered as an **exponential function** defined for $x \in \mathbb{R}$. The figure to the right shows the graphs of $y = 2^x$ and $y = 0.7^x$.

Obviously, $f(x) = a^x$ is **strictly increasing** if $a > 1$; **strictly decreasing** if $0 < a < 1$. In Section 4.4, we will check this property by calculating $f'(x)$. It is also important that a^x can take any positive real value. We will use these facts to define $\log_a x$ below in this section.



Quiz

Q6. Draw the graphs of $y = 2^x$, $y = 0.5^x$, and $y = 1^x$ by hand, without using calculators or computers.

Theorem 4.9 Properties of Exponential Functions

For $a > 0$, $b > 0$, and $x, y \in \mathbb{R}$,

$$a^0 = 1, \quad \frac{1}{a^x} = a^{-x}, \quad \sqrt[n]{a^x} = a^{x/n} \quad (n \in \mathbb{N}, n \geq 2); \quad (4.2)$$

$$a^x a^y = a^{x+y}, \quad \frac{a^x}{a^y} = a^{x-y}, \quad (a^x)^y = a^{xy}, \quad (ab)^x = a^x b^x, \quad \left(\frac{a}{b}\right)^x = \frac{a^x}{b^x}. \quad (4.3)$$

Remark: Usually, a^{x^y} is interpreted as $a^{(x^y)}$, not as $(a^x)^y = a^{xy}$.

Quiz

Q7. Transform the following numbers to the form of $2^\square$.

- (1) 2 (2) 1024 (3) $\sqrt{2}$ (4) $\sqrt[3]{4}$ (5) 0.25 (6) $\sqrt[5]{0.25}$
 (7) 4^x (8) $(\sqrt{2})^x$ (9) $4 \cdot 2^x$ (10) $2^x/16$ (11) $2^x 4^y$ (12) $2^x/4^y$

Q8. Without calculators, arrange the following numbers in order from smallest to largest.

$$2, \quad 1024, \quad 2^5, \quad 8^2, \quad \sqrt{2}, \quad \sqrt[3]{4}, \quad 0.25, \quad 0.5^{0.3}, \quad \sqrt[5]{0.25}, \quad 0, \quad 1$$



Consider the equation $p = a^x$, where $a > 0$, $a \neq 1$ and $p \in \mathbb{R}$. For a given p , how many solutions x does it have? As $a \neq 1$, a^x is strictly increasing or decreasing and it takes any value $0 < a^x < \infty$. Therefore, $p = a^x$ has a unique real solution x for any $p > 0$. We call the solution $x = \log_a p$.

Definition 4.10 Logarithm

For $a > 0$, $a \neq 1$, and $p > 0$, we define the **logarithm** $\log_a p$ as the unique solution x of $a^x = p$.

$$\text{For } a > 0, a \neq 1, \text{ and } p > 0, \quad a^x = p \iff x = \log_a p. \quad (4.4)$$

Notice that $\log_a p$ is undefined if not $(a > 0 \wedge a \neq 1 \wedge p > 0)$, i.e., if $(a < 0 \vee a = 1 \vee p \leq 0)$.

Fail safe note: Go back to Theorem 3.4, de Morgan's theorem, on page 34.

Quiz

Q9. Based only on the above discussion, explain why $k = \log_a(a^k)$ and $a^{\log_a k} = k$.

We here review basic properties of $\log_a x$. Drill problems are prepared later in this section.

Theorem 4.11 Properties of Logarithmic Functions

For $a > 0$ but $a \neq 1$, $b > 0$ but $b \neq 1$, $A > 0$, and $B > 0$,

$$\log_a 1 = 0, \quad \log_a a = 1, \quad \log_a A^k = k \log_a A \quad (k \in \mathbb{R}), \quad (4.5)$$

$$\log_a(AB) = \log_a A + \log_a B, \quad \log_a\left(\frac{A}{B}\right) = \log_a A - \log_a B, \quad (4.6)$$

$$\log_a A = \frac{\log_b A}{\log_b a} \quad [\text{changing the base}]. \quad (4.7)$$

4.3 Napier's number and Natural logarithm

There is a special number for the base, called **Napier's number** e :

$$e = 2.718281828... = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \sum_{k=0}^{\infty} \frac{1}{k!} = 1 + 1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{24} + \frac{1}{120} + \dots \quad (4.8)$$

It is special because of the following theorem:

Theorem 4.12 Napier's number

$$\frac{d}{dx} e^x = e^x, \quad \frac{d}{dx} \ln x = \frac{1}{x}, \quad (4.9)$$

where the **natural logarithm** ($\ln x$) is defined by $\ln x := \log_e x$. Also, we often write e^x as $\exp(x)$.

Quiz

Q10. Draw the graphs of $y = \exp(x)$ and $y = \ln x$, using computers or calculators.

Q11. Calculate the following, where $a > 0$, $b > 0$, and $b \neq 1$.

$$(1) \frac{d}{dx} \exp(x) \quad (2) \frac{d}{dx} e^{-3x} \quad (3) \frac{d}{dx} a^x \quad (4) \frac{d}{dx} \ln(3x) \quad (5) \frac{d}{dx} \log_b x$$

Fail safe note: Review [Problem 3.20](#) for (3) and Eq. (4.7) for (5). Recall e is just a number.

Problems

You may skip these two drill problems, as (Sho expects) you **already have learned** them.

 **4.1** Differentiate them, where a is a positive constant such that $a \neq 1$.

- | | | | | |
|----------------------|-------------------|------------------|-------------------|------------------------|
| (1) $\exp(5x)$ | (2) e^{-x^2} | (3) $3 \exp(2x)$ | (4) $e^{\sin x}$ | (5) $x(e^x)^2$ |
| (6) $\sqrt{e^x + 1}$ | (7) $e^x e^{x^2}$ | (8) $\ln(2x)$ | (9) $\ln 3x $ | (10) $\ln(x^5 + x)$ |
| (11) $\ln \cos x $ | (12) $x \ln x$ | (13) $\ln(1/x)$ | (14) $e^x \ln x$ | (15) $\ln 2x - 1 $ |
| (16) a^{4x} | (17) a^{x^2} | (18) $(2a)^{2x}$ | (19) $\log_a x^5$ | (20) $\log_a(x^5 + x)$ |

 **4.2** Simplify each expression, where a is a positive constant such that $a \neq 1$.

- | | | | |
|-------------------------|-------------------------|------------------------|-------------------------------|
| (1) $9^{-1/2}$ | (2) $16^{3/4}$ | (3) $8^{-2/3}$ | (4) $e^{3 \ln 2}$ |
| (5) $e^{\ln 3 + \ln 4}$ | (6) $e^{\ln(6+7)}$ | (7) $\ln e^{6+7}$ | (8) $e^3(e^4)^4$ |
| (9) $\log_4 8$ | (10) $\log_{10} 0.01$ | (11) $\log_{0.5} 2$ | (12) $\log_3(1/27)$ |
| (13) $\log_6 6^{10}$ | (14) $\ln 6 + \ln(1/6)$ | (15) $2 \ln 3 - \ln 9$ | (16) $\ln 12 - \ln 4 + \ln 2$ |
| (17) 1^{-a} | (18) $\log_a 1$ | (19) $\log_a(2a^2)^4$ | (20) $\log_a(\ln(e^{5a}))$ |

Problems

 **4.3** Some of the following expressions are not defined. Find all the undefined ones.

π^0	π^{-3}	$\pi^{2.5}$	$\pi^{-2.5}$	π^π	$\pi^{-\pi}$	$\pi^{-1/\pi}$
0^0	0^{-3}	$0^{2.5}$	$0^{-2.5}$	0^π	$0^{-\pi}$	$0^{-1/\pi}$
$(-2)^0$	$(-2)^{-3}$	$(-2)^{2.5}$	$(-2)^{-2.5}$	$(-2)^\pi$	$-2^{-\pi}$	$-2^{-1/\pi}$
$\sqrt{4.5}$	$\sqrt{-4}$	$\sqrt[3]{3.5}$	$\sqrt[3.5]{3}$	$^{-4}\sqrt{3}$	$\sqrt[4]{-3}$	$\sqrt[4]{1+\sqrt{2}}$
$\sqrt[1.5]{1}$	$\sqrt[4]{-8}$	$^{-3}\sqrt{8}$	$\sqrt[0]{8}$	$\sqrt[3]{\pi}$	$\sqrt[3]{\pi}$	$\sqrt[4]{1-\sqrt{2}}$

★4.4 Solve these equations.

(1) $2^{x+1} = 64$	(2) $e^{3x-1} = 4$	(3) $e^x = 5$
(4) $\ln x-1 = 0$	(5) $\ln x = 4$	(6) $\log_5(2x+3) = 2$
(7) $2^{-2x} = 7$	(8) $\log_3(2x+1) = 3$	(9) $e^{2x} - 3e^x + 2 = 0$

*****4.5** Draw the graphs of the following functions.

(1) $y = 2^x$ (2) $y = 0.5^x$ (3) $y = \left(\sqrt[3]{2}\right)^x$ (4) $y = 2 \cdot 2^x$ (5) $y = \frac{2^x}{2}$

Fail safe note: Draw $y = 2^x$ first. Rewrite the other functions in the form of $y = 2^\square$.

****4.6** Solve these equations.

(1) $\ln(x^2 - 3) = \ln(2x)$	(2) $\ln x + \ln(x-1) = \ln 6$
(3) $\ln x^2 - 3 = \ln 2x $	(4) $\ln x + \ln x-1 = \ln 6$
(5) $\log_2 x + \log_2(x+2) = 3$	(6) $\ln(x+1) + \ln(x-1) = \ln 3$
(7) $\log_2 x - \log_2(x-2) = 3$	(8) $\ln(x+1) - \ln(x-1) = \ln 4$
(9) $\log_2 x + \log_2 x+2 = 3$	(10) $\ln x+1 + \ln x-1 = \ln 4$

****4.7 Hyperbolic functions** are defined by

$$\cosh x := \frac{e^x + e^{-x}}{2}, \quad \sinh x := \frac{e^x - e^{-x}}{2}, \quad \tanh x := \frac{\sinh x}{\cosh x}, \quad \text{etc.} \quad (4.10)$$

(1) Calculate the derivative of $\cosh x$, $\sinh x$, and $\tanh x$.

(2) Draw the graphs of $\cosh x$, $\sinh x$, and $\tanh x$.

(3) Show the following properties:

(a) $\cosh^2 x - \sinh^2 x = 1$.	(b) $\cosh x \geq 1$ for all x .
(c) $\cosh x$ is an even function.	(d) $\sinh x$ and $\tanh x$ are odd functions.

****4.8** Calculate the derivative of $\cosh 2x^2$ and $3x \tanh x^2$.

****4.9** Calculate the derivative of x^x .

***4.10** Calculate the derivative of x^{x^x} and $(x^x)^x$.

4.4 Exponential growth and Exponential decay

Consider $f_0(t) = \exp(At)$ with $A \in \mathbb{R}$, and imagine $t \in \mathbb{R}$ is the time. You may find

$$f_0(t) = \exp(At) \text{ is \textbf{strictly increasing} if } A > 0, \text{ while \textbf{strictly decreasing} if } A < 0. \quad (4.11)$$

Therefore, $f_0(t) = \exp(At)$ is called exponential growth if $A > 0$ and exponential decay if $A < 0$.

Quiz

Q12. Consider the above function $f_0(t) = \exp(At)$ with $A, t \in \mathbb{R}$.

- (1) What does “strictly increasing” mean?
- (2) Calculate $f'_0(t)$ and confirm the two statements in Eq. (4.11).
- (3) Show that $f_0(0) = 1$ and $f'_0(t) = Af_0(t)$.

Notice $e^{At} = 1$ at $t = 0$. If the number at $t = 0$ is N (we assume $N > 0$), we should use Ne^{At} :

Statement 4.13 Exponential growth and decay

Consider the function $f(t) = N \exp(At)$ with $N > 0$ and $A, t \in \mathbb{R}$.

- If $A > 0$, it is strictly increasing; we call $f(t)$ **exponential growth**.
- If $A < 0$, it is strictly decreasing; we call $f(t)$ **exponential decay**.

The function $f(t)$ is characterized by

$$f(0) = N, \quad f'(t) = Af(t). \quad (4.12)$$

(We will come back to Eq. (4.12) in [♣TODO♣ode-section](#).)

Exponential decay is usually expressed by using a positive number $\Gamma > 0$, called the **decay rate**:

$$f(t) = N \exp(-\Gamma t) = N \exp\left(-\frac{t}{\tau}\right) \quad (4.13)$$

with $\tau := 1/\Gamma$. Since τ has the same *dimension*^{#4} as the time t , we call τ the **lifetime**. We also define the **half-life** $T_{1/2}$ so that, at the time $T_{1/2}$, the number becomes half ($N/2$) of the number N at $t = 0$.

Quiz

- Q13.**
- (1) Show that $f'(t) = -\Gamma f(t)$.
 - (2) Show that $T_{1/2} = \tau \ln 2$. Namely, show $f(t) = N/2$ at the time $t = \tau \ln 2$.
 - (3) Assume $f(0) = 40$. Find $f(T_{1/2})$, $f(2T_{1/2})$, and $f(3T_{1/2})$.
 - (4) Assume $f(t_0) = 40$. Find the time t at which $f(t) = 20$ and $f(t) = 10$.
 - (5) Show that $f(t + T_{1/2}) = f(t)/2$ for any t .

For example, if $f(1) = 12$, then $f(1 + T_{1/2}) = 6$, and $f(1 + 2T_{1/2}) = 3$. Each time one half-life passes, the quantity is reduced by half.

^{#4} → [Chapter 2](#)

Problems

***4.11 An RC circuit has charge $Q(t) = Q_0 e^{-t/\tau}$, where $\tau = RC$.

- (1) Verify that $Q(\tau) = Q_0/e$. Then, find the half-life.
- (2) Assume $R = 2.0 \text{ k}\Omega$ and $C = 50 \text{ }\mu\text{F}$. Calculate τ . Then, find the time t at which Q dropped to 25% of Q_0 .

Chapter 5

Vectors as Arrows

Physicists understand a vector in three ways:

1. an arrow in n -dimensional space ($n \in \mathbb{N}^+$, but usually $n = 3$),
2. a list of n numbers arranged vertically ($n \in \mathbb{N}^+$), and
3. an element of a vector space (such as a Hilbert space).

In this chapter, we will focus on the first two interpretations. The last interpretation, more abstract and mathematical, will be discussed in [♣TODO♣chap:linear-algebra](#).

Remark: This document discusses vectors mainly in terms of mathematics; more detailed *physical* discussion about vectors can be found in Sho's [Vector Boot Camp](#).^{#5}

Be careful: In university, we **never** use the horizontal notation $\vec{v} = (x, y, z)$ to describe vectors. Please **always** use the vertical form $\vec{v} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ to be prepared for [Chapter 7](#).

Remark: We use the arrow notation $\vec{v}$ because this is a first-year lecture. Usually, physicists use boldface ($\mathbf{a}, \mathbf{x}, \mathbf{\beta}$, etc.) to denote vectors.

Advanced note: The discussion in this chapter is only valid for *finite-dimensional* vectors because here we will define vectors as arrows in a “space”. As the “space”, readers are expected to imagine *the 3d space* of our Universe, a 2d-sheet of paper in our Universe, or something like those, and then the **dimension** of the space (defined in Theorem 5.20) will be limited to a finite integer.

5.1 What is a Vector?

You have seen vectors, such as $(5, -1)$, $(1, 0, 2)$, or $(-1, 3)$, or in the vertical form $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, etc. However, it represents a *mathematical* nature of vectors. As a **physicist**, you have to forget about such expressions in this section (we will return to it in Section 5.8). Instead, we will first *define* vectors in a more physical way.

Definition 5.1 Vector for physicists

Consider a space. A **vector** is a straight arrow drawn in the space, or in general, a quantity that has both **magnitude** (= length) and **direction** in the space.

We here do not investigate what “the space” is, but you may well imagine a lecture room as the space.

^{#5}Visit <https://misho104.github.io/LecturePublic> and find gp2_boot2_vector.pdf.

Notation 5.2 Vectors and Scalars

We denote vectors with an arrow over the symbol, such as $\vec{v}$, $\vec{F}$, $\vec{a}$, $\vec{p}$. Their magnitudes are written as $|\vec{v}|$, $|\vec{F}|$, $|\vec{a}|$, $|\vec{p}|$, respectively. Meanwhile, a quantity that has no direction is called a **scalar**. Symbols without arrows, such as v , F , a , p , are scalars.

Be careful: v and $\vec{v}$ are **completely different objects**. They are totally unrelated, just as A and a are unrelated. However, in physics, we are sometimes *lazy* enough to write v to mean $|\vec{v}|$, the magnitude of $\vec{v}$. Please not be confused.

Quiz

Q1. Choose scalar quantities. Choose vector quantities.

- (1) $\vec{a}$ (2) p (3) $|\vec{b}|$ (4) $|g|$ (5) magnitude of $\vec{v}$
 (6) mass (7) velocity (8) speed (9) $\vec{x} + |\vec{x}|$ (10) direction of $\vec{v}$

- Q2.** (1) What do we call $|\vec{v}|$? Also, clearly write down its definition.
 (2) What do we call v ? Also, clearly write down its definition.

Fail safe note: For $x \in \mathbb{R}$, we define $|x| := \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0 \end{cases}$ and call it “the **absolute value** of x ”.

There is a special vector “**zero vector**”, which has magnitude 0 and no direction.

Definition 5.3 Zero vector

There is a vector with magnitude 0 and no direction. We call it **the zero vector** and denote $\vec{0}$.

Any other vectors have a direction and **positive** magnitude. Namely,

$$|\vec{0}| = 0. \quad |\vec{v}| = 0 \iff \vec{v} = \vec{0}. \quad |\vec{v}| > 0 \iff \vec{v} \neq \vec{0}. \quad (5.1)$$

Quiz

Q3. Explain why $\vec{0} \neq 0$. Explain why $|\vec{0}| = 0$.

Advanced note: The uniqueness of $\vec{0}$ is easy to prove, once we clarify the definition of “ $=$ ”. Namely, “ $\vec{a} = \vec{b}$ ” means “ $\vec{a}$ and $\vec{b}$ have the same magnitude and direction”. (Try to complete the proof.)

5.2 Addition and Scalar multiplication

Like $+$ and $\div$ for numbers and $\wedge$ and $\vee$ for true/false, we have two operations for vectors:

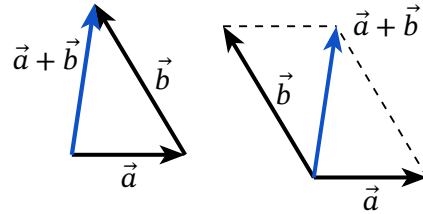
- **addition:** two vectors $\vec{a}$ and $\vec{b}$ can be added; we write the result by $\vec{a} + \vec{b}$.

- **scalar multiplication:** a vector $\vec{a}$ is multiplied by a scalar k ; we write the result by $k\vec{a}$.

Definition 5.4 Vector addition

If $\vec{a}$ and $\vec{b}$ are vectors drawn in the same space, we can define $\vec{a} + \vec{b}$ by the vector obtained by placing the tail of $\vec{b}$ at the head of $\vec{a}$.

The triangle in the figure to the right represents this definition: the tail of $\vec{b}$ is put at the head of $\vec{a}$ to get the blue arrow $\vec{a} + \vec{b}$. You may also use the parallelogram method shown to the right: consider a parallelogram made by $\vec{a}$ and $\vec{b}$ sharing the same tail. Then its diagonal, the blue arrow, is $\vec{a} + \vec{b}$.



Notice that the two horizontal arrows in the figure are both $\vec{a}$, although their locations are different. Similarly, both blue arrows show the vector $\vec{a} + \vec{b}$. Vectors are the same if and only if they have the same magnitude and direction; location does not matter.

Definition 5.5 Scalar multiplication

For a vector $\vec{v}$ and a real number (scalar) k , we define $k\vec{v}$ as follows:

- if $k > 0$, $k\vec{v}$ has the magnitude $|k||\vec{v}|$ and has the same direction as $\vec{v}$.
- if $k < 0$, $k\vec{v}$ has the magnitude $|k||\vec{v}|$ and is anti-parallel to $\vec{v}$.
- if $k = 0$, $k\vec{v} = \vec{0}$.

Remark: The word “**anti-parallel**” means “in the opposite direction”. Meanwhile, **we should avoid** the ambiguous word “**parallel**” for vectors, as it may mean either the same direction or the opposite direction; use the phrase “has the same direction” instead. For more vocabulary to describe directions, please check [the Vector Boot Camp](#).

Advanced note: Watch out we assume $k \in \mathbb{R}$ in Definition 5.5. It restricts the discussion in this chapter to *real vectors*. In Section 6.5, we will discuss *complex vectors* by modifying this definition.

We should carefully digest these definitions. Let’s see an example, and try the next quiz.

Example 5.6

- (1) Explain the meaning of $|k\vec{v}|$ and $|k||\vec{v}|$.
- (2) Prove $|k\vec{v}| = |k||\vec{v}|$, where $\vec{v}$ is a vector and $k \in \mathbb{R}$.
- (3) Based on the above definitions, explain what $-\vec{v}$ is.

Solution

- (1) $|k\vec{v}|$ means the magnitude of a vector $k\vec{v}$, which is a scalar multiplication of $\vec{v}$ by k . Meanwhile, $|k||\vec{v}|$ means the absolute value of k times the magnitude of $\vec{v}$.
- (2) If $k \neq 0$, then, according to the above definition, $k\vec{v}$ has the magnitude $|k||\vec{v}|$ and it means $|k\vec{v}| = |k||\vec{v}|$. If $k = 0$, then $k\vec{v} = \vec{0}$ and thus LHS = $|k\vec{v}| = 0$, while RHS = $0|\vec{v}| = 0$. ■

- (3) $-\vec{v}$ is a shorthand notation of $(-1)\vec{v}$, the scalar multiplication of $\vec{v}$ by -1 . So, $-\vec{v}$ has the same magnitude as $\vec{v}$ but is anti-parallel to $\vec{v}$.

Quiz

Q4. Find out how the following vectors are defined based on the above definitions of addition and scalar multiplication.

- (1) $\vec{a} + \vec{a}$ (2) $2\vec{a}$ (3) $2\vec{a} + 3\vec{b}$ (4) $-\vec{a}$ (5) $\vec{a} - \vec{b}$ (6) $-\vec{a} - 2\vec{b}$

Addition and scalar multiplication have the following properties:

Theorem 5.7

For vectors $\vec{a}$ and $\vec{b}$ in the same space and $p, q \in \mathbb{R}$,

- | | |
|---|--|
| (A) $\vec{a} + \vec{b} = \vec{b} + \vec{a}$, | (E) $p\vec{a} + p\vec{b} = p(\vec{a} + \vec{b})$, |
| (B) $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$, | (F) $p\vec{a} + q\vec{a} = (p + q)\vec{a}$, |
| (C) $\vec{a} + \vec{0} = \vec{a}$, | (G) $(pq)\vec{a} = p(q\vec{a})$, |
| (D) $\vec{a} + (-\vec{a}) = \vec{0}$, | (H) $1\vec{a} = \vec{a}$. |

These properties seem obvious, but in fact, they play a fundamental role in ♣TODO♣....

Problems

***5.1 Consider a vector $\vec{a} \neq \vec{0}$ and a constant $k \in \mathbb{R}$. Consider three vectors $\vec{a}$, $k\vec{a}$, and $k^2\vec{a}$.

- (1) Which are in the same direction? Which are anti-parallel to each other?
- (2) Compare their magnitudes; which are the longest and the shortest?

***5.2 Consider a vector $\vec{a}$ and a constant k .

- (1) Write a vector that has magnitude $|\vec{a}|$ and is anti-parallel to $\vec{a}$.
- (2) Write a vector that has magnitude $3|\vec{a}|$ and is in the same direction as $\vec{a}$.
- (3) Write a vector that has magnitude $3k|\vec{a}|$ and is in the same direction as $\vec{a}$.

Vectors with magnitude 1 are called **unit vectors**.

- (4) Write a vector that has magnitude 1 and is in the same direction as $\vec{a}$.
(Namely, write a unit vector that has the same direction as $\vec{a}$.)
- (5) Write a unit vector that is anti-parallel to $\vec{a}$.

***5.3 Prove the next “**triangle inequality**” geometrically, i.e., only with the above definitions.

$$\text{For any two vectors } \vec{a} \text{ and } \vec{b} \text{ in the same space, } |\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|. \quad (5.2)$$

**5.4 Starting from Definition 5.4 and Definition 5.5, prove the properties in Theorem 5.7.

**5.5 Prove that, for any two vectors $\vec{a}$ and $\vec{b}$ in the same space, $||\vec{a}| - |\vec{b}|| \leq |\vec{a} - \vec{b}|$.

Most of the following discussion is only for *real vectors* and do not apply for *complex vectors*.

5.3 Inner product

Imagine two arrows. Probably you can think the angle θ between the arrows. The angle leads you to the following **geometric** definition of the inner product.

Definition 5.8 Inner product (geometrical definition)

For vectors $\vec{a}$ and $\vec{b}$ drawn in the same space, the **inner product** is defined by

$$\vec{a} \cdot \vec{b} := |\vec{a}| |\vec{b}| \cos \theta, \quad (5.3)$$

where θ is the angle between $\vec{a}$ and $\vec{b}$; if $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, then $\vec{a} \cdot \vec{b} := 0$.

Quiz

Q5. Assume the length of $\vec{a}$ is 3 and the length of $\vec{b}$ is 2.

- (1) If $\vec{a} \perp \vec{b}$, then what is $\vec{a} \cdot \vec{b}$?
- (2) If $\vec{a}$ and $\vec{b}$ are anti-parallel, what is $\vec{a} \cdot \vec{b}$?
- (3) What is the maximum value of $\vec{a} \cdot \vec{b}$? When is it achieved?
- (4) If $\vec{a} \cdot \vec{b} = -3$, what can you say about the angle between $\vec{a}$ and $\vec{b}$?

Fail safe note: $\vec{a} \perp \vec{b}$ means that $\vec{a}$ and $\vec{b}$ are **perpendicular** to each other, or **normal** to each other; in other words, the angle θ between them is $\pi/2 = 90^\circ$.

This inner product has the following properties:

Theorem 5.9 Real-vector inner product

For vectors $\vec{a}$ and $\vec{b}$ drawn in the same space and a constant $k \in \mathbb{R}$,

- | | |
|---|---|
| (A) $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$, | (E) $(\vec{a} + \vec{b}) \cdot \vec{c} = \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c}$, |
| (B) $\vec{a} \cdot \vec{a} \geq 0$ for any vector $\vec{a}$, | (F) $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$, |
| (C) $\vec{a} \cdot \vec{a} > 0$ for any vector $\vec{a} \neq \vec{0}$, | (G) $(k\vec{a}) \cdot \vec{b} = k(\vec{a} \cdot \vec{b})$, |
| (D) $\vec{a} \cdot \vec{a} = 0 \iff \vec{a} = \vec{0}$, | (H) $\vec{a} \cdot (k\vec{b}) = k(\vec{a} \cdot \vec{b})$. |

Be careful: These properties are valid only for “real vectors”.

We will skip their proof. Instead, we focus on these four very important properties. You need to memorize them securely.

Theorem 5.10 Properties of real-vector inner product

$$\text{For a vector } \vec{a}, \quad |\vec{a}| = \sqrt{\vec{a} \cdot \vec{a}}. \quad (5.4)$$

$$\text{For non-zero vectors } \vec{a} \text{ and } \vec{b}, \quad \vec{a} \cdot \vec{b} = 0 \iff \vec{a} \perp \vec{b}. \quad (5.5)$$

$$\text{For vectors } \vec{a} \text{ and } \vec{b}, \quad -|\vec{a}||\vec{b}| \leq \vec{a} \cdot \vec{b} \leq |\vec{a}||\vec{b}| \quad (\text{Cauchy-Schwarz inequality}). \quad (5.6)$$

$$\text{For vectors } \vec{a} \text{ and } \vec{b}, \quad |\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2 \quad (5.7)$$

Be careful: These properties are valid only for “real vectors”.

Quiz

Q6. Prove the following theorems directly from Definition 5.8.

$$(1) \quad \vec{a} \perp \vec{b} \implies \vec{a} \cdot \vec{b} = 0.$$

$$(2) \quad \vec{a} \cdot \vec{b} = 0 \implies (\vec{a} \perp \vec{b}) \vee (\vec{a} = \vec{0}) \vee (\vec{b} = \vec{0}).$$

$$(3) \quad -|\vec{a}||\vec{b}| \leq \vec{a} \cdot \vec{b} \leq |\vec{a}||\vec{b}|.$$

$$(4) \quad |\vec{a} \cdot \vec{b}| \leq |\vec{a}||\vec{b}|. \quad [\text{Hint: Recall that } |x| \leq 3 \text{ means } -3 \leq x \leq 3.]$$

$$(5) \quad |\vec{a}|^2 = \vec{a} \cdot \vec{a}.$$

$$(6) \quad |\vec{a}| = \sqrt{\vec{a} \cdot \vec{a}}. \quad [\text{Hint: Most students make mistakes in this question.}]$$

Q7. Prove the following equation, using Definition 5.8, Theorem 5.9, and the equations in the previous quiz.

$$(1) \quad (\vec{a} + \vec{b}) \cdot \vec{a} = |\vec{a}|^2 + \vec{a} \cdot \vec{b}. \quad (2) \quad |\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + 2\vec{a} \cdot \vec{b} + |\vec{b}|^2.$$

Problems

*****5.6** Two vectors $\vec{a}$ and $\vec{b}$ satisfy $|\vec{a}| = 2$, $|\vec{b}| = 3$, and $\vec{a} \cdot \vec{b} = 3$. Let x, y be real numbers.

(1) Find the angle between $\vec{a}$ and $\vec{b}$.

(2) Calculate $|\vec{a} + \vec{b}|$.

(3) Calculate the magnitude of $4\vec{a} + 3\vec{b}$ and $x\vec{a} + y\vec{b}$.

(4) Explain why $|x\vec{a} + y\vec{b}|^2$ is **not** in general equal to $x^2 + y^2$.

Now let $\vec{e}_1$ and $\vec{e}_2$ satisfy $|\vec{e}_1| = |\vec{e}_2| = 1$ and $\vec{e}_1 \cdot \vec{e}_2 = 0$. Let p, q be real numbers.

(5) Find the angle between $\vec{e}_1$ and $\vec{e}_2$.

(6) Explain why $(p\vec{e}_1 + q\vec{e}_2) \cdot (x\vec{e}_1 + y\vec{e}_2) = px + qy$.

(7) Explain why $|x\vec{e}_1 + y\vec{e}_2| = \sqrt{x^2 + y^2}$.

****5.7** Three vectors $\vec{A}$, $\vec{B}$, $\vec{C}$ satisfy $|\vec{A}| = 2$, $|\vec{B}| = 3$, $\vec{A} \cdot \vec{B} = 3\sqrt{2}$, and $\vec{A} \cdot \vec{C} = -1$.

- (1) Find the angle between $\vec{A}$ and $\vec{B}$.
- (2) Calculate $|\vec{A} + \vec{B}|^2$, $|\vec{A} - \vec{B}|^2$, and $|2\vec{A} + 4\vec{B}|^2$.
- (3) Calculate $(\vec{A} - 2\vec{B}) \cdot (2\vec{A} + \vec{B} + \vec{C}) + 2\vec{B} \cdot \vec{C}$.
- (4) Find k such that $|\vec{A} + k\vec{B}| = \sqrt{10}$.
- (5) Find c such that $\vec{B} + c\vec{C}$ is perpendicular to $\vec{A}$.

5.4 Cross product (only for 3d real-vectors)

For two arrows drawn in three-dimensional space, we can define the **cross product**.

Definition 5.11 Cross product

For $\vec{a}$ and $\vec{b}$ drawn in a *three-dimensional* space, the **cross product** $\vec{a} \times \vec{b}$ is defined as follows:

- It is a vector with magnitude $|\vec{a} \times \vec{b}| = |\vec{a}| |\vec{b}| \sin \theta$, where θ is the angle between $\vec{a}$ and $\vec{b}$.
- If $|\vec{a} \times \vec{b}| \neq 0$, we determine its direction so that $(\vec{a} \times \vec{b}) \perp \vec{a}$ and $(\vec{a} \times \vec{b}) \perp \vec{b}$.

Here, we always have two possible directions. We impose another condition to make it unique:

- If you hold the *right* hand so that your thumb points in the direction of $\vec{a}$ and your index finger in the direction of $\vec{b}$, your middle finger points in the direction of $\vec{a} \times \vec{b}$.
(This is often quoted that “the ordered triple $(\vec{a}, \vec{b}, \vec{a} \times \vec{b})$ obeys the **right-hand rule**”.)

We will not discuss much about the cross product, but only the following properties:

Theorem 5.12

For vectors $\vec{a}$, $\vec{b}$, $\vec{c}$ and real number k ,

- | | |
|---|--|
| (A) $\vec{a} \times \vec{a} = \vec{0}$. | (E) $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$. |
| (B) $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$. | (F) $\vec{a} \times \vec{b} = \vec{0}$ if $\vec{a}$ and $\vec{b}$ are parallel or anti-parallel. |
| (C) $0 \leq \vec{a} \times \vec{b} \leq \vec{a} \vec{b} $. | (G) $\vec{a} \cdot (\vec{a} \times \vec{b}) = \vec{b} \cdot (\vec{a} \times \vec{b}) = 0$. |
| (D) $(k\vec{a}) \times \vec{b} = \vec{a} \times (k\vec{b}) = k(\vec{a} \times \vec{b})$. | (H) $\vec{a} \cdot (\vec{b} \times \vec{c}) = \vec{b} \cdot (\vec{c} \times \vec{a}) = \vec{c} \cdot (\vec{a} \times \vec{b})$. |

The equation (B) is the most important. For (A), notice $\vec{a} \times \vec{a}$ is not “zero”.

Advanced note: Geometrical interpretation of the cross product is sometimes useful:

- $|\vec{a} \times \vec{b}|$ is the area of the parallelogram formed by $\vec{a}$ and $\vec{b}$.
- $(\vec{a} \times \vec{b}) \cdot \vec{c}$ is the (signed) volume of the parallelepiped formed by $\vec{a}$, $\vec{b}$, and $\vec{c}$.

Problems

★5.8 Prove (A), (B), (C), (F), and (G) of Theorem 5.12.

- **5.9** In physics, we **always** use the **right-handed Cartesian coordinate system** to describe our three-dimensional space, which is characterized by three unit vectors $\vec{e}_x, \vec{e}_y, \vec{e}_z$ defined so that they are perpendicular to each other and $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$ obeys the right-hand rule. Calculate their inner products and cross products, such as $\vec{e}_x \cdot \vec{e}_y$ and $\vec{e}_x \times \vec{e}_z$.
- *5.10** Prove (B) of Theorem 5.9 and (E) of Theorem 5.12 geometrically (i.e., based on the definitions given in this chapter). Sho has his own proof but not very confident. Can you find a better proof?

5.5 Position vectors

We have defined vectors as arrows. Arrows are not positions, so vectors are not positions. However, *once we define the origin O* in the space, we can use vectors to represent each position in the space.

Definition 5.13 Position vector

Consider a space and fix a point O as the origin. To each point P, we associate the vector $\vec{OP}$ and call it the **position vector** of P. We often write $\vec{p} = \vec{OP}$, $\vec{q} = \vec{OQ}$, and so on.

Consider $\vec{p} = \vec{OP}$, $\vec{q} = \vec{OQ}$, and $\vec{r} = \vec{OR}$. They are obviously dependent on the choice of the origin O. Also, the point described by $\vec{p} + \vec{q}$ will be different if we chose a different point as the origin. However, we can check that

- the vector $\vec{PQ}$ is given by $\vec{q} - \vec{p}$
- the length of the segment PQ is given by $|\vec{q} - \vec{p}|$.
- the middle point M of the segment PQ has the position vector $\vec{m} = (\vec{p} + \vec{q})/2$.

These properties are independent of the choice of O. Namely, whatever choice we did for origin, $(\vec{p} + \vec{q})/2$ represents the midpoint of PQ. For further discussion, please check [the Vector Boot Camp](#).

Problems

- ***5.11** Show that the length of the segment PQ is given by $|\vec{q} - \vec{p}|$, where $\vec{p}$ and $\vec{q}$ are the position vectors of points P and Q, respectively.
- **5.12** Let $\vec{p}, \vec{q}$, and $\vec{r}$ be the position vectors of points P, Q, R respectively, which are not on the same line.
- (1) Show that $(\vec{p} + \vec{q})/2$ is the position vector of the midpoint of the segment PQ.
 - (2) Show that $(\vec{p} + \vec{q} + \vec{r})/3$ is the position vector of the centroid of the triangle PQR.
- *5.13** Let $\vec{a}, \vec{b}$, and $\vec{p}$ be the position vectors of points A, B, P, respectively. Show the following.
- (1) P is on the segment AB $\iff \vec{p} = t\vec{a} + (1-t)\vec{b}$ with $0 \leq t \leq 1$.
 - (2) P is on the line AB $\iff \vec{p} = t\vec{a} + (1-t)\vec{b}$ with $t \in \mathbb{R}$.

5.6 Linear combination

Remark: This is a bit advanced. You may well come back when you read [Chapter 7](#).

We often consider a **linear combination**. For example, linear combinations of x , y , and z are given by $3x + 2y + 5z$, $x + \sqrt{2}y - 1.4z$, $3x - \pi y + 2z$, $-5z (= 0x + 0y - 5z)$, etc. Similarly, we can consider linear combinations of vectors.

Definition 5.14 Linear combination of vectors

For vectors living in the same space, $\vec{a}, \vec{b}, \vec{c}, \dots$, and scalars $p, q, r, \dots$, we call

$$p\vec{a} + q\vec{b} + r\vec{c} + \dots \quad (5.8)$$

a **linear combination** of $\vec{a}, \vec{b}, \vec{c}, \dots$

Some of the coefficients $p, q, r, \dots$ may be zero. Also, all the coefficients may be zero, where the linear combination results in $\vec{0}$.

Advanced note: It is sometimes important whether the summation allows infinite terms; in this document, we only allow finite summation as a linear combination.

Definition 5.15 Linearly dependent or independent

Consider (a finite number of) vectors $\{\vec{a}, \vec{b}, \vec{c}, \dots\}$ in the same space. Then, the equation

$$p\vec{a} + q\vec{b} + r\vec{c} + \dots = \vec{0} \quad (5.9)$$

always has a trivial solution $p = q = r = \dots = 0$. If there are other solutions, we say that $\{\vec{a}, \vec{b}, \vec{c}, \dots\}$ are **linearly dependent**. Meanwhile, if there is no other solution, we say that they are **linearly independent**.

Example 5.16

1. The set $\left\{\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 4 \end{pmatrix}\right\}$ is linearly independent; the equation $a\begin{pmatrix} 1 \\ 0 \end{pmatrix} + b\begin{pmatrix} 1 \\ 4 \end{pmatrix} = \vec{0}$ has only one solution $a = b = 0$ (*the trivial solution*).
2. The set $\left\{\begin{pmatrix} 2 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix}\right\}$ is linearly dependent because $a\begin{pmatrix} 2 \\ 4 \end{pmatrix} + b\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \vec{0}$ has a non-trivial solution $a = 1, b = -2$.
3. The set $\left\{\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right\}$ is linearly independent; $a\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + b\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + c\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \vec{0}$ is satisfied only by the trivial solution $a = b = c = 0$.
4. The set $\left\{\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}\right\}$ is linearly dependent; $a\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} + b\begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix} + c\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \vec{0}$ has a non-trivial solution $(a, b, c) = (1, 1, -2)$.

If a set is linearly dependent, at least one vector among them can be described as a linear combination of the other vectors. For example, in the example above,

$$\begin{aligned}
 & \text{(for the second example)} \quad \begin{pmatrix} 2 \\ 4 \end{pmatrix} = 2 \times \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \\
 & \text{(for the last example)} \quad \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} = 1 \times \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix} + (-2) \times \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.
 \end{aligned} \tag{5.10}$$

So, we may understand that $\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$ is “dependent” on $\begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. Meanwhile, if a set is linearly independent, it is impossible to do so; every vector in the set is “independent” of the other vectors.

Theorem 5.17

Consider (a finite number of) vectors $\{\vec{a}, \vec{b}, \vec{c}, \dots\}$.

- If they are linearly dependent, at least one vector among them can be described as a linear combination of the other vectors.
- If they are linearly independent, it is impossible to do so. Namely, we cannot express any vector among them as any linear combination of the remaining vectors.

Quiz

Q8. Consider the following sets of vectors. Determine whether they are linearly dependent or independent.

(1) $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \end{pmatrix} \right\}$

(2) $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \end{pmatrix} \right\}$

(3) $\left\{ \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \end{pmatrix} \right\}$

(4) $\left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$

(5) $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 0 \end{pmatrix} \right\}$

(6) $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 3 \\ 3 \end{pmatrix}, \begin{pmatrix} 4 \\ 3 \end{pmatrix} \right\}$

(7) $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$

(8) $\left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ 8 \\ 1 \end{pmatrix} \right\}$

(9) $\left\{ \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ 8 \\ 0 \end{pmatrix} \right\}$

Problems

★5.14 ♣TODO♣prepare

***5.15 ♣TODO♣prepare

**5.16 ♣TODO♣prepare

5.7 Intermission: Vector or Scalar or Not

Let us summarize the operations on vectors. With k a real number and $\vec{a}, \vec{b}$ vectors:

magnitude	$ \vec{a} $	$\rightarrow$ scalar
scalar multiplication	$k\vec{a}$	$\rightarrow$ vector
addition	$\vec{a} + \vec{b}$	$\rightarrow$ vector
inner product	$\vec{a} \cdot \vec{b}$	$\rightarrow$ scalar
cross product	$\vec{a} \times \vec{b}$	$\rightarrow$ vector (only in 3d)

Then, how about them? Try to ensure that you understand the meaning of each operation.

Problems

 **5.17** For each expression, answer **V** if it is a vector, **S** if scalar, and **N** if invalid (not defined). Here, $\vec{a}, \vec{b}, \dots$ are three-dimensional vectors and $a, b, \dots$ are scalars (real numbers).

(1) $\vec{a}$

(2) $\vec{a}^2$

(3) $(\vec{a})^2$

(4) $|\vec{a}|^2$

(5) $|a|^2$

(6) $\vec{0}$

(7) $\vec{0} + 0$

(8) $\vec{0} + \vec{0}$

(9) $0 \cdot \vec{0}$

(10) $\vec{0} \cdot \vec{0}$

(11) $\frac{1}{\vec{a}}$

(12) $\frac{\vec{a}}{\vec{a}}$

(13) $\frac{\vec{a}}{\vec{b}}$

(14) $\frac{1}{|\vec{a}|}$

(15) $\frac{1}{|\vec{a}|^2}$

(16) $\frac{1}{(\vec{a})^2}$

(17) $\frac{\vec{a}}{(\vec{a} \cdot \vec{b})}$

(18) $\frac{\vec{a}}{(\vec{a} \cdot \vec{b})^2}$

(19) $\frac{\vec{a}}{|\vec{a} \cdot \vec{b}|}$

(20) $\frac{\vec{a}}{(\vec{a} \cdot \vec{b})}$

(21) $\sqrt{\vec{a}^2}$

(22) $\sqrt{\vec{a}}$

(23) $\sqrt{|\vec{a}|}$

(24) $\sqrt{|a|}$

(25) $\sqrt{a^2}$

(26) $a + 1$

(27) $a + \vec{a}$

(28) $a \times \vec{a}$

(29) $a \cdot \vec{a}$

(30) $a\vec{a}$

(31) $\vec{a} + 1$

(32) $\vec{a} + \vec{a}$

(33) $\vec{a} \times \vec{a}$

(34) $\vec{a} \cdot \vec{a}$

(35) $\vec{a}\vec{a}$

(36) a^{-2}

(37) $|\vec{a}|^{-2}$

(38) $|\vec{a}|^{-2}$

(39) $|\vec{a}|^{-1} \vec{a}$

(40) $|\vec{a}|^{-1/2} \vec{a}$

(41) $\vec{p} \cdot \vec{q} + \vec{p} \times \vec{q}$

(42) $\vec{p} \times (\vec{q} \times \vec{r})$

(43) $\vec{p} \cdot (\vec{q} \times \vec{r})$

(44) $p(\vec{q} \times \vec{r})$

(45) $(\vec{p} \times \vec{q})^2$

(46) $\frac{1}{|\vec{x} - \vec{r}|^{3/2}}$

(47) $\frac{\vec{x} - \vec{r}}{|\vec{x} - \vec{r}|^{3/2}}$

(48) $\frac{\vec{d} \times (\vec{x} - \vec{r})}{|\vec{x} - \vec{r}|^{3/2}}$

(49) $\frac{d\vec{y}}{dx}$

(50) $\frac{dy}{d\vec{x}}$

5.8 Axes and Components

So far, we have considered vectors as arrows drawn in a space. Vectors are characterized only by the magnitude and direction. Now, we are going to *recall* the component-wise notation such as $\vec{a} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. We define “orthonormal basis vectors” and then see we can *describe an arrow by a list of numbers*.

Definition 5.18 Orthonormal basis vectors

If n vectors $\vec{e}_1, \dots, \vec{e}_n$ satisfy the following properties, we call them **an orthonormal basis**:

- All of them are unit vectors, i.e., $|\vec{e}_i| = 1$ for all i .
- Any of them are perpendicular, i.e., $i \neq j \implies \vec{e}_i \cdot \vec{e}_j = 0$.
- We cannot add any more vectors without violating the above two rules.

Precisely speaking, we call the set $\{\vec{e}_1, \dots, \vec{e}_n\}$ “an orthonormal basis”. Then, if we fix *one* orthonormal basis to use, we call its members “orthonormal basis vectors”.

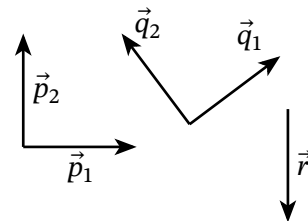
Fail safe note: A woman, two women. A nucleus, two nuclei. A basis, two bases. A matrix, two matrices.

Example 5.19 Basis vectors for arrows on this sheet

Let’s limit our “space” to this 2d sheet and consider the five vectors drawn to the right. Assume their length are all “1”.

If we choose $\vec{p}_1$ and $\vec{p}_2$, they form *an* orthonormal basis $\{\vec{p}_1, \vec{p}_2\}$ because they are unit vectors, perpendicular to each other, and we cannot add any more. Similarly, $\{\vec{q}_1, \vec{q}_2\}$ is *another* orthonormal basis.

Meanwhile, $\{\vec{p}_1, \vec{q}_2\}$ and $\{\vec{p}_1, \vec{p}_2, \vec{q}_1, \vec{q}_2\}$ are not orthonormal bases because the members are not orthogonal. $\{\vec{p}_1\}$ is not an orthonormal basis because we can add one more vector, such as $\vec{r}$, without breaking the conditions; after adding $\vec{r}$, we have an orthonormal basis $\{\vec{p}_1, \vec{r}\}$.



Quiz

Q9. Consider the figure in the above example. Which sets are orthogonal bases?

$$\{\vec{p}_1, -\vec{p}_2\}, \quad \{\vec{p}_1, \vec{r}\}, \quad \{\vec{p}_2, \vec{r}\}, \quad \{\vec{p}_1, \vec{p}_2, \vec{r}\}, \quad \{\vec{q}_1, 2\vec{q}_2\}, \quad \{-\vec{q}_1, -\vec{q}_2\}.$$

This example shows we can find many orthogonal bases, but **the number** of the members is always two. This number is called the **dimension** of the space, and this is why we call “this 2d sheet” in the above example.

In general, a space has *infinitely many* orthonormal bases and we can choose *an* orthonormal basis at our convenience; we will come back to this point in Section 7.2. However, the number of the orthonormal basis vectors, the **dimension**, is fixed by the space we considered.

Theorem 5.20 Properties of orthonormal basis vectors

- Any orthonormal bases of a space have the same number of vectors. We call the number **the dimension** of the space.
- If $\{\vec{e}_1, \vec{e}_2, \dots, \vec{e}_n\}$ is an orthonormal basis, any arrow $\vec{v}$ in the space can be expressed as

$$\vec{v} = c_1\vec{e}_1 + c_2\vec{e}_2 + \dots + c_n\vec{e}_n = \sum_{k=1}^n c_k\vec{e}_k, \quad \text{where } c_k \in \mathbb{R} \quad (5.11)$$

and this expression is *unique*, i.e., if $\vec{v}$ is expressed by

$$\begin{aligned} \vec{v} &= c_1\vec{e}_1 + c_2\vec{e}_2 + \dots + c_n\vec{e}_n \\ &= d_1\vec{e}_1 + d_2\vec{e}_2 + \dots + d_n\vec{e}_n, \end{aligned}$$

then all the coefficients are equal: $c_k = d_k$.

Quiz

Q10. Show that the numbers c_k in Eq. (5.11) are actually determined by $c_k = \vec{e}_k \cdot \vec{v}$.

Advanced note: This theorem seems not difficult to prove because we only think finite-dimensional spaces, but there could be caveats that Sho did not notice. A more rigorous construction is in [♣TODO♣abs-vec](#).

If you choose an orthonormal basis, then it automatically defines the **axes** of the space:

- For a 2d space, we call the directions of the basis vectors as *x*-axis and *y*-axis, respectively. Sho usually writes the basis vectors by $\vec{e}_x$ and $\vec{e}_y$, but other textbooks may write as $\hat{\mathbf{i}}$ and $\hat{\mathbf{j}}$.
- For a 3d space, we call the directions of the basis vectors as *x*-axis, *y*-axis, and *z*-axis, respectively. The basis vectors are expressed by $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$ or $(\hat{\mathbf{i}}, \hat{\mathbf{j}}, \hat{\mathbf{k}})$. Here, physicists **always** choose the axes so that $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$ obeys the **right-hand rule** (see Section 5.4).

They are called **2d Cartesian coordinate system** and **3d right-handed Cartesian coordinate system**, respectively. In general, a coordinate system defined by an orthonormal basis is called Cartesian coordinate system.

Now we are ready to express vectors in their **components** because we have reached Eq. (5.11):

Definition 5.21 Components of a vector

If we fix an orthonormal basis and label the basis vectors by $\vec{e}_1, \dots, \vec{e}_n$, then Theorem 5.20 says any vector $\vec{v}$ can be written as $\vec{v} = c_1\vec{e}_1 + \dots + c_n\vec{e}_n$ with uniquely determined $c_k := \vec{e}_k \cdot \vec{v} \in \mathbb{R}$. We call c_k **the *k*-th component** of $\vec{v}$ and express $\vec{v}$ with the components as

$$\vec{v} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix} = \begin{pmatrix} \vec{e}_1 \cdot \vec{v} \\ \vec{e}_2 \cdot \vec{v} \\ \vdots \\ \vec{e}_n \cdot \vec{v} \end{pmatrix}. \quad (5.12)$$

Vectors in n -dimensional spaces are called **n -dimensional vectors**. Since an orthonormal basis in a n -dimensional space has n basis vectors, a n -dimensional vectors are expressed with n real numbers $c_1, \dots, c_n$.

Advanced note: We have to make sure this representation is *well-defined*; we do not want to have two different expressions for one vector, or two different vectors having the same expressions.

Because c_k is uniquely determined, the component-wise notation is unique for a vector. Meanwhile, if $\vec{a}$ and $\vec{b}$ are different but have the same component-wise notation, it means $\vec{a} - \vec{b} =: \vec{\Delta}$ has the same notation as $\vec{0}$ (why?). It means $\vec{\Delta}/|\vec{\Delta}|$ is a unit vector orthogonal to all of $\vec{e}_k$, which contradicts that $\{\vec{e}_1, \dots, \vec{e}_n\}$ is the basis (why?). Accordingly, different vectors must have different component-wise notation.

Example 5.22

Prove the following equations in a 2d space.

$$(1) \begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} a+p \\ b+q \end{pmatrix} \quad (2) \begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} p \\ q \end{pmatrix} = ap + bq \quad (3) \left| \begin{pmatrix} a \\ b \end{pmatrix} \right| = \sqrt{a^2 + b^2}$$

Solution

(1) Since $\begin{pmatrix} a \\ b \end{pmatrix}$ means $a\vec{e}_x + b\vec{e}_y$ and $\begin{pmatrix} p \\ q \end{pmatrix}$ means $p\vec{e}_x + q\vec{e}_y$ under some orthonormal basis $(\vec{e}_x, \vec{e}_y)$,

$$\text{LHS} = (a\vec{e}_x + b\vec{e}_y) + (p\vec{e}_x + q\vec{e}_y) = (a+p)\vec{e}_x + (b+q)\vec{e}_y = \text{RHS},$$

where we used the equations in Theorem 5.7. ■

(2) Similarly, using the equations in Theorem 5.9 and Definition 5.18,

$$\begin{aligned} \text{LHS} &= (a\vec{e}_x + b\vec{e}_y) \cdot (p\vec{e}_x + q\vec{e}_y) \\ &= ap(\vec{e}_x \cdot \vec{e}_x) + bp(\vec{e}_y \cdot \vec{e}_x) + aq(\vec{e}_x \cdot \vec{e}_y) + bq(\vec{e}_y \cdot \vec{e}_y) = ap + bq. \blacksquare \end{aligned}$$

(3) Thanks to (2), $(\text{LHS})^2 = \begin{pmatrix} a \\ b \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} = a^2 + b^2$. Since $(\text{LHS}) \geq 0$, $\text{LHS} = \sqrt{a^2 + b^2} = \text{RHS}$. ■

Fail safe note: In (3), you must write $\lceil \text{LHS} \geq 0 \rceil$. Without it, you can only claim $\text{LHS} = \pm\sqrt{a^2 + b^2}$.

Notice that we *proved* these equations based on Theorem 5.7 etc. These equations are not *definitions* or *assumptions*, but *derived statements*^{#6} with proofs. There are a few more statements to be proved:

Quiz

Q11. Consider a 2d space.

- (1) Prove $\vec{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$, $\vec{e}_x = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and $\vec{e}_y = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$. [Hint: Recall $c_k = \vec{e}_k \cdot \vec{v}$.]
- (2) Prove $k\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ka \\ kb \end{pmatrix}$, where $k \in \mathbb{R}$.

^{#6} → Section 3.5

Problems

***5.18 Consider a 3d space. An orthonormal basis $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$ is taken according to the right-

hand rule. Let $\vec{a} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ and $\vec{p} = \begin{pmatrix} p \\ q \\ r \end{pmatrix}$.

(1) Prove $\vec{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$, $\vec{e}_x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\vec{e}_y = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, and $\vec{e}_z = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

(2) Prove $k\vec{a} + l\vec{b} = \begin{pmatrix} ka + lp \\ kb + lq \\ kc + lr \end{pmatrix}$.

(3) Prove $\vec{a} \cdot \vec{p} = ap + bq + cr$ and $|\vec{a}| = \sqrt{a^2 + b^2 + c^2}$.

(4) Express $\vec{a} \times \vec{p}$ with using a, b, c, p, q, r .

**5.19 Consider a n -dimensional space ($n \in \mathbb{N}^+$) and an orthonormal basis $(\vec{e}_1, \dots, \vec{e}_n)$ of it. Prove the following.

- $\vec{0} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$ and $(i\text{-th component of } \vec{e}_j) = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$
- Consider $p, q \in \mathbb{R}$ and n -dimensional vectors $\vec{a}$ and $\vec{b}$. Let the k -th component of $\vec{a}$ be a_k and the k -th component of $\vec{b}$ be b_k . Then, the k -th component of $p\vec{a} + q\vec{b}$ is equal to $pa_k + qb_k$. Also, $\vec{a} \cdot \vec{b} = \sum_{k=1}^n a_k b_k$ and $|\vec{a}| = \sqrt{\sum_{k=1}^n a_k^2}$.

Let us summarize the above discussion.

Theorem 5.23 Component-wise interpretation of real-vector arithmetic

Consider n -dimensional real vectors, $\vec{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$, and $k \in \mathbb{R}$. Then, addition, scalar multiplication by $k \in \mathbb{R}$, and inner product are given by, respectively,

$$\vec{a} + \vec{b} = \begin{pmatrix} a_1 + b_1 \\ \vdots \\ a_n + b_n \end{pmatrix}, \quad k\vec{a} = \begin{pmatrix} ka_1 \\ \vdots \\ ka_n \end{pmatrix}, \quad (5.13)$$

$$\vec{a} \cdot \vec{b} = \sum_{k=1}^n a_k b_k = a_1 b_1 + a_2 b_2 + \dots + a_n b_n, \quad (5.14)$$

Be careful: Again, these formulas are *consequences* of the definitions we gave earlier. They are **not** new definitions.

Since you must be familiar with this component-wise notation and calculations based on it, we do not discuss it further. Consult your high-school math textbook or Sho's [Vector Boot Camp](#) if you are not confident in such calculations.

Vectors are closely related to matrices, the main topic of [Chapter 7](#). We will there discuss both real and complex matrices, and thus complex vectors will be introduced there. Since complex vectors are not arrows, we need to *define* complex vectors in a way different from what we did in this chapter.

To prepare for the discussion on complex matrices in [Chapter 7](#), we first review **complex numbers** in the next chapter ([Chapter 6](#)).

Problems

***5.20** Complete the proof in “Advanced Note” after Definition 5.21.

***5.21** When we define a Cartesian coordinate system for a 3d space, we require that $(\vec{e}_x, \vec{e}_y, \vec{e}_z)$ satisfies the right-hand rule. However, similar rules are not required for 2d spaces. Why?

Chapter 6

Complex Numbers

You have learned complex numbers in high school but it is worth reviewing them in a university style. As you learned in Section 3.5, try to distinguish between definitions and derived consequences.

Remark: Geometrical discussions, such as **complex planes**, are not included in this document, mainly because you are expected to be familiar with it.

6.1 Complex numbers

Definition 6.1 Complex numbers

A **complex number** is given by the form

$$z = a + bi, \text{ where } a, b \in \mathbb{R} \text{ and } i \text{ is a symbol satisfying } i^2 = -1. \quad (6.1)$$

We call the newly-introduced symbol i the **imaginary unit**. Also, a is called the **real part** of z and b is called the **imaginary part** of z , written by

$$\operatorname{Re}(z) := a, \quad \operatorname{Im}(z) := b.$$

The set of all complex numbers is written by $\mathbb{C}$.

Be careful: $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$ are both real. If $z = 3 + 5i$, then $\operatorname{Im}(z) = 5$, not $5i$.

Remark: We use i to denote the imaginary unit. You may see other notations such as i , j , j , or $\sqrt{-1}$. However, Sho recommends you **not** to write $\sqrt{-1}$ since it may make you confused. Recall, in [Chapter 4](#), we *avoided* definitions of $\sqrt{x}$ for $x < 0$.

Advanced note: Notice that i is just a symbol and it has only one property $i \cdot i = -1$. You may well ask “ $-i$ also satisfies $(-i)^2 = -1$; which should we choose as i ?” but it sounds a weird question since “ $-i$ ” was not *a priori*: it was not there before we introduced i , so we can discuss $-i$ only after we fix i .

Hereafter, when we write $\boxed{z = a + bi \in \mathbb{C}}$, we implicitly assume a and b are real numbers.

Definition 6.2

Consider $z = a + bi \in \mathbb{C}$.

- z is a real number if $\operatorname{Im} z = b = 0$. Namely, if $b = 0$, then $z = a + 0i = a$.
- z is called **pure-imaginary** if $\operatorname{Re} z = a = 0$ but $z \neq 0$, namely, if $z = bi$ with $b \neq 0$.

Also, for another complex number $c + di$, we define $a + bi = c + di \iff a = c \wedge b = d$.

It is debatable if 0 is pure-imaginary or not, so you need to be careful when you discuss 0 (but usually 0 is not included in pure-imaginary numbers.) Some people write the set of all pure-imaginary number by $i\mathbb{R}$, but then you need to pay attention to this issue.

For complex numbers $z_1 = a_1 + b_1i$ and $z_2 = a_2 + b_2i$, addition and multiplication are defined by

$$\begin{aligned} z_1 + z_2 &= (a_1 + a_2) + (b_1 + b_2)i, & z_1 z_2 &= (a_1 + b_1i)(a_2 + b_2i) \\ & & &= (a_1 a_2 + b_1 b_2 i^2) + (a_1 b_2 + a_2 b_1)i \\ & & &= (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1)i. \end{aligned}$$

and powers of complex numbers z^n for integer n are defined as usual:

- $\alpha^1 = \alpha$, $\alpha^2 = \alpha\alpha$, $\alpha^3 = \alpha\alpha\alpha$, ...
- For $\alpha \neq 0$, $\alpha^0 = 1$, $\alpha^{-1} = \frac{1}{\alpha}$, $\alpha^{-2} = \frac{1}{\alpha^2}$, ...

Remark: Here we define z^n for $n \in \mathbb{Z}$.

Furthermore, two more important operations are defined:

Definition 6.3 Complex conjugate and Absolute value

For $z = a + bi \in \mathbb{C}$, we define

- its **complex conjugate** by $\bar{z} := a - bi$,
- its **absolute value** by $|z| := (\bar{z} z)^{1/2} = (a^2 + b^2)^{1/2}$.

Quiz

- Q1.** Prove that, if $z = a + bi$, then $\bar{z}z$ is a real and non-negative number.
- Q2.** Explain why $|z| \geq 0$. [Hint: See the definition carefully. What does $x^{1/2}$ mean?]
- Q3.** For each z below, find $\text{Re}(z)$, $\text{Im}(z)$, $\bar{z}$, and $|z|$.
- $z = 2 + 7i$, $z = -3 - i$, $z = 5$, $z = -4i$, $z = 0$.
- Q4.** Find $x, y \in \mathbb{R}$ such that $(x + 1) + (2y - 3)i = 4 - i$.

The fact $z\bar{z} \in \mathbb{R}$ is used to calculate fractions of complex numbers:

$$\frac{w}{z} = \frac{w\bar{z}}{z\bar{z}} = \frac{w\bar{z}}{|z|^2}. \quad \text{For example, } \frac{11 + 4i}{1 + 2i} = \frac{(11 + 4i) \times (1 - 2i)}{(1 + 2i) \times (1 - 2i)} = \frac{19 - 18i}{5}. \quad (6.2)$$

Thus, for $z = a + bi \neq 0$ and $n \in \mathbb{Z}$,

$$z^{-n} = \frac{1}{z^n} = \frac{\bar{z}^n}{z^n \bar{z}^n} = \frac{\bar{z}^n}{(z\bar{z})^n} = \frac{\bar{z}^n}{|z|^{2n}} = \frac{(a - bi)^n}{(a^2 + b^2)^n}. \quad (6.3)$$

Problems

 **6.1** [Note: This is a timed practice; just write down the answer, without explanation.]

Consider the following numbers. Assume $a, b, k \in \mathbb{R}$.

$$\begin{aligned} & -2 + i, \quad 1 - i, \quad 3 + 4i, \quad -2(3 + 4i), \quad -2, \quad -2i, \quad -9i, \quad 0, \quad i, \quad 4, \\ & \sqrt{2} + i, \quad -\sqrt{3} + 2i, \quad 3 + i\sqrt{5}, \quad -1 - i\sqrt{7}, \quad -\sqrt{11} + i\sqrt{14}, \\ & \frac{i + \sqrt{5}}{\sqrt{2}}, \quad \frac{i - \sqrt{5}}{\sqrt{2}}, \quad \frac{\sqrt{3}}{\sqrt{2}} + \frac{\sqrt{3}}{\sqrt{2}}i, \quad -\frac{\sqrt{2}}{2} + \frac{\sqrt{6}}{2}i, \quad a + 2i, \quad a - bi, \quad ki. \end{aligned}$$

(1) What are the real part, imaginary part, and complex conjugate of each? (goal time:)

(2) What are their absolute values? (goal time: 120 seconds)

 **6.2** Calculate the following.

(1) $(3 + 2i) + (4 - 5i)$	(2) $(-2 + 7i) - (5 + 3i)$	(3) $(6 - 4i) - (-1 + 2i)$
(4) $(2 + 3i)(4 - i)$	(5) $(-1 + 5i)(2 + 3i)$	(6) $(3 - 2i)(3 + 2i)$
(7) $i(3 + i) - i(1 - 4i)$	(8) $(1 - 2i) + \overline{4 + i}$	(9) $\overline{(3 - 4i) + 1 + 2i}$
(10) $(1 + 2i)^4$	(11) $(1 - 2i)^4$	(12) $(2 + 3i)^3(2 - 3i)^3$
(13) $\frac{2 + i}{-i}$	(14) $\frac{11 + 4i}{1 + 2i}$	(15) $\frac{1 - i}{(1 + i)^2}$

****6.3** Find $z \in \mathbb{C}$ that satisfies $z^2 = 1 + 2i$.

The following relations hold for these operations. They are somewhat obvious, but with using them you can speed up your calculations.

Theorem 6.4 Properties of Complex-number arithmetic

For $\alpha, \beta, \gamma \in \mathbb{C}$ and $n, m \in \mathbb{Z}$,

$$\begin{aligned} & (\alpha + \beta) + \gamma = \alpha + (\beta + \gamma), \quad \alpha + \beta = \beta + \alpha, \quad 0 + \alpha = \alpha, \\ & (\alpha\beta)\gamma = \alpha(\beta\gamma), \quad \alpha\beta = \beta\alpha, \quad 1\alpha = \alpha, \quad \alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma. \end{aligned} \tag{6.4}$$

$$\alpha^n \alpha^m = \alpha^{n+m}, \quad (\alpha^n)^m = \alpha^{nm}, \quad (\alpha\beta)^n = \alpha^n \beta^n, \quad \alpha^{-1} = \frac{\bar{\alpha}}{|\alpha|^2}, \quad \left(\frac{\alpha}{\beta}\right)^n = \frac{\alpha^n}{\beta^n}; \tag{6.5}$$

$$\overline{\overline{\alpha}} = \alpha, \quad \overline{\alpha + \beta} = \bar{\alpha} + \bar{\beta}, \quad \overline{\alpha\beta} = \bar{\alpha} \bar{\beta}, \quad \overline{\alpha^n} = (\bar{\alpha})^n, \quad \overline{\left(\frac{\alpha}{\beta}\right)} = \frac{\bar{\alpha}}{\bar{\beta}}; \tag{6.6}$$

$$|\bar{\alpha}| = |\alpha|, \quad |\alpha\beta| = |\alpha||\beta|, \quad \left|\frac{\alpha}{\beta}\right| = \frac{|\alpha|}{|\beta|}, \quad |\alpha^n| = |\alpha|^n, \tag{6.7}$$

$$|\alpha| - |\beta| \leq |\alpha - \beta|, \quad |\alpha + \beta| \leq |\alpha| + |\beta|, \tag{6.8}$$

Theorem 6.5

$$\operatorname{Re}(z) = \frac{z + \bar{z}}{2}, \quad \operatorname{Im}(z) = \frac{z - \bar{z}}{2i}. \quad (6.9)$$

Example 6.6

Express $\operatorname{Re}(z)^2$, $\operatorname{Re}(z^2)$, and $\operatorname{Re}(z) \operatorname{Im}(z)$ without using “Re” and “Im”, i.e., only with z and $\bar{z}$.

Solution

$$\operatorname{Re}(z)^2 = \left(\frac{z + \bar{z}}{2} \right)^2 = \frac{(z + \bar{z})^2}{4}; \quad \operatorname{Re}(z^2) = \frac{z^2 + \bar{z}^2}{2} = \frac{z^2 + \bar{z}^2}{2}; \quad \operatorname{Re}(z) \operatorname{Im}(z) = \frac{z^2 - \bar{z}^2}{4i}.$$

Quiz

Q5. Calculate them.

$$(1) \left| (\sqrt{2} + i\sqrt{7})^4 \right| \quad (2) |11 + 22i|^2 \quad (3) \left| \frac{1+i}{1-i} \right| \quad (4) (1+i)\overline{(1+i)^2}$$

Q6. Express following expressions without “Re” and “Im”, i.e., only with z and $\bar{z}$.

$$(1) (\operatorname{Im}(z))^2 \quad (2) \operatorname{Im}(2z) \quad (3) \operatorname{Re}(z) + \operatorname{Im}(z)$$

Problems

★6.4 Let $z = 2 + i$ and $w = 1 - 2i$. Calculate:

$$\begin{array}{lllll} (1) z + w & (2) \bar{z} & (3) \bar{w} & (4) |z| & (5) |w| \\ (6) \bar{z} + \bar{w} & (7) \overline{z + w} & (8) z\bar{w} & (9) \bar{z}w & (10) |zw| \\ (11) \left| \frac{z}{w} \right| & (12) |z + w| & (13) |z| + |\bar{w}| & (14) |zw + z\bar{w}| & (15) |\bar{z}w + z\bar{w}| \end{array}$$

★6.5 Calculate i^2 , i^4 , and i^{127} . Also, for $z = (1 + i)/\sqrt{2}$, calculate z^{127} .

★6.6 Prove the following statements, where $z, \alpha, \beta \in \mathbb{C}$.

$$(1) \overline{z_1 z_2} = \bar{z}_1 \bar{z}_2. \quad (2) |z_1 z_2| = |z_1| |z_2|. \quad (3) \operatorname{Im}(z) = (z - \bar{z})/(2i).$$

****6.7** Prove that, for $a, b, c \in \mathbb{R}$ and $z \in \mathbb{C}$, the solution of $az^2 + bz + c = 0$ is given by

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \text{if } b^2 \geq 4ac; \quad z = \frac{-b \pm i\sqrt{|b^2 - 4ac|}}{2a} \quad \text{if } b^2 < 4ac. \quad (6.10)$$

****6.8** Find all complex numbers z satisfying the following equations.

$$(1) z^2 + 16 = 0 \quad (2) z^2 + 4z + 5 = 0 \quad (3) z^2 + 16i = 0 \quad (4) z^2 + 4z + 5i = 0$$

****6.9** Let's prove $0 \leq |\alpha + \beta| \leq |\alpha| + |\beta|$ step by step, where $\alpha, \beta \in \mathbb{C}$.

- (1) For $z \in \mathbb{C}$, prove the next: $z \neq 0 \implies z\bar{z} > 0$. [Hint: Let $z = a + bi$.]
- (2) For $z \in \mathbb{C}$, prove $|z| \geq \operatorname{Re}(z)$. Then, let $z = \alpha\bar{\beta}$ and simplify $2|z| - 2\operatorname{Re}(z)$.
- (3) Calculate $(|\alpha| + |\beta|)^2 - |\alpha + \beta|^2$ and prove it is not negative (≥ 0).
- (4) Prove $0 \leq |\alpha + \beta| \leq |\alpha| + |\beta|$. Also, find the condition for $|\alpha + \beta| = |\alpha| + |\beta|$.

****6.10** Let $p(z) = a_n z^n + a_{n-1} z^{n-1} + \dots + a_1 z + a_0$ be a polynomial with *real* coefficients ($a_k \in \mathbb{R}$). Prove that if $z_0 \in \mathbb{C}$ is a solution of $p(z) = 0$, then so is $\bar{z}_0$.

6.2 Polar form

Theorem 6.7 Polar form

A complex number $z = a + bi$, where $a, b \in \mathbb{R}$, can be expressed in the following **polar form**:

$$z = r(\cos \theta + i \sin \theta), \quad \text{where } r = |z| \text{ is the absolute value of } z, \quad (6.11)$$

$\theta \in \mathbb{R}$ is called the **argument** of z .

For $z \neq 0$, this expression is unique up to the 2π -periodicity of θ .

Proof: Let $z = a + bi$ with $a, b \in \mathbb{R}$. For $z = 0$, the statement is valid because $r = |z| = 0$. Now we assume $z \neq 0$ and, with $r := |z|$, consider a point P at $(a/r, b/r)$ on xy -plane. Since $(a/r)^2 + (b/r)^2 = 1$, P is on the unit circle around the origin O and thus can be expressed by $(\cos \theta, \sin \theta)$ with θ being the angle from the positive x -axis to $\overrightarrow{OP}$ and the choice of θ is unique up to the 2π -periodicity. It means $a/r = \cos \theta$ and $b/r = \sin \theta$, i.e., $z = r \cos \theta + ir \sin \theta$. ■

Advanced note: In this document, the **trigonometric functions** $\cos \theta$ and $\sin \theta$ are defined using the unit circle in xy -plane and this proof employs that definition. In advanced mathematics, we often take another approach, where we first define e^x for $x \in \mathbb{R}$ by a **power series**. Then we extend e^x to $\mathbb{C}$ and, from this complex function e^z , we define $\cos x$ and $\sin x$ for $x \in \mathbb{R}$ by Euler's formula. See ♠**TODO**♠**Lorem ipsum dolor sit.** for further discussion.

Quiz

Q7. Find the complex number expressed by the following polar form:

- | | | |
|-------------------------------------|-------------------------------|-----------------------------|
| (1) $r = \sqrt{2}, \theta = \pi/4$ | (2) $r = 2, \theta = -\pi/6$ | (3) $r = 4, \theta = \pi/2$ |
| (4) $r = \sqrt{2}, \theta = 9\pi/4$ | (5) $r = 2, \theta = -5\pi/6$ | (6) $r = 4, \theta = -\pi$ |

Similar to $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$, we may consider a function $\operatorname{Arg}(z)$ to return its argument. Here, however, we need to care the **multivaluedness** in θ . For example, (1) and (4) in the above quiz express the same number $z = 1 + i$, so there can be multiple values of θ for $z = 1 + i$. So we need to choose one; a convenient choice is to define $-\pi < \operatorname{Arg}(z) \leq \pi$:

Definition 6.8

For $z \in \mathbb{C}$, $z \neq 0$,

$\text{Arg}(z) :=$ the argument of z , chosen between $-\pi$ (exclusive) and π (inclusive). (6.12)

This strategy is called “choosing the **principle value** of the argument of z so that $-\pi < \text{Arg}(z) \leq \pi$ ”.

Quiz

Q8. Find $\text{Arg}(5i)$, $\text{Arg}(2 + 2i)$, $\text{Arg}(-1 - i)$, and $\text{Arg}(1 - i)$.

Due to the subtlety of $\text{Arg}(z)$, we do not discuss $\text{Arg}(z)$ in the rest of this document.

Advanced note: You can see the subtlety in the next example. Consider $z = e^{2i}$. Then, $\text{Arg}(z) = 2$, but $\text{Arg}(z^2) = \text{Arg}(e^{4i}) \approx -2.3$ (Why?). Similarly, $\text{Arg}(z^3) \approx -0.28$ and $\text{Arg}(z^4) \approx 1.72$.

6.3 Euler's formula

We have seen five types of operations on complex numbers: $\bar{z}$, $|z|$, $z + w$, zw , and z^n for $n \in \mathbb{Z}$. The next step is to define e^z for $z \in \mathbb{C}$, but we want to define it nicely: we want to keep the key properties of e^x such as $de^{ax}/dx = ae^{ax}$ and $e^{a+b} = e^a e^b$ even for $a, b \in \mathbb{C}$. So,

Definition 6.9

We **define**, for a complex number $z = a + bi$ ($a, b \in \mathbb{R}$),

$$e^{a+bi} := e^a (\cos b + i \sin b). \quad (6.13)$$

In particular, for $\theta \in \mathbb{R}$, $e^{i\theta} = \cos \theta + i \sin \theta$ (**Euler's formula**). (6.14)

Advanced note: Let $a, b, x \in \mathbb{R}$. Then, we want to keep $e^{(a+bi)x} = e^{ax} e^{bix}$ and thus we just have to define $e^{i\theta}$ for a real number $\theta = bx$. Letting $e^{i\theta} = f(\theta) + ig(\theta)$, the requirement of derivative is given by $de^{ibx}/dx = ibe^{ibx}$, where the LHS is $b \cdot de^{i\theta}/d\theta = bf'(\theta) + ibg'(\theta)$ and the RHS is $ib(f(\theta) + ig(\theta))$. It means $f'(\theta) = -g(\theta)$ and $g'(\theta) = f(\theta)$. Together with $f(0) = 1$ and $g(0) = 0$, we get $f(\theta) = \cos \theta$ and $g(\theta) = \sin \theta$ and now we have understand our wanted definition is $e^{a+ib} = e^a (\cos b + i \sin b)$.

Accordingly, the polar form of complex numbers is given by

$$z = a + bi = r(\cos \theta + i \sin \theta) = r \exp(i\theta) = |z| \exp(i \text{Arg}(z)). \quad (6.15)$$

As we defined e^z so that the key properties are kept, the following formulae are valid. (Compare with Theorem 4.9).

Theorem 6.10 Properties of Exponential Functions (2)

$$\text{For } z, w \in \mathbb{C}, \quad \frac{1}{e^z} = e^{-z}, \quad e^z e^w = e^{z+w}, \quad \frac{e^z}{e^w} = e^{z-w}, \quad (e^z)^w = e^{zw}. \quad (6.16)$$

$$\text{For } z \in \mathbb{C} \text{ and } a > 0, \quad a^z = (e^{\ln a})^z = e^{z \ln a}. \quad (6.17)$$

Currently, we know $\ln a$ only for $a > 0$. So, the second statement is only for $a > 0$ (see Table 2). Discussions on a^z for $a \leq 0$ and $a \notin \mathbb{R}$ will be given in **♣TODO♣later?**.

Quiz

Q9. Express the following number in the form $a + bi$:

- | | | | | | |
|-------------------|--------------|---------------|-----------------|----------------|----------------|
| (1) e^i | (2) e^{2i} | (3) e^{-i} | (4) e^{1+i} | (5) $(e^i)^4$ | (6) e^{i^4} |
| (7) $e^{i \ln 3}$ | (8) 2^i | (9) 2^{2+i} | (10) 2^{2+2i} | (11) 4^{1+i} | (12) 5^{i^2} |

The next theorem is extremely important:

Theorem 6.11

$$\text{For } \theta \in \mathbb{R}, \quad \cos \theta = \operatorname{Re}(e^{i\theta}) = \frac{e^{i\theta} + e^{-i\theta}}{2}, \quad \sin \theta = \operatorname{Im}(e^{i\theta}) = \frac{e^{i\theta} - e^{-i\theta}}{2i}. \quad (6.18)$$

Then for $\theta \in \mathbb{R}$ and $n \in \mathbb{Z}$, **de Moivre's theorem** $(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$ holds, but we do not need to memorize it. Just use the equation $e^{in\theta} = (e^{i\theta})^n$, e.g.,

Example 6.12

Express $\cos 4\theta$ in terms of $\cos \theta$.

Solution

Write $c = \cos \theta$ and $s = \sin \theta$ for brevity. Then.

$$\cos 4\theta = \operatorname{Re}(e^{4i\theta}) = \operatorname{Re}(e^{i\theta})^4 = \operatorname{Re}(c + is)^4 = \operatorname{Re}(c^4 + 4c^3is + 6c^2i^2s^2 + 4ci^3s^3 + i^4s^4).$$

As $s^2 = 1 - c^2$, we obtain $\cos 4\theta = c^4 - 6c^2(1 - c^2) + (1 - c^2)^2 = 8\cos^4 \theta - 8\cos^2 \theta + 1$.

Quiz

Q10. Express $\cos 3\theta$, $\sin 3\theta$, $\cos(\theta + \varphi)$, and $\sin(\theta + \varphi)$ in terms of $\cos \theta$, $\sin \theta$, $\cos \varphi$, and $\sin \varphi$. [Hint: Expand both sides of $e^{i(\theta+\varphi)} = e^{i\theta}e^{i\varphi}$.]

Problems

6.11 Compute the following and express in the form $a + bi$.

- | | | | |
|---------------------------|--------------------------------|-----------------|------------------------|
| (1) $e^{i\pi/6}$ | (2) $e^{i\pi}$ | (3) $e^{2i\pi}$ | (4) $2e^{-i\pi/3}$ |
| (5) $\sqrt{2}e^{-i\pi/4}$ | (6) $e^{i\pi/2} + e^{-i\pi/2}$ | (7) $(1+i)^4$ | (8) $(\sqrt{3} + i)^6$ |

★6.12 Convert to polar form $re^{i\theta}$, choosing $\text{Arg}(z) \in (-\pi, \pi]$.

(1) $1 + \sqrt{3}i$

(2) -2

(3) $-i$

(4) $3 - 3i$

(5) $-1 + i$

(6) $-\sqrt{3} - i$

(7) $4i$

(8) $1 - \sqrt{3}i$

***6.13 Using the polar form, compute the following. Express in the form $a + bi$.

(1) $(1 + i)^8$

(2) $((\sqrt{3} + i)/2)^{12}$

(3) $(1 - i)^{10}$

(4) $(1 + \sqrt{3}i)^{-3}$

(5) $((1 + i)/(1 - i))^{20}$

(6) $((\sqrt{3} + i)/(1 + i))^6$

***6.14 Use de Moivre's theorem to express $\cos(4\theta)$ and $\sin(4\theta)$ only with $\cos \theta$ (without $\sin \theta$).

**6.15 Express $\cos^3 \theta$ and $\sin^3 \theta$ as a linear combination of $\cos(n\theta)$ and $\sin(n\theta)$ for various n .

**6.16 Using the polar form $z = re^{i\theta}$, find all solutions of the following equations in $\mathbb{C}$.

(1) $z^3 - 8 = 0$

(2) $z^4 + 16 = 0$

(3) $z^6 - 1 = 0$

(4) $z^3 + i = 0$

(5) $z^4 - 4z^2 + 3 = 0$

(6) $z^6 + z^3 - 2 = 0$

**6.17 For $n \in \mathbb{N}^+$ the equation $z^n = 1$ has n solutions. They are called the n -th roots of unity.

(1) Let $\omega = e^{2\pi i/n}$. Show that ω^k with $k = 0, 1, \dots, n-1$ are the solutions of $z^n = 1$.

(2) Show the following equations for $n \geq 2$:

$$(a) \sum_{k=0}^{n-1} \omega^k = 0 \quad (b) \prod_{k=0}^{n-1} \omega^k = (-1)^{n+1} \quad (c) \sum_{k=0}^{n-1} \cos\left(\frac{2\pi k}{n}\right) = \sum_{k=0}^{n-1} \sin\left(\frac{2\pi k}{n}\right) = 0$$

6.4 Trigonometric and Hyperbolic functions

For real numbers, we have defined the trigonometric and hyperbolic functions (see page 44) by

$$\cos x = \text{Re } e^{ix} = \frac{e^{ix} + e^{-ix}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2},$$

$$\sin x = \text{Im } e^{ix} = \frac{e^{ix} - e^{-ix}}{2i}, \quad \sinh x = \frac{e^x - e^{-x}}{2}.$$

We naturally extend this definition to complex numbers:

$$\cos z := \frac{e^{iz} + e^{-iz}}{2}, \quad \sin z := \frac{e^{iz} - e^{-iz}}{2i}, \quad \cosh z := \frac{e^z + e^{-z}}{2}, \quad \sinh z := \frac{e^z - e^{-z}}{2}.$$

Then, almost obviously,

Theorem 6.13

For $x \in \mathbb{R}$,

$$\cos ix = \cosh x, \quad \sin ix = i \sinh x, \quad \cosh ix = \cos x, \quad \sinh ix = i \sin x. \quad (6.19)$$

Other functions, such as $\tan z$, $\cot z$, $\tanh z$, ... are defined similarly.

6.5 Complex vectors

The next step is “complex vectors”. Let us recall three interpretations of vectors, given in [Chapter 5](#):

1. an arrow in n -dimensional space ($n \in \mathbb{N}^+$, but usually $n = 3$),
2. a list of n numbers arranged vertically ($n \in \mathbb{N}^+$), and
3. an element of a vector space (such as a Hilbert space).

There we started from the first interpretation (Definition 5.1) and reached the second interpretation (Definition 5.21). For complex vectors, the first interpretation seems not nice, but Definition 5.21 looks nice to *define* complex vectors: we just extend real numbers to complex numbers.

Definition 6.14 Inner product of Complex vectors

We define n -dimensional **complex vectors** by a list of n complex number: (cf. Definition 5.21)

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}; \quad v_k \in \mathbb{C}, \quad n \in \mathbb{N}^+. \quad (6.20)$$

With this definition, we can analyze complex vectors similarly as real vectors, except for one caveat. We would like to use Eq. (5.14) to define inner products, but then it would break nice properties such as Eq. (5.4) and (C) of Theorem 5.9.

Quiz

- Q11.** Confirm that we cannot use Eq. (5.14) for complex vectors since it would break Eq. (5.4) and (C) of Theorem 5.9 in some cases such as $\begin{pmatrix} 1 \\ i \end{pmatrix}$ or $\begin{pmatrix} 0 \\ i \end{pmatrix}$.

So, we define inner products for complex vectors by (compare with Eq. (5.14))

Definition 6.15 Complex vectors

Consider n -dimensional complex vectors $\vec{a}$ and $\vec{b}$. We define the **inner product** by

$$\langle \vec{a} | \vec{b} \rangle = \overline{a_1} b_1 + \overline{a_2} b_2 + \dots + \overline{a_n} b_n = \sum_{k=1}^n \overline{a_k} b_k \quad (\text{for complex vectors}). \quad (6.21)$$

We use a different notation $\langle \vec{a} | \vec{b} \rangle$ to indicate it is *complex* inner product.

Quiz

Be careful: Important quizzes!

- Q12.** Check that $\langle \vec{a} | \vec{b} \rangle = \langle \vec{b} | \vec{a} \rangle$ is false (incorrect). [Hint: Find a counterexample.]
- Q13.** Let $\vec{v} = k\vec{a}$ with $k \in \mathbb{C}$. Check that $\langle \vec{v} | \vec{b} \rangle = k\langle \vec{a} | \vec{b} \rangle$ is false.

Remark: Cross products are not defined for complex vectors.

For completeness, we give the definition of addition and scalar multiplication of complex vectors:

Definition 6.16 Addition and Scalar multiplication of complex vectors

$$\text{For complex vectors } \vec{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \text{ and } \vec{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}, \quad \vec{a} + \vec{b} = \begin{pmatrix} a_1 + b_1 \\ \vdots \\ a_n + b_n \end{pmatrix}, \quad k\vec{a} = \begin{pmatrix} ka_1 \\ \vdots \\ ka_n \end{pmatrix}, \quad (6.22)$$

where $k \in \mathbb{C}$. (Compare with Eq. (5.13): there we only considered $k \in \mathbb{R}$, but here $k \in \mathbb{C}$.)

With these definitions, Theorem 5.7 holds **as is** for complex vectors (compare!):

Theorem 6.17

For n -dimensional complex vectors $\vec{a}$ and $\vec{b}$ and $p, q \in \mathbb{C}$,

- | | |
|---|--|
| (A) $\vec{a} + \vec{b} = \vec{b} + \vec{a}$, | (E) $p\vec{a} + p\vec{b} = p(\vec{a} + \vec{b})$, |
| (B) $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$, | (F) $p\vec{a} + q\vec{a} = (p + q)\vec{a}$, |
| (C) $\vec{a} + \vec{0} = \vec{a}$, | (G) $(pq)\vec{a} = p(q\vec{a})$, |
| (D) $\vec{a} + (-\vec{a}) = \vec{0}$, | (H) $1\vec{a} = \vec{a}$. |

Properties in Theorem 5.9 are also valid for complex vectors *with slight modifications*.

Theorem 6.18 Complex-vector inner product

For n -dimensional complex vectors $\vec{a}$ and $\vec{b}$ and a constant $k \in \mathbb{C}$,

- | | |
|--|---|
| (A) $\langle \vec{a} \vec{b} \rangle = \overline{\langle \vec{b} \vec{a} \rangle}$, | (E) $\langle \vec{a} + \vec{b} \vec{c} \rangle = \langle \vec{a} \vec{c} \rangle + \langle \vec{b} \vec{c} \rangle$, |
| (B) $\langle \vec{a} \vec{a} \rangle \geq 0$ for any vector $\vec{a}$, | (F) $\langle \vec{a} \vec{b} + \vec{c} \rangle = \langle \vec{a} \vec{b} \rangle + \langle \vec{a} \vec{c} \rangle$, |
| (C) $\langle \vec{a} \vec{a} \rangle > 0$ for any vector $\vec{a} \neq \vec{0}$, | (G) $\langle k\vec{a} \vec{b} \rangle = k\langle \vec{a} \vec{b} \rangle$, |
| (D) $\langle \vec{a} \vec{a} \rangle = 0 \iff \vec{a} = \vec{0}$, | (H) $\langle \vec{a} k\vec{b} \rangle = k\langle \vec{a} \vec{b} \rangle$. |

Using (B) of the above, we can define the **magnitude** of complex vectors by

Theorem 6.19 Magnitude of complex vectors

$$\text{For a complex vector } \vec{a}, \text{ we define its } \mathbf{magnitude} \text{ by } |\vec{a}| := \sqrt{\langle \vec{a} | \vec{a} \rangle}. \quad (6.23)$$

Theorem 6.20 Properties of complex-vector magnitude

For complex vectors $\vec{a}$ and $\vec{b}$ and a constant $k \in \mathbb{C}$,

$$|\vec{a}| \geq 0, \quad |\vec{a}| = 0 \iff \vec{a} = \vec{0}, \quad |k\vec{a}| = |k| |\vec{a}|, \quad |\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|. \quad (6.24)$$

Quiz

- Q14.** In Eq. (6.24), what do $|k|$ and $|\vec{a}|$ mean, respectively? What are their definitions?
- Q15.** Compare Theorem 6.18 with Theorem 5.9 and find all the differences.
- Q16.** Prove (A) and (G) of Theorem 6.18 using component-wise notation.
- Q17.** Prove (B) and (H) of Theorem 6.18. [Hint: (A) and (G) might be useful.]

Problems

Compare these problems with Theorem 5.10.

★6.18 (1) Verify $|\vec{a} + \vec{b}|^2 = |\vec{a}|^2 + 2\operatorname{Re}\langle \vec{a} | \vec{b} \rangle + |\vec{b}|^2$.

(2) Simplify $|(3 + 4i)\vec{v}| + |5\vec{v}|$.

(3) Expand $|\vec{v} - \vec{w}|^2$, $|\vec{v} + i\vec{w}|^2$, and $|\vec{v} + k\vec{w}|^2$ with $k \in \mathbb{C}$.

***6.19 Prove (C), (D), (E), (F) of Theorem 6.18.

6.20 Prove Theorem 6.20. Also, prove **Cauchy-Schwarz inequality, $|\langle \vec{a} | \vec{b} \rangle| \leq |\vec{a}||\vec{b}|$.

Linear combinations of complex vectors are defined similarly as real vectors (see Section 5.6); the only difference is that we can use complex coefficients instead of real coefficients. Linear dependence and independence are defined in the same manner, and Theorem 5.17 holds, but the solutions now allow complex numbers.

Recall that the component-wise notation was introduced in Section 5.8 with respect to a *specific* orthonormal basis vectors. Our discussion in this section is also built over a *specific* orthonormal basis vectors,

$$\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \quad \vec{e}_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}, \quad \text{with which } \vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \sum_{k=1}^n v_k \vec{e}_k. \quad (6.25)$$

Definition 6.21 Orthonormal basis vectors for complex vectors

Assume that we are considering n -dimensional *complex* vectors. If n complex vectors $\vec{e}_1, \dots, \vec{e}_n$ satisfy the following properties, we call them **an orthonormal basis** (for complex vectors):

- All of them are unit vectors, i.e., $|\vec{e}_i| = 1$ for all i .
- Any of them are perpendicular, i.e., $i \neq j \implies \langle \vec{e}_i | \vec{e}_j \rangle = 0$.
- We cannot add any more vectors without violating the above two rules.

Quiz

- Q18.** Compare this definition with Definition 5.18 and find the differences.

Q19. Check that $\{\vec{e}_1, \dots, \vec{e}_n\}$ in Eq. (6.25) is a basis.

Advanced note: In mathematical literature, you may find different notations for the inner product. Most physicists write $\langle \vec{a} | \vec{b} \rangle = \sum \overline{a_k} b_k$, which we use in this document. Meanwhile, mathematicians tend to denote inner products by $(\vec{a}, \vec{b})$ and define it by $(\vec{a}, \vec{b}) = \sum a_k \overline{b_k}$, or even $\langle \vec{a} | \vec{b} \rangle = \sum a_k \overline{b_k}$.

Chapter 7

Matrices

We discuss matrices^{#7}, which are deeply tied to vectors, so we will begin with a review of vectors.

The space in which the vectors “live” can be with *infinite dimension*. However, as in [Chapter 5](#) and [Chapter 6](#), we only consider finite-dimensional spaces in this chapter.

Since a matrix is described as a two-dimensional list of numbers, e.g., $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 1+i \\ 1-i & 0 \end{pmatrix}$, we will first focus on the list-of-numbers interpretation^{#8} of vectors; in ♣TODO♣section, we will use the arrow interpretation of vectors.

7.1 Review: Vectors as Lists of numbers

Let us introduce some notation for vectors. $\mathbb{R}^n$ is the set of all real n -dimensional vectors. i.e., $\vec{a} \in \mathbb{R}^n$ means “ $\vec{a}$ is a real n -dimensional vector.” Similarly, $\mathbb{C}^n$ is the set of all complex n -dimensional vectors. i.e., $\vec{a} \in \mathbb{C}^n$ means “ $\vec{a}$ is a complex n -dimensional vector.” This notation is consistent with $\mathbb{R}^{m,n}$ and $\mathbb{C}^{m,n}$, which we will see later (the set of all $m \times n$ real or complex matrices). The k -th component of $\vec{a}$ is denoted by $(\vec{a})_k$, but we may write it as a_k if not confusing.

We will also use a shorthand notation, $\mathbb{K}$, to express either $\mathbb{R}$ or $\mathbb{C}$, and you need to identify its meaning from the context. For example, “If $\vec{v} \in \mathbb{K}^3$, then $v_k \in \mathbb{K}$ but $|\vec{v}| \in \mathbb{R}$ ” is to mean

If $\vec{v}$ is a (real / complex) 3-dimensional vector, then its k -th component is a (real / complex) number and its magnitude $|\vec{v}|$ is always a real number.



First, we summarize what we have discussed about vectors. A n -dimensional (real / complex) vector, $\vec{v} \in \mathbb{K}^n$, is a column array of n (real / complex) numbers $v_k \in \mathbb{K}$:

$$\vec{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}; \quad \vec{v} + \vec{w} = \begin{pmatrix} v_1 + w_1 \\ \vdots \\ v_n + w_n \end{pmatrix}, \quad k\vec{v} = \begin{pmatrix} kv_1 \\ \vdots \\ kv_n \end{pmatrix} \quad (k \in \mathbb{K}). \quad (7.1)$$

^{#7}One dog, two dogs. One matrix, two matrices. One vertex, two vertices. One index, two indices.

^{#8}→ The beginning of [Chapter 5](#).

Notice that $k \in \mathbb{K}$ means k can only be a real number if we are considering real vectors, while k is complex if we are considering complex vectors.

The inner product and magnitude are given by

$$\begin{aligned} \text{for } \vec{v}, \vec{w} \in \mathbb{R}^n, \quad \vec{v} \cdot \vec{w} &= \sum_{k=1}^n v_k w_k, \quad |\vec{v}| = \sqrt{\vec{v} \cdot \vec{v}} = \sum_{k=1}^n v_k^2; \\ \text{for } \vec{v}, \vec{w} \in \mathbb{C}^n, \quad \langle \vec{v} | \vec{w} \rangle &= \sum_{k=1}^n \overline{v_k} w_k, \quad |\vec{v}| = \sqrt{\langle \vec{v} | \vec{v} \rangle} = \sum_{k=1}^n |v_k|^2. \end{aligned} \quad (7.2)$$

Remark: We **must** use the notation $\langle \vec{v} | \vec{w} \rangle$ for complex vectors. Meanwhile, for real vectors, we usually use $\vec{v} \cdot \vec{w}$ but also we may write $\langle \vec{v} | \vec{w} \rangle$.

For their properties, review [Chapter 5](#) and [Chapter 6](#). Cross products are only defined for $\mathbb{R}^3$:

$$\text{for } \vec{v}, \vec{w} \in \mathbb{R}^3, \quad \vec{v} \times \vec{w} = \begin{pmatrix} v_2 w_3 - v_3 w_2 \\ v_3 w_1 - v_1 w_3 \\ v_1 w_2 - v_2 w_1 \end{pmatrix}. \quad (7.3)$$

For complex vectors or real vectors other than 3d, the cross product is not defined.

We also discussed orthogonal bases of $\mathbb{K}^n$. An orthogonal basis is made of n vectors $\vec{e}_1, \dots, \vec{e}_n$, where $\vec{e}_k \in \mathbb{K}^n$, that satisfy

$$\left(|\vec{e}_k| = 1 \right) \wedge \left(j \neq k \implies \langle \vec{e}_j | \vec{e}_k \rangle = 0 \right) \quad \text{for all } j, k = 1, \dots, n, \quad (7.4)$$

or more simply,

$$\langle \vec{e}_j | \vec{e}_k \rangle = \delta_{jk} \quad \text{for all } j, k = 1, \dots, n, \quad (7.5)$$

where we introduced the **Kronecker delta** δ_{jk} , which is defined as

$$\delta_{jk} := \begin{cases} 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases} \quad (7.6)$$

Problems

★7.1 Prove Eq. (7.4) and Eq. (7.5) are equivalent.

*7.2 This discussion assumes an orthonormal basis of $\mathbb{K}^n$ always have n vectors, which we have not proved yet. So, prove that the number of basis vectors in an orthonormal basis of $\mathbb{K}^n$ is always n . [Hint: Linear dependency, discussed after this problem, will be useful.]

Let us define linear combinations (see Section 5.6) of (real / complex) vectors in a formal way:

Definition 7.1 Linear combination of vectors

Consider (real / complex) vectors $\vec{v}_1, \dots, \vec{v}_m$ and numbers $c_1, \dots, c_m$, where $\vec{v}_k \in \mathbb{K}^n$ and $c_k \in \mathbb{K}$.

$$\sum_{k=1}^m c_k \vec{v}_k = c_1 \vec{v}_1 + c_2 \vec{v}_2 + \dots + c_m \vec{v}_m \quad (7.7)$$

is called a **linear combination** of $\vec{v}_1, \dots, \vec{v}_m$.

If an equation $\sum_{k=1}^m c_k \vec{v}_k = \vec{0}$ has a solution with at least one $c_k \neq 0$, then we say that $\vec{v}_1, \dots, \vec{v}_m$ are **linearly dependent**. Otherwise, i.e., if the equation has only the trivial solution $c_1 = \dots = c_m = 0$, we say that $\vec{v}_1, \dots, \vec{v}_m$ are **linearly independent**.

Let $S = \{\vec{v}_1, \dots, \vec{v}_m\}$ with $\vec{v}_k \in \mathbb{K}^n$. If S is linearly dependent, then some vector $\vec{v}_p$ in S can be expressed by a linear combination of the other vectors in S . Meanwhile, if S is linearly independent, then it is impossible to do so. Namely, none of the vectors in S can be expressed by a linear combination of the other vectors in S .

Advanced note: Consider an orthonormal basis S of $\mathbb{K}^n$. It is obvious that

- S is linearly independent, and
- any vectors in $\mathbb{K}^n$ can be expressed as a linear combination of the vectors in S .

It is also easy to prove that $\{\vec{v}_1, \vec{v}_2, \dots, \vec{v}_m\}$, where $\vec{v}_k \in \mathbb{K}^n$, is always linear dependent if $m > n$.

7.2 Change of basis

For vectors $\mathbb{K}^n$, we have many (actually infinite) orthonormal bases. For example, for $\mathbb{C}^2$,

$$\{\vec{e}_1, \vec{e}_2\} \text{ with } \vec{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \vec{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \{\vec{f}_1, \vec{f}_2\} \text{ with } \vec{f}_1 = \begin{pmatrix} 0 \\ i \end{pmatrix}, \quad \vec{f}_2 = \begin{pmatrix} -i \\ 0 \end{pmatrix} \quad (7.8)$$

are both orthonormal bases.

Quiz

Q1. Show $\langle \vec{e}_j | \vec{e}_k \rangle = \delta_{jk}$ and $\langle \vec{f}_j | \vec{f}_k \rangle = \delta_{jk}$.

Recall that the component-wise notation is based on a specific orthonormal basis. If we used $\{\vec{f}_1, \vec{f}_2\}$ instead of $\{\vec{e}_1, \vec{e}_2\}$, the component-based notation would be different. For example, $\vec{v} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$ actually means

$$\vec{v} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} = 3\vec{e}_1 + 2\vec{e}_2, \quad \text{which is equal to} \quad -2i\begin{pmatrix} 0 \\ i \end{pmatrix} + 3i\begin{pmatrix} -i \\ 0 \end{pmatrix} = -2i\vec{f}_1 + 3i\vec{f}_2. \quad (7.9)$$

So, if we had chosen $\{\vec{f}_1, \vec{f}_2\}$ as our favorite basis, we would write

$$\vec{v} = \begin{pmatrix} -2i \\ 3i \end{pmatrix}_f, \quad (7.10)$$

where the subscript f emphasizes that the components are under the basis $\{\vec{f}_1, \vec{f}_2\}$.



In physics, we often switch from one basis to another. Choosing a basis means choosing the axes of the space^{#9}. We should choose the basis so that we can analyze the problem easily. The axes belong to the whole space, not to a single vector. Therefore, we cannot change the basis of just one vector. We have to rewrite all the vectors in the new basis.

Suppose we have vectors $\vec{a}, \vec{b}, \dots$, written under the basis $\{\vec{e}_1, \dots, \vec{e}_n\}$: (For some reason, it is useful to write $\vec{e}_1 a_1$ instead of $a_1 \vec{e}_1$ when we discuss basis transformation.)

$$\vec{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}_e = \vec{e}_1 a_1 + \dots + \vec{e}_n a_n, \quad \text{etc.} \quad (7.11)$$

How can we change the basis of all of them at once, to the new basis $\{\vec{f}_1, \dots, \vec{f}_n\}$? The key idea is that the old basis vectors $\vec{e}_k$ can be expressed in terms of the new basis vectors $\vec{f}_k$:

$$\vec{e}_k = \vec{f}_1 u_{1k} + \vec{f}_2 u_{2k} + \dots + \vec{f}_n u_{nk} = \sum_{j=1}^n \vec{f}_j u_{jk} = \begin{pmatrix} u_{1k} \\ \vdots \\ u_{nk} \end{pmatrix}_f \quad \text{for } k = 1, \dots, n, \quad (7.12)$$

where, again, we use the subscript f to emphasize that the components are under the new basis. So,

$$\vec{a} = \sum_{k=1}^n \vec{e}_k a_k = \sum_{k=1}^n \left(\sum_{j=1}^n \vec{f}_j u_{jk} \right) a_k = \sum_{j=1}^n \vec{f}_j \sum_{k=1}^n u_{jk} a_k = \sum_{j=1}^n \vec{f}_j a'_j \quad \text{with } a'_j := \sum_{k=1}^n u_{jk} a_k \quad (7.13)$$

and this equation already shows the expression of $\vec{a}$ under the new basis: we have got $\vec{a} = \begin{pmatrix} a'_1 \\ \vdots \\ a'_n \end{pmatrix}_f$.

Quiz

- Q2.** Explain each equal symbol “=” in Eq. (7.13). Also, explain why the equation means $\vec{a} = \begin{pmatrix} a'_1 \\ \vdots \\ a'_n \end{pmatrix}_f$.
- Q3.** The coefficient u_{jk} in Eq. (7.12) is given by $u_{jk} = \langle \vec{f}_j | \vec{e}_k \rangle$. Explain why. [Hint: Calculate the inner product, using Theorem 6.18.]

Theorem 7.2 Change of basis

Consider $\mathbb{K}^n$ and two bases on it: $\{\vec{e}_1, \dots, \vec{e}_n\}$ (“old” basis) and $\{\vec{f}_1, \dots, \vec{f}_n\}$ (“new” basis). If a vector $\vec{a}$ is written under the old basis as

$$\vec{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}_e = \sum_{k=1}^n \vec{e}_k a_k, \quad (7.14)$$

then it can be rewritten under the new basis as

$$\vec{a} = \begin{pmatrix} a'_1 \\ \vdots \\ a'_n \end{pmatrix}_f = \sum_{k=1}^n \vec{f}_k a'_k, \quad (7.15)$$

where the new components a'_j are given by

$$a'_j := \sum_{k=1}^n u_{jk} a_k, \quad u_{jk} = \langle \vec{f}_j | \vec{e}_k \rangle. \quad (7.16)$$

We will come back to this discussion later.

7.3 What is a Matrix?

Remark: This is a quick review; you should already be familiar with matrix arithmetic.

We introduce matrices as a 2d extension of vectors.

Definition 7.3 Matrix

A (complex / real) $m \times n$ **matrix** is defined as a 2d array of (complex / real) numbers,

$$A = \begin{pmatrix} A_{11} & A_{12} & \dots & A_{1n} \\ A_{21} & A_{22} & \dots & A_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} & A_{m2} & \dots & A_{mn} \end{pmatrix}; \quad a_{jk} \in \mathbb{K}. \quad (7.17)$$

In particular; we name a **row**, a **column** and a **component** as follows:

$$j\text{-th row: } (A_{j1} \ A_{j2} \ \dots \ A_{jn}); \quad k\text{-th column: } \begin{pmatrix} A_{1k} \\ \vdots \\ A_{mk} \end{pmatrix}; \quad (j, k)\text{-component: } A_{jk}; \quad (7.18)$$

“ $n \times m$ ” is called the **size** of the matrix. We write the set of all $n \times m$ matrix as $\mathbb{K}^{m,n}$.

For example, we can write $[M \text{ is a complex } 3 \times 2 \text{ matrix}]$ by $[M \in \mathbb{C}^{3,2}]$.

Remark:

- The last component of a $m \times n$ matrix is A_{mn} .

- Columns are tall because “**l**” is tall; rows are flat because “r-o-w” is flat.

Quiz

- Q4.** Write a 5×2 matrix. Answer the number of its components. Answer how many rows and columns does it have. Answer the number of the components in its first row. Answer the number of the components in its first column.
- Q5.** Consider a $a \times b$ matrix. Answer the number of its components. Answer how many rows and columns does it have. Answer the number of the components in its first row. Answer the number of the components in its first column.

Elements at the diagonal, $A_{11}, A_{22}, \dots, A_{nn}$ are called the **diagonal elements**. Meanwhile, other elements, i.e., A_{ij} with $i \neq j$, are called the **off-diagonal elements**. Also, vectors $\vec{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$ are called **column vectors**. We will later see **row vectors** $\vec{v}^T = (v_1 \dots v_n)$.

Quiz

- Q6.** $1 \times n$ matrices and $n \times 1$ matrices can be identified as column vectors or row vectors. Which is which?

Be careful: $\begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$ and $(v_1 \dots v_n)$ are different objects.

You will consider complex matrices in quantum mechanics courses, while in other physics we usually use real matrices. A real matrix is (of course) a complex matrix, so it is almost enough if you learn about complex matrices.



Zero matrices, denoted by $O^{m,n}$, are matrices with all components being zero, such as

$$O^{2,3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad O^{3,3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad O^{1,4} = (0 \ 0 \ 0 \ 0), \quad \text{etc.} \quad (7.19)$$

Sometimes O_n is used to denote $O^{n,n}$. Also, we may simply write O instead of $O^{m,n}$ when the size is clear from context.

Identity matrices, denoted by I_n , are $n \times n$ matrices with diagonal components being one and off-diagonal components being zero. Namely, $n \times n$ matrices with $(I_n)_{ij} = \delta_{ij}$:

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \text{etc.} \quad (7.20)$$

We may simply write I instead of I_n when the size is clear from context.

Square matrices are matrices that have the same number of rows and columns, i.e., $n \times n$ matrices. If A is square, A^T , $\bar{A}$, and $A^\dagger$ are also square matrices.

7.4 Matrix arithmetic

You need to learn following operations for matrices:

- (i) addition and scalar multiplication
- (ii) transposition, complex conjugate, and Hermitian conjugate
- (iii) matrix multiplication
- (iv) trace and determinant

We first review the definitions of these operation. Their properties are postponed to the next section.

Remark: This note does not discuss **rank**, **elementary row operations**, matrix as **linear equations**, **eigenvalues** and **eigenvectors**, and **diagonalization of matrices**, which will be discussed in sophomore courses.

Addition and **scalar multiplication** are defined just like vectors:

Definition 7.4 (i) Addition and Scalar multiplication

Addition and **scalar multiplication** of $m \times n$ matrices are defined by

$$A + B := \begin{pmatrix} A_{11} + B_{11} & A_{12} + B_{12} & \dots & A_{1n} + B_{1n} \\ A_{21} + B_{21} & A_{22} + B_{22} & \dots & A_{2n} + B_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1} + B_{m1} & A_{m2} + B_{m2} & \dots & A_{mn} + B_{mn} \end{pmatrix}, \quad kA := \begin{pmatrix} kA_{11} & \dots & kA_{1n} \\ kA_{21} & \dots & kA_{2n} \\ \vdots & \ddots & \vdots \\ kA_{m1} & \dots & kA_{mn} \end{pmatrix} \quad (7.21)$$

for the matrices A and B with same size. If A and B are in different size, $A + B$ is *not defined*.

Transposition, **complex conjugate**, and **Hermitian conjugate** are defined as follows:

Definition 7.5 (ii) Transposition, Complex conjugate, and Hermitian conjugate

Let A be a $m \times n$ matrix. Its **transposition** A^T is defined by a $n \times m$ matrix with $(A^T)_{ij} = A_{ji}$. The **complex conjugate** $\bar{A}$ of a matrix A is defined by $(\bar{A})_{ij} := \overline{A_{ij}}$. The **Hermitian conjugate** $A^\dagger$ of a matrix A is defined by $A^\dagger := \overline{A^T} = \overline{A}^T$.

Be careful: Memorize these words: “complex conjugate” and “Hermitian conjugate”.

Example 7.6

Let $A = \begin{pmatrix} 1 & 2 + i & 3i \\ -4i & 5 + 6i & -7i \end{pmatrix}$. Then its transposition is $A^T = \begin{pmatrix} 1 & -4i \\ 2 + i & 5 + 6i \\ 3i & -7i \end{pmatrix}$, complex conjugate is $\bar{A} = \begin{pmatrix} 1 & 2 - i & -3i \\ 4i & 5 - 6i & 7i \end{pmatrix}$, and Hermitian conjugate is $A^\dagger = \begin{pmatrix} 1 & 4i \\ 2 - i & 5 - 6i \\ -3i & 7i \end{pmatrix}$.

Quiz

- Q7. (1) Let $A = \begin{pmatrix} 1 & 2+i \\ -3i & 4 \end{pmatrix}$. Compute A^T , $\bar{A}$, and $A^\dagger$.
 (2) Write a 3×2 matrix A whose elements are given by $A_{ij} = i + j$.
 (3) For a matrix A , determine the condition under which $A + A^T$ is defined.
 (4) Find a matrix A that is *not* real and satisfies $A = A^\dagger$.

Matrix multiplications, defined next, is the most important operation. The definition is somewhat strange, but has a very nice property as we will see in the Remark box on page 90.

Definition 7.7 (iii) Matrix multiplication

Matrix multiplication AB is defined *iff* the number of columns of A is equal to the number of rows of B . In other words, if A is a $s \times t$ matrix and B is a $u \times v$ matrix, then AB is defined *iff* $t = u$. The product AB is a $s \times v$ matrix defined by

$$(AB)_{jk} := \sum_{n=1}^t A_{jn} B_{nk}. \quad (7.22)$$

Be careful: You need a lot of practice to get used to this operation. So...

Quiz

Q8. Calculate the following products if defined. Answer “undefined” if the product is not defined.

- | | | | | |
|------------|------------|-------------|--------------|---------------|
| (1) AB | (2) BA | (3) AC | (4) CA | (5) AD |
| (6) DA | (7) BC | (8) CB | (9) BD | (10) DB |
| (11) CD | (12) DC | (13) A^2 | (14) A^3 | (15) B^2 |
| (16) C^2 | (17) D^2 | (18) DD^T | (19) $D^T D$ | (20) $DD^T D$ |

with $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 3 & 1 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$.

Fail safe note: A^2 means AA . Also, $A^3 = AAA$, $(AB)^2 = ABAB$, and $(AB)^3 = ABABAB$.

You can calculate the product of a matrix and a vector, such as

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \vec{v} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad A\vec{v} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \quad (7.23)$$

as if $\vec{v}$ is a 2×1 matrix.

For square matrices, we can define the **trace** and **determinant** as follows:

Definition 7.8 (iv) Trace and Determinant

Consider a square matrix $A \in \mathbb{K}^{n,n}$, i.e., its size is $n \times n$. Its **trace** is defined by

$$\text{tr}(A) := \sum_{k=1}^n A_{kk} = A_{11} + A_{22} + \cdots + A_{nn}. \quad (7.24)$$

Its **determinant** is defined by

$$\text{For } 1 \times 1 \text{ matrices, } \det \begin{pmatrix} a \end{pmatrix} := a$$

$$\text{For } 2 \times 2 \text{ matrices, } \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} := ad - bc, \quad (7.25)$$

$$\text{For } 3 \times 3 \text{ matrices, } \det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} := aei + bfg + cdh - ceg - bdi - afh.$$

We will not discuss the determinant of larger matrices in this course.

Obviously, the trace and determinant of real matrices are real numbers.

Remark: Sometimes $\det A$ is written by $|A|$.

7.5 Theorems about Matrix arithmetic

Now we discuss the properties of the operations

- (i) addition and scalar multiplication
- (ii) transposition, complex conjugate, and Hermitian conjugate
- (iii) matrix multiplication
- (iv) trace and determinant

one by one.

Recall [Chapter 5](#), where addition and scalar multiplication are the fundamental operations of vectors and we saw Theorem 5.7. Here, similar properties hold for matrices.

Theorem 7.9 Properties (i) Addition and Scalar multiplication

Consider $A, B, C \in \mathbb{K}^{m,n}$ and scalars $p, q \in \mathbb{K}$. (So, A, B , and C are in the same size.) Then,

- | | |
|-----------------------------------|----------------------------|
| (A) $A + B = B + A$, | (E) $pA + pB = p(A + B)$, |
| (B) $(A + B) + C = A + (B + C)$, | (F) $pA + qA = (p + q)A$, |
| (C) $A + O^{m,n} = A$, | (G) $(pq)A = p(qA)$, |
| (D) $A + (-A) = O^{m,n}$, | (H) $1A = A$. |

Quiz

Q9. Check these properties and compare them with Theorem 5.7.

Matrix multiplications are easy to understand, except for the “be careful” box below.

Theorem 7.10 Properties (iii) Matrix multiplication

Consider $A \in \mathbb{K}^{m,n}$, $B \in \mathbb{K}^{n,l}$, $C \in \mathbb{K}^{l,k}$. (Notice that we have chosen them so that AB and BC are defined.) Also, consider $P, Q \in \mathbb{K}^{m,n}$, $R, S \in \mathbb{K}^{n,l}$, and $p \in \mathbb{K}$. Then,

- | | |
|----------------------------|-------------------------------|
| (A) $(AB)C = A(BC)$, | (D) $p(AB) = (pA)B = A(pB)$, |
| (B) $A(R + S) = AR + AS$, | (E) $AI_n = I_m A = A$, |
| (C) $(P + Q)B = PB + QB$, | (F) $AO = O$, $OA = O$. |

Be careful: The following statements are all incorrect (false).

- $AB = BA$.
- If $AB = A$, then $B = I$.
- If $A^2 = O$, then $A = O$.
- If $A^2 = A$, then $A = I$.
- If $AB = O$, then $A = O$ or $B = O$.
- If $AB = AC$, then $B = C$.
- If $AB = B$, then $A = I$.

Also, $AO = OA$ is incorrect; imagine A is a 3×2 matrix and calculate AO and OA .

Advanced note: Precisely speaking, we need the above property (A) to define A^k for a *square* (why?) matrix $A \in \mathbb{N}^{n,n}$ and $k \in \mathbb{N}^0$. Anyway, it is defined by $A^k := AA^{k-1}$ with $A^0 := I_n$.

Transpositions and conjugations obviously satisfy the following equalities:

$$\begin{aligned}
 (A^T)^T &= A, & \overline{\overline{A}} &= A, & (A^\dagger)^\dagger &= A; \\
 \overline{A^\dagger} &= \overline{A}^\dagger = A^T, & (A^T)^\dagger &= (A^\dagger)^T = \overline{A}, & \overline{A^T} &= \overline{A}^T = A^\dagger; \\
 (A + B)^T &= A^T + B^T, & \overline{A + B} &= \overline{A} + \overline{B}, & (A + B)^\dagger &= A^\dagger + B^\dagger;
 \end{aligned} \tag{7.26}$$

but the following properties are of important. *You need to check them carefully.*

Theorem 7.11 Properties (ii) Transposition and Conjugation

For matrices A, B and $p \in \mathbb{K}$,

- | | |
|---|---|
| (A) $(pA)^T = pA^T$, | (D) $(AB)^T = B^T A^T$, |
| (B) $\overline{(pA)} = \overline{p} \overline{A}$, | (E) $\overline{(AB)} = \overline{A} \overline{B}$, |
| (C) $(pA)^\dagger = \overline{p} A^\dagger$, | (F) $(AB)^\dagger = B^\dagger A^\dagger$. |

Finally, trace and determinant satisfy the following properties:

Theorem 7.12 Properties (iv) Trace and Determinant

For $A, B \in \mathbb{K}^n$ and $p \in \mathbb{K}$,

- | | |
|--|--|
| (A) $\text{tr}(A + B) = \text{tr} A + \text{tr} B$, | (E) $\det(AB) = \det(A) \det(B) = \det(BA)$, |
| (B) $\text{tr}(pA) = p \text{tr} A$, | (F) $\det(pA) = p^n \det A$; |
| (C) $\text{tr}(AB) = \text{tr}(BA)$, | (G) $\text{tr} A^T = \text{tr} A$, $\text{tr} \bar{A} = \text{tr} A^\dagger = \overline{\text{tr} A}$, |
| (D) $\text{tr}(ABC) = \text{tr}(CAB) = \text{tr}(BCA)$, | (H) $\det A^T = \det A$, $\det \bar{A} = \det A^\dagger = \overline{\det A}$. |

(These properties are correct for any $n \times n$ matrices.)

Also, $\text{tr}(I_n) = n$, $\det(I_n) = 1$, and $\text{tr}(O^{n,n}) = \det(O^{n,n}) = 0$.

Be careful: The following statements are all incorrect (false).

- $(AB)^T = A^T B^T$ • $\text{tr}(ABC) = \text{tr}(CBA)$ • $\det(-A) = -\det(A)$
- $(AB)^\dagger = A^\dagger B^\dagger$ • $\text{tr}(AB) = \text{tr}(A) \text{tr}(B)$ • $\det(A + B) = \det A + \det B$

Also, if A and B are not square matrices, then $\det(AB) = \det(BA)$ is false.

Now it's time to do (a lot of) exercises.

Problems

Let $A = \begin{pmatrix} -i & 1 \\ 0 & 2i \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, $C = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $D = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{pmatrix}$, $\vec{v} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$, $\vec{w} = \begin{pmatrix} 1+i \\ 0 \\ 1-i \end{pmatrix}$, and $\vec{x} = \begin{pmatrix} 2i \\ -1 \end{pmatrix}$.

★7.3 Calculate them. If the result is not defined, answer “undefined”.

- | | | | | | | |
|----------------------------------|--------------------------------|--------------------------------|--------------------------------|------------------------------------|----------------------------------|--|
| (1) A^2 | (2) AB | (3) BA | (4) $A^\dagger$ | (5) $A^\dagger B^\dagger$ | (6) ABD | (7) $AB^T D$ |
| (8) C^2 | (9) C^3 | (10) C^9 | (11) $D^T D$ | (12) DD^T | (13) D^2 | (14) $A\vec{x}$ |
| (15) $\vec{x}^\dagger A^\dagger$ | (16) $\vec{x}^\dagger \vec{x}$ | (17) $\vec{v}^\dagger \vec{v}$ | (18) $\vec{w}^\dagger \vec{w}$ | (19) $\vec{v}^\dagger \vec{w}$ | (20) $\vec{w}^\dagger \vec{v}$ | (21) $\langle \vec{v} \vec{w} \rangle$ |
| (22) $ \vec{v} ^2$ | (23) $\vec{v}^2$ | (24) $\vec{v} \vec{v}^\dagger$ | (25) $ A\vec{x} ^2$ | (26) $\vec{x}^\dagger A^2 \vec{x}$ | (27) $\vec{x}^\dagger B \vec{x}$ | (28) $\vec{x}^\dagger D \vec{v}$ |

★7.4 Calculate them. If the result is not defined, answer “undefined”.

- | | | | |
|-------------------------|---------------------------|-----------------------------|-----------------------------|
| (1) $\text{tr} A$ | (2) $\text{tr} B$ | (3) $\text{tr} C$ | (4) $\text{tr} D$ |
| (5) $\text{tr}(B^2)$ | (6) $\text{tr}(C^2)$ | (7) $\text{tr}(C^3)$ | (8) $\text{tr}(D^2)$ |
| (9) $\text{tr}(ABC)$ | (10) $\text{tr}(BCA)$ | (11) $\text{tr}(CBA)$ | (12) $\text{tr}(ACB)$ |
| (13) $\text{tr}(ABAB)$ | (14) $\text{tr}(A^2 B^2)$ | (15) $\text{tr}(A^\dagger)$ | (16) $\text{tr}(B^\dagger)$ |
| (17) $\det A$ | (18) $\det B$ | (19) $\det C$ | (20) $\det D$ |
| (21) $\det(A^3)$ | (22) $\det(A^9)$ | (23) $\det(ABAB)$ | (24) $\det(A^2 B^2)$ |
| (25) $\det(A^\dagger)$ | (26) $\det(C^3)$ | (27) $\det(C^9)$ | (28) $\det(D^2)$ |
| (29) $\text{tr}(AC)$ | (30) $\det(AC)$ | (31) $\text{tr}(-3AC)$ | (32) $\det(-3AC)$ |
| (33) $\text{tr}(D^T D)$ | (34) $\det(D^T D)$ | (35) $\text{tr}(-3D^T D)$ | (36) $\det(-3D^T D)$ |
| (37) $\text{tr}(DD^T)$ | (38) $\det(DD^T)$ | (39) $\text{tr}(-3DD^T)$ | (40) $\det(-3DD^T)$ |

★7.5 Calculate $\det(I_n)$, $\text{tr}(I_n)$, $\det(-I_n)$, $\text{tr}(-I_n)$, $\det(5I_n)$, and $\text{tr}(5I_n)$ for $n \in \mathbb{N}^+$.

Problems

★7.6 We have seen several “false” equations in the above “be careful” boxes. Let’s find counterexamples for them!

- (1) Find 2×2 real matrices A and B such that $AB \neq BA$.
- (2) Find 2×2 real matrices A and B such that $(A \neq O_2) \wedge (B \neq O_2) \wedge (AB = O_2)$.
- (3) Find 2×2 real matrices A and B such that $(AB = A) \wedge (B \neq I_2) \wedge (A \neq O_2)$.
- (4) Find a 2×2 real matrix A such that $A^2 = A$ and $A_{11} = A_{12} = 5$.
- (5) Find a 2×2 real matrix A such that $A^2 = O$ and $A_{11} = -A_{22} \neq 0$.

★7.7 Let us accept the equality $\text{tr}(AB) = \text{tr}(BA)$. Starting from this fact, prove $\text{tr}(ABC) = \text{tr}(CBA)$ and $\text{tr}(ABCD) = \text{tr}(DBCA)$.

***7.8 Try to prove (A)(B)(G) of Theorem 7.12.

**7.9 Prove (A)(D) of Theorem 7.11 and (C) of Theorem 7.12.

*7.10 Consult textbooks to find the definition of determinant for general $n \times n$ matrix. Write a 4×4 matrix A with components $A_{jk} = j + k + \delta_{jk}$ and calculate $\det A$.

7.6 Inverse of a matrix

If x is a (complex) number, an equation $ax = b$ with $a \neq 0$ can be solved by

$$(a \cdot x) \cdot a^{-1} = b \cdot a^{-1}$$

and then, since $(a \cdot x) \cdot a^{-1} = (a \cdot a^{-1}) \cdot x$ and $a \cdot a^{-1} = 1$, we have $x = ba^{-1}$. Can we do the same thing for matrices?

First, consider an equation $AX = B$ with $A, B, X \in \mathbb{R}^{2,2}$. If there is a matrix P such that $PA = I_2$, then we can multiply both sides of the equation by P from the left:

$$PAX = PB, \quad \text{thus} \quad PAX = I_2X = X = PB.$$

(Notice we cannot do it from the right: we get $AXP = BP$ but we cannot proceed from it.)

So, the question is whether there exists a matrix P such that $PA = I_2$, and the answer is given in the next theorem.

Quiz

- Q10. (1) Assume $PA = I_2$ and $\det A = 3$. Find $\det P$.
- (2) Assume $\det A = 0$. It is impossible to find P such that $PA = I_2$, but why?

Theorem 7.13 Inverse of a matrix

Consider a square matrix $A \in \mathbb{K}^{n,n}$. Then

$$\begin{aligned} \det A \neq 0 &\iff \text{there exists a matrix } P \text{ such that } PA = I_n \\ &\iff \text{there exists a matrix } Q \text{ such that } AQ = I_n \end{aligned} \quad (7.27)$$

and $P = Q$ if exists. Here, we call $P = Q$ the **inverse** of A and denote it by A^{-1} .
In particular, if $\det A \neq 0$, then A^{-1} exists such that $A^{-1}A = AA^{-1} = I_n$.

So, if $\det A \neq 0$ and $AX = B$, then we can solve it as $X = A^{-1}B$.

For 1×1 and 2×2 matrices, we can easily find the inverse matrices as follows:

Theorem 7.14 Explicit formulae of inverse matrices

$$\text{For } 1 \times 1 \text{ matrix, } \begin{pmatrix} a \end{pmatrix}^{-1} = \begin{pmatrix} 1/a \end{pmatrix}. \quad \text{For } 2 \times 2 \text{ matrix, } \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}. \quad (7.28)$$

Theorem 7.15 Properties of inverse matrices

Consider a square matrix $A, B \in \mathbb{K}^{n,n}$ with $\det A \neq 0$ and $\det B \neq 0$.

- | | |
|----------------------------------|---|
| (A) $(A^{-1})^{-1} = A$, | (D) $(A^T)^{-1} = (A^{-1})^T$, |
| (B) $(AB)^{-1} = B^{-1}A^{-1}$. | (E) $(\overline{A})^{-1} = \overline{A^{-1}}$, |
| (C) $\det(A^{-1}) = 1/(\det A)$ | (F) $(A^\dagger)^{-1} = (A^{-1})^\dagger$, |

7.7 Matrices with Special names

Square matrices can have various special properties. A square matrix A is called

- **symmetric** if $A = A^T$,
- **skew-symmetric** if $A = -A^T$,
- **Hermitian** if $A = A^\dagger$,
- **skew-Hermitian** if $A = -A^\dagger$,
- **unitary** if $AA^\dagger = I$,
- **orthogonal** if $AA^T = I$ and A is real [Pay attention to the second condition!],
- **upper-triangular** if $A_{jk} = 0$ for all $j > k$,
- **lower-triangular** if $A_{jk} = 0$ for all $j < k$,
- **diagonal** if all off-diagonal elements are zero.
- **normal** if it satisfies $AA^\dagger = A^\dagger A$. However, we will not discuss them in this course,

Furthermore, a square matrix A is called

- **regular** if $\det A \neq 0$; it may also be called **invertible** or **non-singular**.
- **singular** if $\det A = 0$; it may also be called **non-invertible** or **irregular**.

Be careful: Too many names? Yes, but you need to know them all!

Example 7.16

Consider the following square matrices.

$$\begin{array}{llllll}
 (1) \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} & (2) \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} & (3) \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix} & (4) \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} & (5) \begin{pmatrix} 1 & i \\ -i & 1 \end{pmatrix} & (6) \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\
 (7) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} & (8) \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} & (9) \begin{pmatrix} 0 & 2i \\ 2i & 0 \end{pmatrix} & (10) \begin{pmatrix} 1 & 0 \\ 2i & 1 \end{pmatrix} & (11) \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} & (12) \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}
 \end{array}$$

(2)(3)(7)(9) are symmetric and (4)(6) are skew-symmetric; (2)(3)(5)(6)(7) are Hermitian and (4)(9) is skew-Hermitian. (4)(6)(7)(8) are unitary; (4)(7)(8) are orthogonal. (2)(7)(11)(12) are upper-triangular, (2)(7)(10) are lower-triangular, and (2)(7) are diagonal. Normal matrices are (2)(3)(4)(5)(6)(7)(8)(9).

These special matrices have the following important properties:

Theorem 7.17 Matrices with special names

We here assume all matrices are $n \times n$ matrices.

- Properties about determinant:
 1. If A is Hermitian, $\det A \in \mathbb{R}$.
 2. If A is skew-Hermitian, $\det A$ is zero or pure-imaginary.
 3. If A is unitary, $|\det A| = 1$.
 4. If A is orthogonal, $\det A = \pm 1$.
- Properties about Hermitian and skew-Hermitian matrices:
 5. If A is Hermitian, then A^n is also Hermitian for $n \in \mathbb{N}^+$.
 6. If A and B are both Hermitian, then, AB is Hermitian $\iff AB = BA$.
 7. For any matrices M , $M + M^\dagger$ is Hermitian and $M - M^\dagger$ is skew-Hermitian.
- Properties about unitary and orthogonal matrices:
 8. A unitary matrix U is invertible; $U^{-1} = U^\dagger$.
 9. An orthogonal matrix P is invertible; $P^{-1} = P^T$.
- Properties about symmetric and skew-symmetric matrices:
 10. For any matrices M , $M + M^T$ is symmetric and $M - M^T$ is skew-symmetric.
- Properties about triangular matrices:
 11. If A is (upper or lower) triangular, $\det A = \prod_{k=1}^n A_{kk} = A_{11}A_{22} \cdots A_{nn}$.
- Finally,
 12. Hermitian, skew-Hermitian, unitary, and orthogonal matrices are all normal. Real symmetric matrices and real skew-symmetric matrices are also normal.

The property 7. and 10. can be used for **decomposition of matrices**.

- Any matrix M can be *uniquely* decomposed as a sum of a Hermitian matrix H and a skew-Hermitian matrix K , i.e.,

$$M = H + K, \quad \text{where} \quad H = \frac{M + M^\dagger}{2} \quad \text{and} \quad K = \frac{M - M^\dagger}{2}. \quad (7.29)$$

- Any matrix M can be *uniquely* decomposed as a sum of a symmetric matrix S and a skew-symmetric matrix T , i.e.,

$$M = S + T, \text{ where } S = \frac{M + M^T}{2} \text{ and } T = \frac{M - M^T}{2}. \quad (7.30)$$

These **decompositions** are important in linear algebra.

Quiz

Q11. Prove the following facts:

- (1) For any square matrix M , $M + M^\dagger$ is Hermitian, $M - M^\dagger$ is skew-Hermitian, $M + M^T$ is symmetric, and $M - M^T$ is skew-symmetric.
- (2) Hermitian, skew-Hermitian, symmetric, and skew-symmetric matrices are all normal.
- (3) If A is Hermitian, then A^n is also Hermitian for $n \in \mathbb{N}^+$.

Advanced note: The following facts will be important when you learn **diagonalization of matrices**.

- A normal matrix is diagonalizable by a unitary matrix.
- A real symmetric matrix is diagonalizable by an orthogonal matrix.

Problems

****7.11** Prove the following facts:

- (1) 1–10 and 12 of Theorem 7.17.
- (2) The decompositions of matrix, given in Eq. (7.29) and Eq. (7.30), are unique.

Problems

*****7.12** A square matrix A is called **idempotent** if $A^2 = A$. Let A be a $n \times n$ idempotent matrix. Prove the following facts.

- (1) $I_n - A$ is also idempotent. [Hint: Calculate $(I_n - A)^2$.]
- (2) If A is regular, then $A = I_n$. Namely, $(A \text{ is idempotent}) \Rightarrow (A = I_n) \vee (\det A = 0)$.

*****7.13** A square matrix A is called **nilpotent** if $A^k = O$ for some positive integer k . Let A be a $n \times n$ nilpotent matrix. Prove the following facts.

- (1) A is singular. [Hint: Find its determinant.]
- (2) $(I_n - A)(I_n + A + A^2 + \cdots + A^{k-1}) = I_n$. (It means $I - A$ is invertible.)

- **7.14**
- (1) Find all 2×2 idempotent matrices.
 - (2) Find all 2×2 nilpotent matrices.
 - (3) Find all 2×2 matrices that satisfy $A^2 = I_2$.

7.8 Geometric interpretation of Matrices

Consider 2×1 real matrices and 3×1 real matrices, such as $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}$. Because Sho asked you to write a vector in the vertical form, it is natural to identify $n \times 1$ real matrices as n -dimensional real vectors. And thus,

Statement 7.18

We can identify a matrix $M \in \mathbb{R}^{n,1}$ as a n -dimensional real vector $\vec{v} \in \mathbb{R}^n$, and furthermore, as *an arrow* in a n -dimensional space.

The cases with $n = 2$ and 3 are important in physics because our space is three-dimensional. So, let's consider a few 2×2 matrices:

$$V = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad H = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} \cos 90^\circ & -\sin 90^\circ \\ \sin 90^\circ & \cos 90^\circ \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (7.31)$$

Quiz

- Q12.** (1) Draw $\vec{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ as an arrow on a xy -plane.
 (2) Calculate $V\vec{v}$, $H\vec{v}$, and $C\vec{v}$ and draw them on the plane.
 (3) Think about the meanings of these matrices V , H , and C .
 (4) Calculate H^2 , V^2 , C^2 , C^4 , HV , and VH , and think about their meanings.

Obviously, V corresponds to a vertical **reflection** (reflection with respect to x -axis), H to a horizontal **reflection** (reflection with respect to y -axis), and C corresponds to a 90° -rotation around the origin. Therefore, V^2 , H^2 , and C^4 must be the identity matrix I_2 , and $C^2 = HV = VH$.

Quiz

- Q13.** (1) Calculate V^{-1} , H^{-1} , and C^{-1} .
 (2) Show that V , H , C are orthogonal. (Also, show that they are unitary.)
- Q14.** Let us define $[A, B] := AB - BA$, which we call it **commutator**. Show that
 (1) $[H, V] = 0$, which we say “the matrix H and the matrix V **commutes**”.
 (2) $[H, C] \neq 0$, which we say “ H and C do not commute.”

When we operate H and V on an arrow, we do not have to care about the order because $HV = VH$, i.e., H and V commute. Meanwhile, when we operate reflection H and 90° -rotation C , the order matters because the operators do not commute.

Remark: You may think it is natural or obvious that the operation $C^2 = CC$ is equivalent to the operation “first do C , and then C ”, or that HC is equal to the operation “first C , and then H ”. However, it is not either obvious, fortune, or accident; rather, it is *by definition*. In fact, the matrix multiplication (Definition 7.7) is defined so that the product AB becomes equivalent to “operate B and then A ”.

For 2d arrows, the following matrices are important.

Theorem 7.19 Matrices as operators on 2d arrows

For arrows in 2d space,

- $H = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ corresponds to horizontal **reflection**, or reflection about y-axis, of the arrow.
- $V = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$ corresponds to vertical **reflection**, or reflection about x-axis, of the arrow.
- $R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ corresponds to **rotation** by the angle θ about the origin.
- $M_{s,t} = \begin{pmatrix} s & 0 \\ 0 & t \end{pmatrix}$ with $s > 0$ and $t > 0$ corresponds to **scaling** by a factor s along x -axis and by a factor t along y -axis, or in other words, horizontally by s and vertically by t . In particular, $M_{s,s} = sI_2$ magnifies the whole space by s .

Be careful: You need to memorize R_θ .

Quiz

- Q15.** (1) Show that $R_\theta R_\varphi = R_{\theta+\varphi}$. Also, show that R_θ and R_φ commute.
 (2) Show that R_θ is orthogonal (and unitary).
 (3) Show that the inverse of R_θ is $R_{-\theta}$ and the inverse of $M_{s,t}$ is $M_{1/s, 1/t}$.
 [Remark: It means they are invertible.]
- Q16.** Rotation and reflection should not modify the magnitude of a vector, while $M_{s,s}$ should scale the magnitude by a factor s . Let us confirm these properties as follows. First, consider $\vec{v} = \begin{pmatrix} a \\ b \end{pmatrix}$ and calculate $H\vec{v}$, $V\vec{v}$, $R_\theta\vec{v}$, and $M_{s,s}\vec{v}$. Then, calculate their magnitude and compare them with $|\vec{v}|$.

For 3d arrows, the following matrices are important:

Theorem 7.20 Matrices as operators on 3d arrows

For arrows in 3d space,

- $P_x = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ corresponds to **reflection** about yz -plane. Similarly, $P_y = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and $P_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ correspond to reflection about, respectively, zx -plane and xy -plane.
- $P = P_x P_y P_z = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ corresponds to the reflection about all the axes. This operation is called **parity transformation**.
- $R_{x;\theta} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & -\sin \theta \\ 0 & \sin \theta & \cos \theta \end{pmatrix}$, $R_{y;\theta} = \begin{pmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{pmatrix}$, and $R_{z;\theta} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}$ corresponds to **rotation** by the angle θ about x -axis, y -axis, and z -axis, respectively.
- $M_{s,t,u} = \begin{pmatrix} s & 0 & 0 \\ 0 & t & 0 \\ 0 & 0 & u \end{pmatrix}$ with $s, t, u > 0$ corresponds to **scaling** by a factor s along x -axis, t along y -axis, and u by z -axis. In particular, $M_{s,s,s} = sI_3$ magnifies the whole space by s .

Notice that rotations in 3d-space are much more complicated than in 2d. In 2d, we only have one parameter θ , while you need three parameters to describe a 3d rotation:

- Any 3d rotation can be expressed by rotation around one axis $\vec{n}$ by the angle θ (Euler's rotation theorem). Here, two parameters are to determine the axis $\vec{n}$ and one parameter is the angle θ .
- Any 3d rotation can be expressed by a combination of three rotations around different axes, such as $R_{z;\gamma}R_{y;\beta}R_{x;\alpha}$. The three angles α, β, γ are called **Euler angles**.

We do not discuss it in details except for one *be-careful* box.

Be careful: Notice the order of the operations in $R_{z;\gamma}R_{y;\beta}R_{x;\alpha}$ because they are operating on a vector $\vec{v}$ from the left. The first rotation is around x -axis; the second one is about y -axis; the last one is about z -axis.

Advanced note: Also notice that, in $R_{z;\gamma}R_{y;\beta}R_{x;\alpha}$, the second rotation is not about the original y -axis, but about y -axis after the first rotation (sometimes called y' -axis). Similarly, the third rotation is not about the original z -axis, but about z'' -axis after the two rotations. This is called **extrinsic rotations**. There is also a concept of **intrinsic rotations**, which is defined in a different way.

For more discussions, see Section 4.4–4.6 of [2]. Euler's rotation theorem is “physically” obvious, but for rigorous proof, see Theorem 6.6.15 of [3].

Quiz

- Q17.** Show that reflections P_x, P_y, P_z do not modify the magnitude of a vector. [Hint: Consider $\vec{v}$ and calculate the magnitude of $P_x\vec{v}$ etc.]

Now we want to confirm that the rotation matrices do not modify the magnitude of a vector, but it is a bit messy to calculate the magnitude of $R_{x;\theta}\vec{v}$ etc. Here, instead, we use the following theorem:

Theorem 7.21

Consider a vector $\vec{v} \in \mathbb{K}^n$ and a matrix $A \in \mathbb{K}^{n,n}$. Then,

$$|A\vec{v}|^2 = (A\vec{v})^\dagger A\vec{v} = \vec{v}^\dagger A^\dagger A\vec{v}. \quad (7.32)$$

Fail safe note: Recall $(AB)^\dagger = B^\dagger A^\dagger$. Also, if A and $\vec{v}$ are real, we may write $|A\vec{v}|^2 = (A\vec{v})^T (A\vec{v}) = \vec{v}^T A^T A\vec{v}$.

Proof: $A\vec{v}$ is a vector with components $(A\vec{v})_k = \sum A_{kj}v_j$. So, its magnitude squared is given by

$$|A\vec{v}|^2 = \sum_{k=1}^n |(A\vec{v})_k|^2 = \sum_{k=1}^n \overline{(A\vec{v})_k} (A\vec{v})_k = \sum_{k=1}^n \sum_{j=1}^n \overline{A_{kj}} \overline{v_j} \sum_{j'=1}^n A_{kj'} v_{j'} = \sum_{k=1}^n \sum_{j=1}^n \sum_{j'=1}^n \overline{v_j} \overline{A_{kj}} A_{kj'} v_{j'}.$$

Meanwhile,

$$\vec{v}^\dagger A^\dagger A\vec{v} = \sum_{a=1}^n \sum_{b=1}^n \sum_{c=1}^n \overline{v_a} (A^\dagger)_{ab} A_{bc} v_c = \sum_{a=1}^n \sum_{b=1}^n \sum_{c=1}^n \overline{v_a} \overline{A_{ba}} A_{bc} v_c,$$

so they are equal. ■

With this theorem, we can easily show that rotations do not modify the magnitude of a vector.

Quiz

Q18. Show that rotations $R_{x;\theta}$ etc. do not modify the magnitude of a vector by the following procedure.

- (1) Show that $(R_{x;\theta})^\dagger R_{x;\theta} = I_3$. (Notice that it means $R_{x;\theta}$ is unitary.)
- (2) Show that $|R_{x;\theta}\vec{v}| = |\vec{v}|$.

Problems

★7.15 Show the following facts, using Theorem 7.21.

- (1) If $R \in \mathbb{C}^{n,n}$ is unitary, then $|R\vec{v}| = |\vec{v}|$ for any vector $\vec{v} \in \mathbb{C}^n$.
- (2) If $R \in \mathbb{C}^{n,n}$ is normal, then $|R\vec{v}| = |R^\dagger \vec{v}|$ for any vector $\vec{v} \in \mathbb{C}^n$.

***7.16 Show that, for $A, B, C \in \mathbb{K}^{n,n}$,

- (1) $[B, A] = -[A, B]$
- (2) $[A + B, C] = [A, C] + [B, C]$
- (3) $[AB, C] = A[B, C] + [A, C]B$
- (4) $[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = O^{n,n}$.

7.9 Matrix as an Operator on Vectors

In the previous sections, we have done the identification

a $n \times 1$ real matrix = a n -dimensional real vector = an arrow in n -d space.

We have seen, for $M \in \mathbb{R}^{n,n}$ and $\vec{v} \in \mathbb{R}^n$, $M\vec{v}$ is again a n -dimensional vector. So, we could identify

a $n \times n$ real matrix = an operator on a n -dimensional real vector.

In this section, we will do a similar discussion for complex vectors and matrices.

a $n \times 1$ complex matrix = a n -dimensional complex vector,

a $n \times n$ complex matrix = an operator on a n -dimensional complex vector,

Advanced note: In this document, we have introduced matrices as a 2d list of numbers and identified them as operators. However, in formal mathematics, the operator nature is more fundamental and the 2d list are secondary. We will briefly see this topic in [♣TODO♣chap](#).

The starting point is the inner product of $\vec{v}, \vec{w} \in \mathbb{C}^n$. If we regard the vectors as matrices, we can describe the inner product by

$$\langle \vec{v} | \vec{w} \rangle = \sum_{k=1}^n \overline{v_k} w_k = (\overline{v_1} \ \overline{v_2} \ \dots \ \overline{v_n}) \begin{pmatrix} w_1 \\ w_2 \\ \dots \\ w_n \end{pmatrix} = (\vec{v})^\dagger \vec{w}.$$

The first box is just Eq. (7.2). In the second box, the vectors are identified as matrices, where $(\vec{v})^\dagger$ is the Hermitian conjugate of $\vec{v}$. This equality motivates us to describe

a vector $\vec{v}$ as $|\vec{v}\rangle$ (or simply written by $|v\rangle$), which is called a **ket**,
 an Hermitian conjugate $(\vec{v})^\dagger$ as $\langle \vec{v} |$ (or simply written by $\langle v |$), which is called a **bra**,

and this is why we denote the inner product as $\langle \vec{v} | \vec{w} \rangle$.

Theorem 7.22

For any vectors $\vec{v} \in \mathbb{K}^n$ and $\vec{w} \in \mathbb{K}^n$, we can express the inner product $\langle \vec{v} | \vec{w} \rangle$ and the magnitude $|\vec{v}|$ by matrix multiplication:

$$\langle \vec{v} | \vec{w} \rangle = (\vec{v})^\dagger \vec{w}, \quad |\vec{v}|^2 = (\vec{v})^\dagger \vec{v},$$

where the vectors $\vec{v}$ and $\vec{w}$ are identified as $n \times 1$ matrices.

Since $A\vec{v}$ is a vector, we can write $|A\vec{v}\rangle$, or even

$$\begin{aligned} |A\vec{v}\rangle &= A|\vec{v}\rangle : \text{the vector } A\vec{v} \text{ as a ket, or equivalently, a ket } \vec{v} \text{ with an operation } A, \\ \langle A\vec{v} | &= \langle \vec{v} | A^\dagger : \text{the Hermitian conjugate of } |A\vec{v}\rangle. \end{aligned}$$

We can also combine them. For example, if we let $\vec{w} = A\vec{v}$,

$$\begin{aligned} \langle \vec{v} | \vec{w} \rangle &= \langle \vec{v} | A\vec{v} \rangle = \langle \vec{v} | A | \vec{v} \rangle \\ \langle \vec{w} | \vec{v} \rangle &= \langle A\vec{v} | \vec{v} \rangle = \langle \vec{v} | A^\dagger | \vec{v} \rangle \\ \langle \vec{w} | \vec{w} \rangle &= \langle A\vec{v} | A\vec{v} \rangle = \langle \vec{v} | A^\dagger A | \vec{v} \rangle \quad \text{etc.} \end{aligned}$$

Quiz

Q19. Consider vectors $\vec{v}, \vec{w} \in \mathbb{C}^n$ and matrices $A, B \in \mathbb{C}^{n,n}$.

(1) Show that the Hermitian conjugate of $A\vec{v}$ is $\langle \vec{v}|A^\dagger$, i.e., $\langle A\vec{v}| = \langle \vec{v}|A^\dagger$.

(2) Show that $\overline{\langle \vec{v}|A|\vec{w}\rangle} = \langle \vec{w}|A^\dagger|\vec{v}\rangle$.

Fail safe note: Let $\vec{u} = A\vec{w}$. Recall $\overline{\langle \vec{v}|\vec{u}\rangle} = \langle \vec{u}|\vec{v}\rangle$.

(3) Show that $\langle \vec{v}|A|\vec{v}\rangle$ is real if A is Hermitian.

(4) Express $|A\vec{v}|^2$ as the expression $\langle \vec{v}|\dots|\vec{v}\rangle$.

(5) Show that $|A\vec{v}|^2 = |\vec{v}|^2$ if A is unitary.



We saw $\langle \vec{v}|\vec{w}\rangle$ is just a complex number. How about $|\vec{v}\rangle\langle \vec{w}|$?

$$|\vec{v}\rangle\langle \vec{w}| = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} (\overline{w_1} \ \overline{w_2} \ \dots \ \overline{w_n}) = \begin{pmatrix} v_1\overline{w_1} & v_1\overline{w_2} & \dots & v_1\overline{w_n} \\ v_2\overline{w_1} & v_2\overline{w_2} & \dots & v_2\overline{w_n} \\ \vdots & \vdots & \ddots & \vdots \\ v_n\overline{w_1} & v_n\overline{w_2} & \dots & v_n\overline{w_n} \end{pmatrix} \quad (7.33)$$

and this is a $n \times n$ matrix. Recall that v_k and w_k , the components, are given under some basis vectors $\{\vec{e}_k\}$. The matrix expression is therefore basis-dependent: the value of each element will be different if we use another basis.

The next theorem is useful in your future lectures:

Theorem 7.23 Projection operators

If $\{\vec{f}_1, \dots, \vec{f}_n\}$ is an orthonormal basis of $\mathbb{K}^n$,

- $P_1 = |\vec{f}_1\rangle\langle \vec{f}_1|$, $P_2 = |\vec{f}_2\rangle\langle \vec{f}_2|$, ..., $P_n = |\vec{f}_n\rangle\langle \vec{f}_n|$ are called **projection operators**.
- The sum of all projection operators are identity. Namely,

$$\sum_{k=1}^n P_k = \sum_{k=1}^n |\vec{f}_k\rangle\langle \vec{f}_k| = I_n. \quad (7.34)$$

Example 7.24

In this example, we consider $\mathbb{K}^2 = \mathbb{C}^2$, assuming that the component-wise notation $\vec{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ is given under the basis $\{\vec{e}_1, \vec{e}_2\}$, i.e., $\vec{v} = v_1\vec{e}_1 + v_2\vec{e}_2$ and $v_k = \langle \vec{e}_k | \vec{v} \rangle$ for $i = 1, 2$. Then, $\vec{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\vec{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

$\{\vec{e}_1, \vec{e}_2\}$ is an orthonormal basis. The projection operators associated to these basis vectors are given by $P_1 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ and $P_2 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$, so $P_1 + P_2 = I_2$.

If we define $\vec{f}_1 = \begin{pmatrix} 3/5 \\ 4/5 \end{pmatrix}$ and $\vec{f}_2 = \begin{pmatrix} -4/5 \\ 3/5 \end{pmatrix}$, then $\{\vec{f}_1, \vec{f}_2\}$ is an orthonormal basis. Projection operators for these basis vectors are given by

$$|\vec{f}_1\rangle\langle\vec{f}_1| = \begin{pmatrix} 9/25 & 12/25 \\ 12/25 & 16/25 \end{pmatrix}, \quad |\vec{f}_2\rangle\langle\vec{f}_2| = \begin{pmatrix} 16/25 & -12/25 \\ -12/25 & 9/25 \end{pmatrix}, \quad (7.35)$$

and thus $\sum |\vec{f}_k\rangle\langle\vec{f}_k| = I_2$.

Fail safe note: If we used the basis vectors $\{\vec{f}_1, \vec{f}_2\}$ for the component-wise notation, we would obtain $|\vec{f}_1\rangle\langle\vec{f}_1| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}_f$ and $|\vec{f}_2\rangle\langle\vec{f}_2| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}_f$.

Quiz

Q20. Show that

$$\{\vec{g}_1, \vec{g}_2\}, \quad \text{where} \quad \vec{g}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} e^i \\ 1 \end{pmatrix}, \quad \vec{g}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -e^i \end{pmatrix}$$

is an orthonormal basis for $\mathbb{C}^2$. Calculate the projection operators for these basis vectors and check that their sum is I_2 .

With this identity, Theorem 7.2 can be understood in a more intuitive way. Consider a vector $\vec{a} \in \mathbb{K}^n$. Then,

$$|\vec{a}\rangle = I_n |\vec{a}\rangle = \sum_{k=1}^n |\vec{e}_k\rangle\langle\vec{e}_k| \vec{a}\rangle = \sum_{k=1}^n |\vec{f}_k\rangle\langle\vec{f}_k| \vec{a}\rangle \quad (7.36)$$

with two bases $\{\vec{e}_1, \dots, \vec{e}_n\}$ and $\{\vec{f}_1, \dots, \vec{f}_n\}$. The component-wise expressions are, for each basis,

$$\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}_e \quad \text{with } a_k = \langle \vec{e}_k | \vec{a} \rangle, \quad \text{and} \quad \begin{pmatrix} a'_1 \\ \vdots \\ a'_n \end{pmatrix}_f \quad \text{with } a'_k = \langle \vec{f}_k | \vec{a} \rangle, \quad (7.37)$$

and the coefficients are related by

$$\begin{aligned} a_k &= \langle \vec{e}_k | \vec{a} \rangle = \langle \vec{e}_k | \cdot \left[\sum_{m=1}^n |\vec{f}_m\rangle\langle\vec{f}_m| \right] \vec{a}\rangle = \sum_{m=1}^n \langle \vec{e}_k | \vec{f}_m \rangle \langle \vec{f}_m | \vec{a} \rangle = \sum_{m=1}^n \langle \vec{e}_k | \vec{f}_m \rangle a'_m, \\ a'_k &= \langle \vec{f}_k | \vec{a} \rangle = \langle \vec{f}_k | \cdot \left[\sum_{m=1}^n |\vec{e}_m\rangle\langle\vec{e}_m| \right] \vec{a}\rangle = \sum_{m=1}^n \langle \vec{f}_k | \vec{e}_m \rangle \langle \vec{e}_m | \vec{a} \rangle = \sum_{m=1}^n \langle \vec{f}_k | \vec{e}_m \rangle a_m. \end{aligned} \quad (7.38)$$

We can write $\langle \vec{f}_k | \vec{e}_m \rangle$ as a $n \times n$ matrix U :

$$U_{km} = \langle \vec{f}_k | \vec{e}_m \rangle. \quad (7.39)$$

Then, $\langle \vec{e}_m | \vec{f}_k \rangle = (U_{km})^* = (U^\dagger)_{mk}$ and we found

$$\begin{pmatrix} a'_1 \\ \vdots \\ a'_n \end{pmatrix} = U \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}, \quad \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = U^\dagger \begin{pmatrix} a'_1 \\ \vdots \\ a'_n \end{pmatrix}. \quad (7.40)$$

This equation actually means U is a unitary matrix, but we are not going to dive there in this document.

Problems

****7.17** Prove Eq. (7.34).

Chapter 8

Vector Calculus

After we introduced vectors in [Chapter 5](#), we took a long journey through complex numbers, complex vectors, and matrices. Those abstract structures are essential for modern physics, but at the end of the day, we physicists live in the real world: the three-dimensional space $\mathbb{R}^3$.

In this chapter, we return to that world. We will focus on vectors in $\mathbb{R}^3$, i.e., arrows in our three-dimensional space, and study **vector calculus**: how to apply the tools of calculus (derivatives and integrals) to such vectors. The central objects will be **scalar fields** and **vector fields**, which we introduce first.

In this chapter, vectors are real and three-dimensional ($\in \mathbb{R}^3$) unless otherwise stated. Also, we will assume all functions are smooth enough (or “friendly to physicists”); we do not consider functions with singularities, discontinuities, or other “pathological” behaviors.

Advanced note: Mathematically, all functions in this chapter are assumed to be **class C^2** . A function is called C^p if we can calculate its p -th partial derivative in any combinations and the derivatives are all continuous. So, we assume $\partial_x^2 f$, $\partial_x \partial_y f$, $\partial_y \partial_x f$, ..., and $\partial_z^2 f$ are all existent and continuous. This is a sufficient condition to ensure the symmetry of second derivatives, $\partial_i \partial_j f = \partial_j \partial_i f$ [4, pp. 732–733].

In this section, we introduce the short-hand notation for partial derivatives,

$$\partial_x = \frac{\partial}{\partial x}, \quad \partial_y = \frac{\partial}{\partial y}, \quad \partial_z = \frac{\partial}{\partial z}; \quad \text{we also use} \quad \partial_1 = \partial_x, \quad \partial_2 = \partial_y, \quad \partial_3 = \partial_z. \quad (8.1)$$

For example, $\partial_x(3x + 2xy + y + 1) = \frac{\partial}{\partial x}(3x + 2xy + y + 1) = 3 + 2y$.

Quiz

Q1. With $f(x, y) = x^3 + 4x^2 + 5xy^2 + 7y^2$, calculate the following.

- (1) $\partial_x f$ (2) $\partial_y f$ (3) $\partial_x \partial_y f$ (4) $\partial_y \partial_x f$ (5) $\partial_x^2 f$

8.1 Scalar fields and Vector fields

You already know functions like $f(x) = x^2$: one number in, one number out. It is natural to extend the idea to “many numbers in, many numbers out”. For example,

- $v(t) = \begin{pmatrix} 2t \\ 1-t^2 \\ 0 \end{pmatrix}$... one number in, three numbers (= a vector) out.
- $T(x, y, z) = \sqrt{x^2 + y^2 + z^2}$... three numbers in, one number out.

- $\vec{F}(x, y, z) = \begin{pmatrix} 3x+2y \\ y+z \\ x+y+z \end{pmatrix} = \begin{pmatrix} 3 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}$... three numbers in, three numbers out.

Here, if you recall the concept of position vectors ($\rightarrow$ Section 5.5), you notice that the input (x, y, z) may represent a point in space. Namely, we can understand the above functions as

- $T(x, y, z) = \sqrt{x^2 + y^2 + z^2}$... a function that assigns a scalar value T for each position (x, y, z) ,
- $\vec{F}(x, y, z) = \begin{pmatrix} 3x+2y \\ y+z \\ x+y+z \end{pmatrix}$... a function that assigns a vector value $\vec{F}$ for each position (x, y, z) ,

and they are called **scalar field** and **vector field**, respectively. Or, more formally,

Definition 8.1 Scalar field and Vector field

A **scalar field** is a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, $(x, y, z) \mapsto f(x, y, z)$. (8.2)

A **vector field** is a function $\vec{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $(x, y, z) \mapsto \vec{F}(x, y, z) = \begin{pmatrix} F_1(x, y, z) \\ F_2(x, y, z) \\ F_3(x, y, z) \end{pmatrix}$. (8.3)

They assign a scalar or a vector, respectively, to every point in some region of the space.

Here we have used a mathematical notation, $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, which means f accepts an element of $\mathbb{R}^3$ and returns an element of $\mathbb{R}$. For physicists, however, it is more important to notice the physical meaning: $f(x, y, z)$ and $\vec{F}(x, y, z)$ **assigns** some value to each point. For example...

Quiz

Q2. Choose scalar fields. Choose vector fields.

- | | |
|------------------------------------|-------------------------------------|
| (1) Velocity of a car | (2) Wind velocity in Kaohsiung |
| (3) Water velocity in a sea | (4) PM2.5 density in Kaohsiung air |
| (5) Air temperature in a room | (6) Age of people in Kaohsiung |
| (7) Population in cities in Taiwan | (8) Air pressure in the atmosphere |
| (9) Electric field in a space | (10) Voltage at points in a circuit |

Among them, wind velocity in a city, water velocity in a sea, and electric field in a space are vector fields; PM2.5 density and air temperature, and air pressure are scalar fields. Velocity of a car is a vector but not a vector field since it is just for a single point, not for every point in a space. Similarly, age of people, population of cities, and voltage at points in a circuit are scalars but not scalar fields.

Quiz

- Q3. (1) Assume $\vec{F}(x, y, z)$ is a vector field. How about its x -component $F_1(x, y, z)$? [Hint: It is a scalar field. Explain why.]
- (2) How about “wind speed in Kaohsiung”? Is it a scalar field or a vector field?

Definition 8.2 Fields and Position vectors

For a field, we often use position vector $\vec{r}$ to express the input numbers:

- a scalar field $f(x, y, z)$ is also written $f(\vec{r})$.
- a vector field $\vec{F}(x, y, z)$ is also written $\vec{F}(\vec{r})$, $\begin{pmatrix} F_1(\vec{r}) \\ F_2(\vec{r}) \\ F_3(\vec{r}) \end{pmatrix}$, or $\begin{pmatrix} F_1(x, y, z) \\ F_2(x, y, z) \\ F_3(x, y, z) \end{pmatrix}$.

In addition, we often write F_x, F_y, F_z instead of F_1, F_2, F_3 . So,

- a vector field $\vec{F}(x, y, z)$ is written $\vec{F}(\vec{r})$, $\begin{pmatrix} F_x(\vec{r}) \\ F_y(\vec{r}) \\ F_z(\vec{r}) \end{pmatrix}$, or $\begin{pmatrix} F_x(x, y, z) \\ F_y(x, y, z) \\ F_z(x, y, z) \end{pmatrix}$.

Be careful: Do not confuse the two “x”s in $F_x(x, y, z)$. The first x is just a function name ($= F_1$), while the second x is a variable: the x -coordinate of a position. In other

words, $\vec{F}(5, 7, 9) = \begin{pmatrix} F_5(5, 7, 9) \\ F_7(5, 7, 9) \\ F_9(5, 7, 9) \end{pmatrix}$ is incorrect, but $\vec{F}(5, 7, 9) = \begin{pmatrix} F_x(5, 7, 9) \\ F_y(5, 7, 9) \\ F_z(5, 7, 9) \end{pmatrix}$.

Advanced note: In mathematics, we use the following notation to express a function f :

$$\begin{array}{ccc} f : \mathbb{R}^3 & \longrightarrow & \mathbb{R} \\ \Downarrow & & \Downarrow \\ \vec{r} & \longmapsto & f(\vec{r}) = |\vec{r}|^2 \end{array} \quad \text{or equivalently,} \quad \boxed{f : \mathbb{R}^3 \longrightarrow \mathbb{R}, \quad \vec{r} \longmapsto f(\vec{r}) = |\vec{r}|^2.}$$

This means that the function f

- can accept any element of $\mathbb{R}^3$ as an input,
- returns an element of $\mathbb{R}$ as an output,
- returns $f(\vec{r}) = |\vec{r}|^2$ if an input $\vec{r} \in \mathbb{R}^3$ is given.

As declared, the output $|\vec{r}|^2$ is an element of $\mathbb{R}$.

In general, for a function $f : A \longrightarrow B$, we call A the **domain** f needs to accept any elements of A . Meanwhile, the output does not have to cover all of B , but a subset of B , called **image**. For example, if we consider $h(x) = x^{-2}$, we cannot write $h : \mathbb{R} \longrightarrow \mathbb{R}$ because we cannot accept $x = 0$. Instead, we write

$$h : \mathbb{R} \setminus \{0\} \longrightarrow \mathbb{R}, \quad x \longmapsto x^{-2}.$$

The **domain** of h is $\mathbb{R} \setminus \{0\}$ and the **image** is $\mathbb{R}^+ = \{y \in \mathbb{R} \mid y > 0\}$ since $1/x^2$ is always positive.

Problems

***8.1** Assume $x \in \mathbb{R}$ and $z \in \mathbb{C}$. Write the following functions as given in the first box of the above “advanced note”. Find its domain and image.

- (1) x^2 (2) $\ln(x)$ (3) $\sqrt{x}$ (4) $(x + 2)^2$ (5) $\overline{z^2}$ (6) $|z|^2$

8.2 Gradient, Divergence, Curl, and Laplacian

We can analyze scalar and vector fields by derivatives and integrals. For derivatives, the following operations are useful.

Definition 8.3 Gradient

For a scalar field $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, the **gradient** of f , $\boxed{\text{grad } f}$ or $\boxed{\vec{\nabla} f}$, is defined by

$$\text{grad } f := \begin{pmatrix} \partial f / \partial x \\ \partial f / \partial y \\ \partial f / \partial z \end{pmatrix} = \begin{pmatrix} \partial_x f \\ \partial_y f \\ \partial_z f \end{pmatrix}. \quad (8.4)$$

Quiz

- Q4.** (1) In this definition, we did not write the argument (x, y, z) for simplicity. Write the equation Eq. (8.4) with the argument (x, y, z) explicitly.
- (2) $\text{grad } f$ is a vector field. Explain why.
- (3) Let $f(x, y, z) = x(x + y)^2 + 2z$. Calculate $\text{grad } f$. Also, calculate $\text{grad } f$ at the origin $(0, 0, 0)$.

Definition 8.4 Divergence and Curl

For a vector field $\vec{F} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, the **divergence**, $\boxed{\text{div } \vec{F}}$ or $\boxed{\vec{\nabla} \cdot \vec{F}}$, is defined by

$$\text{div } \vec{F} := \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z} = \sum_{k=1}^3 \partial_k F_k \quad (8.5)$$

and the **curl**, $\boxed{\text{curl } \vec{F}}$ or $\boxed{\vec{\nabla} \times \vec{F}}$, is defined by

$$\text{curl } \vec{F} := \begin{pmatrix} \partial F_z / \partial y - \partial F_y / \partial z \\ \partial F_x / \partial z - \partial F_z / \partial x \\ \partial F_y / \partial x - \partial F_x / \partial y \end{pmatrix} = \begin{pmatrix} \partial_y F_z - \partial_z F_y \\ \partial_z F_x - \partial_x F_z \\ \partial_x F_y - \partial_y F_x \end{pmatrix}. \quad (8.6)$$

Curl is also called **rotation** and written by $\boxed{\text{rot } \vec{F}}$ in some textbooks.

Quiz

- Q5.** (1) Again, write Eq. (8.5) and Eq. (8.6) with the argument (x, y, z) explicitly.
- (2) $\text{grad } f$ is a vector field. How about $\text{div } \vec{F}$ and $\text{curl } \vec{F}$?
- (3) Let $\vec{F}(x, y, z) = \begin{pmatrix} x^2 + y^2 \\ xyz \\ z^2 \end{pmatrix}$. Calculate $\text{div } \vec{F}$ and $\text{curl } \vec{F}$. Also, calculate $\text{div } \vec{F}$ and $\text{curl } \vec{F}$ at the point $(1, 1, 1)$.

Definition 8.5 Laplacian

For a scalar field $f(\vec{r})$, the **Laplacian** of f , $\boxed{\Delta f}$ or $\boxed{\nabla^2 f}$, is defined by

$$\Delta f := \operatorname{div}(\operatorname{grad} f) = \operatorname{div} \begin{pmatrix} \partial_x f \\ \partial_y f \\ \partial_z f \end{pmatrix} = \partial_x^2 f + \partial_y^2 f + \partial_z^2 f. \quad (8.7)$$

We may also consider laplacian for a vector field $\vec{F}$:

$$\Delta \vec{F} := \begin{pmatrix} \Delta F_x \\ \Delta F_y \\ \Delta F_z \end{pmatrix} = \begin{pmatrix} \partial_x^2 F_x + \partial_y^2 F_x + \partial_z^2 F_x \\ \partial_x^2 F_y + \partial_y^2 F_y + \partial_z^2 F_y \\ \partial_x^2 F_z + \partial_y^2 F_z + \partial_z^2 F_z \end{pmatrix}. \quad (8.8)$$

Quiz

- Q6.** (1) Again, write Eq. (8.7) with the argument (x, y, z) explicitly.
 (2) Calculate $\operatorname{div}(\operatorname{grad} f)$ and check that it is equal to $\partial_x^2 f + \partial_y^2 f + \partial_z^2 f$.
 (3) Calculate Δf for $f(x, y, z) = x(x + y)^2 + z^2$.

We are going to analyze the properties of these operations and their physical meanings, but before that, we prepare some notations in the next section.

8.3 Kronecker, Levi-Civita, and Nabla

In [Chapter 7](#) we defined the **Kronecker delta** (see Eq. (7.6))

$$\delta_{ij} := \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j, \end{cases} \quad \text{where } i, j \in \{1, 2, 3\} \quad (\text{recall that we are focusing on } \mathbb{R}^3).$$

Here we introduce the **Levi-Civita symbol** ε_{ijk} :

$$\begin{aligned} \varepsilon_{123} &= \varepsilon_{231} = \varepsilon_{312} = +1, \\ \varepsilon_{132} &= \varepsilon_{213} = \varepsilon_{321} = -1, \\ \varepsilon_{ijk} &= 0 \text{ if any two indices are equal.} \end{aligned} \quad (8.9)$$

Quiz

Q7. Calculate the following.

$$(1) \varepsilon_{123} \quad (2) \varepsilon_{321} \quad (3) \sum_{a=1}^3 \varepsilon_{a23} \quad (4) \sum_{a=1}^3 \sum_{b=1}^3 \varepsilon_{ab3} \quad (5) \sum_{a=1}^3 a^2 \delta_{2a}$$

Fail safe note: Just expand the summation: $\sum_{a=1}^3 \varepsilon_{a23} = \varepsilon_{123} + \varepsilon_{223} + \varepsilon_{323}$.

This symbol allows us to write the cross product in a simple way:

$$\vec{A} \times \vec{B} = \sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^3 \varepsilon_{ijk} \vec{e}_i A_j B_k \quad (8.10)$$

or, considering the i -th component of $\vec{A} \times \vec{B}$,

$$(\vec{A} \times \vec{B})_i = \sum_{j=1}^3 \sum_{k=1}^3 \varepsilon_{ijk} A_j B_k. \quad \left(\text{very similar to } \vec{A} \cdot \vec{B} = \sum_{j=1}^3 \sum_{k=1}^3 \delta_{jk} A_j B_k. \right) \quad (8.11)$$

Proof: According to Definition 5.21, the i -th component of $\vec{v}$ is given by $v_i = \vec{e}_i \cdot \vec{v}$; here $i = 1, 2, 3$ means $i = x, y, z$, respectively. So, the i -th component of $\vec{A} \times \vec{B}$ is given by

$$\begin{aligned} (\vec{A} \times \vec{B})_i &= \vec{e}_i \cdot \left(\sum_{p=1}^3 \sum_{j=1}^3 \sum_{k=1}^3 \varepsilon_{pjk} \vec{e}_p A_j B_k \right) = \sum_{p=1}^3 \sum_{j=1}^3 \sum_{k=1}^3 \varepsilon_{pjk} (\vec{e}_i \cdot \vec{e}_p) A_j B_k = \sum_{p=1}^3 \sum_{j=1}^3 \sum_{k=1}^3 \varepsilon_{pjk} \delta_{ip} A_j B_k \\ &= \sum_{j=1}^3 \sum_{k=1}^3 \varepsilon_{ijk} A_j B_k. \blacksquare \end{aligned}$$

Quiz

- Q8.** (1) Show Eq. (8.10). [Hint: Just expand the right-hand side!]
 (2) Show Eq. (8.11). The proof is given above, but can you do it by yourself?

The Levi-Civita symbol satisfies the following identities:

Theorem 8.6 Properties of Levi-Civita symbol

- | | |
|--|--|
| (A) $\varepsilon_{abc} = \varepsilon_{bca} = \varepsilon_{cab} = -\varepsilon_{cba} = -\varepsilon_{bac} = -\varepsilon_{acb}$ | (D) $\sum_{i,j,k=1}^3 \varepsilon_{ijk} \varepsilon_{ijk} = 6.$ |
| (B) $\sum_{i=1}^3 \varepsilon_{iab} \varepsilon_{ixy} = \delta_{ax} \delta_{by} - \delta_{ay} \delta_{bx}.$ | (E) $\varepsilon_{abc} \varepsilon_{xyz} = \det \begin{pmatrix} \delta_{ax} & \delta_{ay} & \delta_{az} \\ \delta_{bx} & \delta_{by} & \delta_{bz} \\ \delta_{cx} & \delta_{cy} & \delta_{cz} \end{pmatrix}$ |
| (C) $\sum_{i,j=1}^3 \varepsilon_{ija} \varepsilon_{ijx} = 2\delta_{ax}.$ | (F) $\varepsilon_{abc} = \det \begin{pmatrix} \delta_{a1} & \delta_{a2} & \delta_{a3} \\ \delta_{b1} & \delta_{b2} & \delta_{b3} \\ \delta_{c1} & \delta_{c2} & \delta_{c3} \end{pmatrix}$ |

Quiz

Q9. (1) Simplify the following expressions:

- | | |
|--|--|
| (a) $\varepsilon_{123} \delta_{11} + \varepsilon_{121} \delta_{11}$ | (b) $\sum_{a=1}^3 \varepsilon_{12a} (\delta_{1a} + \delta_{a3})$ |
| (c) $\sum_{a=1}^3 \varepsilon_{13a} \varepsilon_{13a}$ | (d) $\varepsilon_{abc} - \varepsilon_{bac} + 2\varepsilon_{cba}$ |
| (e) $\sum_{a=1}^3 \varepsilon_{axy} (\delta_{ax} + \delta_{ay} + \delta_{ab})$ | (f) $\sum_{a=1}^3 \varepsilon_{abc} \varepsilon_{a23}$ |

Problems

****8.2** (1) Prove (A)–(D) of Theorem 8.6.

(2) Show that $\det M = \sum_{i,j,k=1}^3 \varepsilon_{ijk} M_{1i} M_{2j} M_{3k}$ for a 3×3 matrix $M \in \mathbb{C}^{3,3}$.

(3) Show (E). [Hint: Use (2). What M should we use?]

(4) Show (F). [Hint: Substitute (E) with $x = 1, y = 2, z = 3$.]

 **8.3**

$$(1) \sum_{a=1}^3 \sum_{b=1}^3 \delta_{ab}$$

$$(2) \sum_{a=1}^3 \sum_{b=1}^3 a \delta_{ab}$$

$$(3) \sum_{a=1}^3 \sum_{b=1}^3 |\varepsilon_{ab3}|$$

Advanced note: The Levi-Civita symbol is related to the sign of a **permutation**. A permutation is a rearrangement of elements in a list $(1, 2, \dots, n)$, such as $(1234) \rightarrow (4132)$. The simplest permutations are transpositions, a swap of two elements with the rest unchanged, such as $(1234) \rightarrow (1432)$. Any permutation can be decomposed into a sequence of transpositions, and although neither the decomposition nor the number of transpositions is unique, it always turns out that the number is either always even or always odd. So, we can define the “sign” of a permutation σ as $\text{sgn } \sigma = +1$ (-1) for an even (odd) permutation. Then, can define the Levi-Civita symbol by $\varepsilon_{ijk} = \text{sgn } \sigma_{(123) \rightarrow (ijk)}$.

With the above properties, it is easy to prove the following:

Theorem 8.7 Properties of Levi-Civita symbol

$$(1) \vec{A} \cdot (\vec{B} \times \vec{C}) = \vec{B} \cdot (\vec{C} \times \vec{A}) = \vec{C} \cdot (\vec{A} \times \vec{B}) \quad (\text{scalar triple product})$$

$$(2) \vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C} \quad (\text{vector triple product})$$

$$(3) \vec{A} \times (\vec{B} \times \vec{C}) + \vec{B} \times (\vec{C} \times \vec{A}) + \vec{C} \times (\vec{A} \times \vec{B}) = 0 \quad (\text{Jacobi identity})$$

Proof: (1) is obvious (why?) once we write $\vec{A} \cdot (\vec{B} \times \vec{C}) = \sum A_i (\vec{B} \times \vec{C})_i = \sum A_i \varepsilon_{ijk} B_j C_k$. For (2), we write

$$\vec{A} \times (\vec{B} \times \vec{C}) = \sum \varepsilon_{ijk} A_i (\vec{B} \times \vec{C})_j \vec{e}_k = \sum \varepsilon_{ijk} A_i (\varepsilon_{jlm} B_l C_m) \vec{e}_k$$

using (B) of Theorem 8.6, and then use (D) of Theorem 8.6:

$$\sum \varepsilon_{ijk} A_i (\varepsilon_{jlm} B_l C_m) \vec{e}_k = \sum \varepsilon_{jki} A_i (\varepsilon_{jlm} B_l C_m) \vec{e}_k = \sum (\delta_{kl} \delta_{im} - \delta_{km} \delta_{il}) A_i B_l C_m \vec{e}_k;$$

so, please continue and complete these proofs in the next problems.

Problems

★8.4 Using Theorem 8.6, prove (A) and (B) of Theorem 8.7.

****8.5** Prove (C) of Theorem 8.7. Also, prove

$$(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = (\vec{A} \cdot \vec{C})(\vec{B} \cdot \vec{D}) - (\vec{A} \cdot \vec{D})(\vec{B} \cdot \vec{C}). \quad (8.12)$$

We also introduce one more notation, called **nabla**:

$$\vec{\nabla} = \begin{pmatrix} \partial_x \\ \partial_y \\ \partial_z \end{pmatrix} \quad \text{with an abbreviation} \quad \partial_x = \frac{\partial}{\partial x}, \quad \partial_y = \frac{\partial}{\partial y}, \quad \partial_z = \frac{\partial}{\partial z}. \quad (8.13)$$

This notation is helpful to recall the definitions of gradient, divergence, and curl:

$$\text{grad } f = \begin{pmatrix} \partial_x f \\ \partial_y f \\ \partial_z f \end{pmatrix} = \vec{\nabla} f, \quad \text{div } \vec{F} = \partial_x F_x + \partial_y F_y + \partial_z F_z = \vec{\nabla} \cdot \vec{F}, \quad \text{curl } \vec{F} = \vec{\nabla} \times \vec{F}.$$

Be careful: Like ∂_x , ∂_y , and ∂_z , we can understand grad, div, and curl as operators. Meanwhil, it is a bit dangerous to regard $\vec{\nabla}$ itself as an operator; Sho recommends you to consider it a “symbol” or “way of writing”. Also, please *do not* think $\vec{\nabla}$ is a vector. It just looks like a vector, but it is not!

However, the $\vec{\nabla}$ -notation is often abused, such as $(\vec{A} \cdot \vec{\nabla})$: we can understand it as

$$(\vec{A} \cdot \vec{\nabla})f = \begin{pmatrix} A_x \\ A_y \\ A_z \end{pmatrix} \cdot \begin{pmatrix} \partial_x \\ \partial_y \\ \partial_z \end{pmatrix} f = \vec{A} \cdot (\vec{\nabla} f) = A_x \partial_x f + A_y \partial_y f + A_z \partial_z f,$$

but sometimes you may be confused by such abuse. In general, when you see $\vec{\nabla}$, you should carefully identify the author’s intention.

We can express them in terms of Kronecker delta and Levi-Civita symbol:

Theorem 8.8

For a scalar field f and a vector field $\vec{F}$,

$$\begin{aligned} \text{grad } f &= \sum_i \vec{e}_i \partial_i f, & \text{div } \vec{F} &= \sum_i \partial_i F_i, & \text{curl } \vec{F} &= \sum_{i,j,k} \varepsilon_{ijk} \vec{e}_i \partial_j F_k, \\ (\text{grad } f)_i &= \partial_i f, & & & (\text{curl } \vec{F})_i &= \sum_{j,k} \varepsilon_{ijk} \partial_j F_k. \end{aligned} \quad (8.14)$$

Quiz

Q10. Check Eq. (8.14). [Hint: You just expand all the terms.]

Now, we are going to combine all the above to understand the properties and physical meanings of gradient, divergence, curl, and Laplacian.

8.4 Geometrical Interpretations

First, let us recall what are the input and output types of the vector-calculus operators.

Summary 8.9 Type rules for vector-calculus operators

- grad turns a scalar field into a vector field.
- div turns a vector field into a scalar field.
- curl turns a vector field into a vector field.
- Δ turns a scalar field into a scalar field and a vector field into a vector field.

Example 8.10

Consider $\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, and let $r = |\vec{r}|$. Then, $r = \sqrt{x^2 + y^2 + z^2}$ and

$$\text{grad} \frac{1}{r} = \begin{pmatrix} \partial_x \\ \partial_y \\ \partial_z \end{pmatrix} \frac{1}{\sqrt{x^2 + y^2 + z^2}} = \begin{pmatrix} -x/(x^2 + y^2 + z^2)^{3/2} \\ -y/(x^2 + y^2 + z^2)^{3/2} \\ -z/(x^2 + y^2 + z^2)^{3/2} \end{pmatrix} = -\frac{\vec{r}}{r^3},$$

i.e., $\vec{\nabla}(1/r) = -\vec{r}/r^3$. In this calculation, $\vec{r}$ is not a vector; rather, we should understand it as a vector field, which receives a position and returns a vector. Similarly, r and $1/r$ should be regarded as scalar fields. The operation “grad” turns a scalar field $1/r$ to a vector field $-\vec{r}/r^3$.

Similarly, we can calculate $\text{div} \vec{r}$ and $\text{curl} \vec{r}$:

$$\text{div} \vec{r} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3, \quad \text{curl} \vec{r} = \vec{0}.$$

These 3 and $\vec{0}$ are understood as constant fields returning 3 and $\vec{0}$ for any position, respectively.

Quiz

Q11. Let $\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, $r = |\vec{r}|$, and $\vec{a} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$. Calculate the following if defined:

- | | | | |
|--|---|--|--------------------------|
| (1) $\text{grad}(1/r)$ | (2) $\text{grad} r^2$ | (3) $\text{grad}(\vec{a} \cdot \vec{r})$ | (4) $\text{div} \vec{a}$ |
| (5) $\text{div}(\vec{a} \times \vec{r})$ | (6) $\text{curl}(\vec{a} \times \vec{r})$ | (7) $\Delta(\vec{r} \cdot \vec{r})$ | (8) $\Delta(1/r)$ |

Consider a scalar function $f(x, y, z)$ and its gradient $(\text{grad} f)(x, y, z)$; recall $\text{grad} f$ is a vector field, which means $\text{grad} f$ for a given point $\vec{p}$ is a vector.

Theorem 8.11 Meaning of Gradient

Consider a scalar field $f(\vec{r})$. Its **gradient** at a point $\vec{p}$, which we write $(\text{grad} f)(\vec{p})$, is a vector with

- direction: the direction in which f increases fastest from the point.
- magnitude: the fastest rate of the increase.

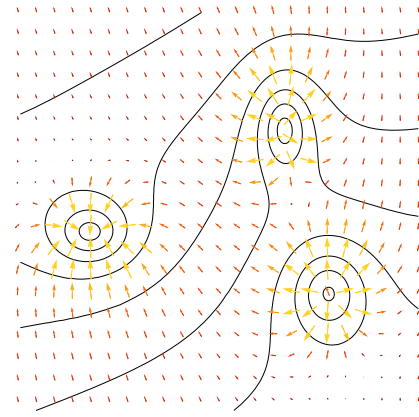
Also, if we consider the *isosurface* (the two-dimensional surface on which the value of $f(\vec{x})$ is the same as $f(\vec{p})$), then $(\text{grad} f)(\vec{p})$ is perpendicular to the isosurface at the point $\vec{p}$.

The isosurface is also called *equivalue surface* or, in mathematics, *level surface*.

To understand gradient, it is easier to consider a scalar field which has no dependence on z . Then, $(\text{grad } f)_z = \partial_z f = 0$ and we can draw the vectors on the xy -plane, as in the figure to the right. There, the black contours show the isosurface of f , i.e., the value of f is the same at all the point on each line. The arrows show $\text{grad } f$ at representative points.

You should observe the following properties:

- f is smaller in the central region; larger in the outer area.
- $\text{grad } f$ is always perpendicular to the isosurface.
- $|\text{grad } f|$ is larger in the left-hand side than the right-hand side, which means the “slope” is steeper in the left-hand side.

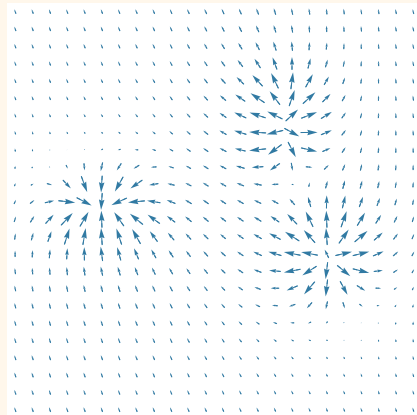


Quiz

Q12. The figure to the right shows $\text{grad } f$ for a scalar field $f(x, y, z)$ drawn on xy -plane. We assume that f is independent of z , so that $\text{grad } f$ has no z -component.

- (1) Try to draw contour lines describing the isosurface of f .
- (2) At what point is f the largest?
- (3) At what point is f the smallest?

Q13. In Example 8.10 we got $\vec{\nabla}(r^{-1}) = -\vec{r}/r^3$. Try to visualize this vector field, $-\vec{r}/r^3$.



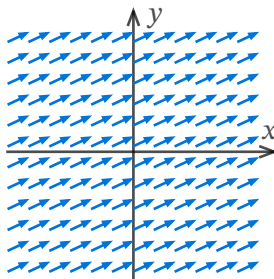
Be careful: Vector calculus is always on the three-dimensional space, so your imagination is crucial. If you feel difficulties in 3D visualization, search for videos, animations, or interactive applications on the internet; they will help your imagination.

The divergence and curl are considered for a vector field $\vec{F}$, and thus it is difficult to visualize them here. Rather, it is important to notice when the divergence and curl are positive, negative, or zero.

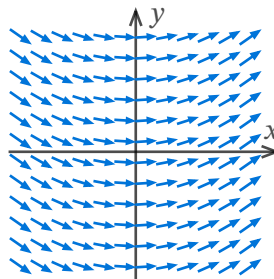
Example 8.12

Consider the following vector fields. *Because we use the right-handed Cartesian coordinate system, the z-axis is perpendicular to the page, running toward you.*

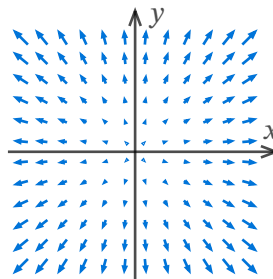
(A) $\vec{F} = \begin{pmatrix} 1.5 \\ 0.7 \\ 0 \end{pmatrix}$



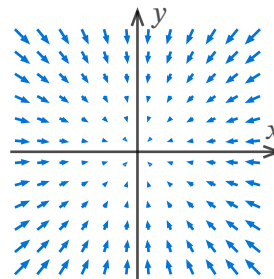
(B) $\vec{F} = \begin{pmatrix} 1.5 \\ x \\ 0 \end{pmatrix}$



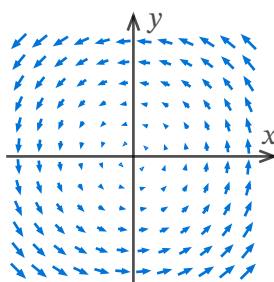
(C) $\vec{F} = \begin{pmatrix} x \\ y \\ 0 \end{pmatrix}$



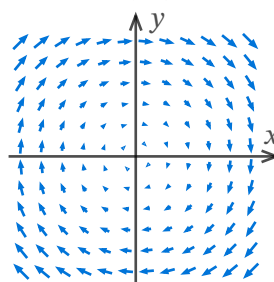
(D) $\vec{F} = \begin{pmatrix} -x \\ -y \\ 0 \end{pmatrix}$



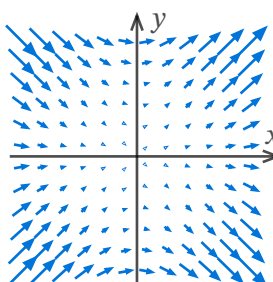
(E) $\vec{F} = \begin{pmatrix} -y \\ x \\ 0 \end{pmatrix}$



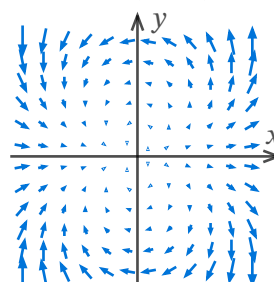
(F) $\vec{F} = \begin{pmatrix} y \\ -x \\ 0 \end{pmatrix}$



(G) $\vec{F} = \begin{pmatrix} x^2+y^2 \\ 2xy \\ 0 \end{pmatrix}$



(H) $\vec{F} = \begin{pmatrix} x^2-y^2 \\ 2xy \\ 0 \end{pmatrix}$



If you calculate $\text{div } \vec{F}$, you should notice

- (A) and (B) has no divergence.
- (C) has positive divergence at every point of xy -plane.
- (D) has negative divergence at every point of xy -plane.

These (C) and (D) are the typical situations for positive and negative divergence, respectively.

For curl, you should notice

- (A), (C), and (D) has no curl.
- For (E), $\text{curl } \vec{F} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} = 2\vec{e}_z$. Meanwhile, (F) has $\text{curl } \vec{F} = \begin{pmatrix} 0 \\ 0 \\ -2 \end{pmatrix} = -2\vec{e}_z$.

You see some *rotation* in (E) and (F). What is the direction of the rotations? If you recall that we use the right-hand rule to describe the rotation, you can observe that $\text{curl } \vec{F}$ gives the direction of rotation.

These are the typical cases and you can guess the divergence and curl from the visualizations. However, in general, it is difficult to see the divergence and curl. As a first learner, you should focus on the following points:

Summary 8.13 Geometric properties of divergence and curl

- $\text{div} = 0$ means the flow is conserved *at the point*, without any source or sink.
- $\text{div} > 0$ means the point has a source; some flow is coming out from the point.
- $\text{div} < 0$ means the point has a sink; some flow is absorbed at the point.
- $\text{curl} \neq 0$ means the vector field has some rotational motion. The rotation direction is given by the right-hand rule with respect to the direction of curl.

Quiz

- Q14.** Calculate $\text{div } \vec{F}$ and $\text{curl } \vec{F}$ for each vector field in Example 8.12. Can you find the following features?
- (B) has some rotational movement.
 - (E) and (F) has no source or sink in this region.
 - In (G) and (H), some regions have sources of the flow and some regions have sinks of the flow.
 - In (H), there is one region that has a rotational movement, and there is another region that has a rotational movement in the opposite direction of the first region. Meanwhile, (G) has no rotational movement.

Theorem 8.14 Vanishing combination

For any scalar field f , $\text{curl}(\text{grad } f) = \vec{0}$.
 For any vector field $\vec{F}$, $\text{div}(\text{curl } \vec{F}) = 0$. (8.15)

Quiz

- Q15.** Prove this theorem. You may assume $\partial_x \partial_y f = \partial_y \partial_x f$, etc.

Advanced note: The equality $\partial_i \partial_j f = \partial_j \partial_i f$ holds because we have assumed the fields are in class C^2 .

8.5 Formulae in Vector calculus

Since $\text{grad } f$ is a vector field, we can apply “div” and “curl” to it; but then what will happen?

Theorem 8.15 Double derivatives

For a scalar field f and a vector field $\vec{F}$,

- curl grad: $\vec{\nabla} \times (\vec{\nabla} f) = \vec{0}$.
- div grad: $\vec{\nabla} \cdot (\vec{\nabla} f) = \Delta f = \partial_x^2 f + \partial_y^2 f + \partial_z^2 f$.
- div curl: $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{F}) = 0$.
- grad div: $\vec{\nabla}(\vec{\nabla} \cdot \vec{F}) = \begin{pmatrix} \partial_x(\partial_x F_x + \partial_y F_y + \partial_z F_z) \\ \partial_y(\partial_x F_x + \partial_y F_y + \partial_z F_z) \\ \partial_z(\partial_x F_x + \partial_y F_y + \partial_z F_z) \end{pmatrix}$.
- curl curl: $\vec{\nabla} \times (\vec{\nabla} \times \vec{F}) = \vec{\nabla}(\vec{\nabla} \cdot \vec{F}) - \Delta \vec{F}$.

Quiz

Q16. We have three operations (div, curl, grad), so nine combinations are possible, but only the five are listed above. Why are the remaining four not shown?

Q17. Show the above five equations.

Fail safe note: Use Eq. (8.14). Notice that $\vec{e}_i$ is a constant, so $\partial_i \vec{e}_j = \vec{0}$.

We may have other formulae such as

$$\vec{\nabla}(fg) = f\vec{\nabla}g + g\vec{\nabla}f \quad (8.16)$$

$$\vec{\nabla} \cdot (f\vec{F}) = f\vec{\nabla} \cdot \vec{F} + (\vec{\nabla}f) \cdot \vec{F} \quad (8.17)$$

$$\vec{\nabla} \times (f\vec{F}) = f\vec{\nabla} \times \vec{F} + (\vec{\nabla}f) \times \vec{F} \quad (8.18)$$

$$\vec{\nabla} \cdot (\vec{F} \times \vec{G}) = (\vec{\nabla} \times \vec{F}) \cdot \vec{G} - \vec{F} \cdot (\vec{\nabla} \times \vec{G}) \quad (8.19)$$

but usually you will not need them; instead, it is better to use Eq. (8.14) and the chain rule.

Example 8.16

Recalling $(\vec{F} \times \vec{G})_i = \varepsilon_{ijk}F_jG_k$, we can calculate

$$\vec{\nabla} \cdot (\vec{F} \times \vec{G}) = \sum \partial_i (\vec{F} \times \vec{G})_i = \sum \partial_i (\varepsilon_{ijk}F_jG_k) = \sum \varepsilon_{ijk}[(\partial_i F_j)G_k + F_j(\partial_i G_k)].$$

Then, looking at Eq. (8.14) carefully, we may obtain $\sum \varepsilon_{ijk}(\partial_i F_j)G_k = \sum (\text{curl } F)_k G_k$ and $\sum \varepsilon_{ijk}F_j(\partial_i G_k) = -\sum F_j(\text{curl } G)_j$, which leads to

$$\vec{\nabla} \cdot (\vec{F} \times \vec{G}) = (\vec{\nabla} \times \vec{F}) \cdot \vec{G} - \vec{F} \cdot (\vec{\nabla} \times \vec{G}).$$

Similarly, we can expand $\vec{\nabla} \times (\vec{F} \times \vec{G})$ by

$$\begin{aligned} [\vec{\nabla} \times (\vec{F} \times \vec{G})]_a &= \sum \varepsilon_{abc} \partial_b (\vec{F} \times \vec{G})_c \\ &= \sum \varepsilon_{abc} \partial_b (\varepsilon_{cjk} F_j G_k) \\ &= \sum \varepsilon_{cab} \varepsilon_{cjk} [(\partial_b F_j) G_k + F_j (\partial_b G_k)] \\ &= \sum (\delta_{aj} \delta_{bk} - \delta_{ak} \delta_{bj}) [(\partial_b F_j) G_k + F_j (\partial_b G_k)] \\ &= \sum [(\partial_b F_a) G_b + F_a (\partial_b G_b) - (\partial_b F_b) G_a - F_b (\partial_b G_a)] \\ &= \sum [(\partial_b F_a) G_b + F_a (\vec{\nabla} \cdot \vec{G}) - (\vec{\nabla} \cdot \vec{F}) G_a - F_b (\partial_b G_a)]. \end{aligned}$$

Problems

★8.6 Repeating Example 8.16, prove Eq. (8.16) – Eq. (8.19).

Example 8.17 Divergence of the position field

Compute $\nabla \cdot \vec{r}$ for $\vec{r} = (x, y, z)^T$.

Solution

$$\nabla \cdot \vec{r} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 1 + 1 + 1 = 3. \quad (8.20)$$

Every point in space is a source: the field $\vec{r}$ points away from the origin everywhere and grows in magnitude, so it “spreads out” at every point, consistently with the constant, positive divergence.

Example 8.18 Curl of the position field

Compute $\nabla \times \vec{r}$ for $\vec{r} = (x, y, z)^T$.

Solution

$$(\nabla \times \vec{r})_x = \frac{\partial z}{\partial y} - \frac{\partial y}{\partial z} = 0 - 0 = 0, \quad (8.21)$$

and the other two components vanish the same way, since each component of $\vec{r}$ depends on only one coordinate. So $\nabla \times \vec{r} = \vec{0}$: the position field does not rotate around any point, which matches the intuition that $\vec{r}$ points straight out from the origin everywhere, with no swirl.

$\nabla^2 f$ is a scalar field: it is the divergence of the gradient, i.e. “the gradient of f , then take its divergence.” It appears everywhere in physics: the diffusion equation, the wave equation, and Laplace’s equation $\nabla^2 V = 0$ for the electric potential in charge-free regions all use it.

Example 8.19 Laplacian of $1/r$

Show that $\nabla^2(1/r) = 0$ for $r \neq 0$.

Solution

$\nabla(1/r) = -\vec{r}/r^3$, with components $-x/r^3, -y/r^3, -z/r^3$. Differentiate the x -component again with respect to x (using the product rule, since r depends on x):

$$\frac{\partial}{\partial x} \left(-\frac{x}{r^3} \right) = -\frac{1}{r^3} - x \cdot (-3)r^{-4} \frac{\partial r}{\partial x} = -\frac{1}{r^3} + \frac{3x}{r^4} \cdot \frac{x}{r} = -\frac{1}{r^3} + \frac{3x^2}{r^5}, \quad (8.22)$$

using $\frac{\partial r}{\partial x} = x/r$ (same calculation as in the gradient example, applied to r itself instead of $1/r$). Summing the analogous results for y and z :

$$\nabla^2 \left(\frac{1}{r} \right) = -\frac{3}{r^3} + \frac{3(x^2 + y^2 + z^2)}{r^5} = -\frac{3}{r^3} + \frac{3r^2}{r^5} = -\frac{3}{r^3} + \frac{3}{r^3} = 0, \quad (8.23)$$

for all $\vec{r} \neq \vec{0}$. This fact is the mathematical heart of why the electric potential of a point charge solves Laplace’s equation everywhere except at the charge itself.

Quiz

★Q18. For $f(x, y, z) = x^2y + z$, compute ∇f .

★Q19. For $\vec{F} = (x^2, y^2, z^2)^T$, compute $\nabla \cdot \vec{F}$.

★Q20. For $\vec{F} = (-y, x, 0)^T$, compute $\nabla \times \vec{F}$. What direction does the curl point in, and what does that tell you about the rotation of this field?

★Q21. True or false: $\nabla \cdot \vec{F}$ is a vector field. [Hint: Check Definition 8.4 again.]

Problems

★8.7 Compute ∇f , $\nabla \cdot \vec{F}$, $\nabla \times \vec{F}$, and $\nabla^2 f$ for $f = xyz$ and $\vec{F} = (yz, xz, xy)^T$.

***8.8 Show directly (without using thm:curl-grad-zero) that $\nabla \times (\nabla f) = \vec{0}$ for $f(x, y, z) = x^3y^2 + \sin(z)$, by computing both ∇f and its curl.

***8.9 Let $\vec{F} = \vec{r}/r^3$ for $\vec{r} \neq \vec{0}$. Show $\nabla \cdot \vec{F} = 0$ for $r \neq 0$. [Hint: This is the divergence-free electric field of a point charge, away from the charge itself.]

**8.10 Compute $\nabla^2 f$ for $f = \ln r$ (with $r = \sqrt{x^2 + y^2}$, i.e. treat this as a field on $\mathbb{R}^2$, ignoring z), and comment on where it is undefined.

🐞8.11 (drill: which operator, which output type) For each expression below, state whether it is defined, and if so, whether the result is a scalar field or a vector field: (a) ∇f , (b) $\nabla \times f$, (c) $\nabla \cdot \vec{F}$, (d) $\nabla \times (\nabla \cdot \vec{F})$, (e) $\nabla(\nabla \cdot \vec{F})$, (f) $\nabla \cdot \vec{F} \times \nabla g$.

Be careful: A very common slip is to write $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \times \vec{c}$, i.e. to assume the cross product is associative.

Problems

★8.12 Compute $\vec{a} \times \vec{b}$ for $\vec{a} = (1, 0, 2)^T$ and $\vec{b} = (0, 3, -1)^T$ using eq:vc-cross-index directly (sum over j, k by hand for each i), then check against the determinant formula from [Chapter 7](#).

***8.13 Prove the scalar triple product formula $\vec{a} \cdot (\vec{b} \times \vec{c}) = \sum_{i,j,k} \epsilon_{ijk} a_i b_j c_k$, and use it to show $\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c}$.

**8.14 Use thm:eps-delta to prove Lagrange's identity $|\vec{a} \times \vec{b}|^2 = |\vec{a}|^2 |\vec{b}|^2 - \vec{a} \cdot \vec{b}^2$.

🐞8.15 (drill: reading ϵ) Evaluate without a table: (a) ϵ_{132} , (b) ϵ_{313} , (c) ϵ_{321} , (d) ϵ_{111} .

🐞8.16 (drill: contracting one index) Simplify each sum: (a) $\sum_k \epsilon_{ijk} \delta_{jk}$, (b) $\sum_{j,k} \epsilon_{ijk} \epsilon_{ljk}$. [Hint: For (b), use thm:eps-delta and then set $m = k$, summing over k at the end.]

Theorem 8.20 Vector calculus identities

$$\begin{aligned}
\nabla \times (\nabla f) &= \vec{0} && \text{(proved above, thm:curl-grad-zero)} \\
\nabla \cdot (\nabla \times \vec{F}) &= 0 && \text{(proved above, thm:div-curl-zero)} \\
\nabla(fg) &= f\nabla g + g\nabla f \\
\nabla \cdot (f\vec{F}) &= f(\nabla \cdot \vec{F}) + \vec{F} \cdot (\nabla f) \\
\nabla \times (f\vec{F}) &= f(\nabla \times \vec{F}) + (\nabla f) \times \vec{F} \\
\nabla(\vec{A} \cdot \vec{B}) &= (\vec{A} \cdot \nabla)\vec{B} + (\vec{B} \cdot \nabla)\vec{A} + \vec{A} \times (\nabla \times \vec{B}) + \vec{B} \times (\nabla \times \vec{A}) \\
\nabla \cdot (\vec{A} \times \vec{B}) &= \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B}) \\
\nabla \times (\vec{A} \times \vec{B}) &= \vec{A}(\nabla \cdot \vec{B}) - \vec{B}(\nabla \cdot \vec{A}) + (\vec{B} \cdot \nabla)\vec{A} - (\vec{A} \cdot \nabla)\vec{B} \\
\nabla^2 \vec{F} &= \nabla(\nabla \cdot \vec{F}) - \nabla \times (\nabla \times \vec{F}).
\end{aligned} \tag{8.24}$$

Here $(\vec{A} \cdot \nabla)$ denotes the scalar differential operator $A_x \frac{\partial}{\partial x} + A_y \frac{\partial}{\partial y} + A_z \frac{\partial}{\partial z}$, applied componentwise to the vector field that follows it; it is **not** the same object as $\nabla \cdot \vec{A}$ (a scalar field), so keep the parentheses.

Quiz

- ★Q22. Using the fourth identity in Eq. (8.24), simplify $\nabla \cdot (r^2 \vec{r})$ where $r = |\vec{r}|$, given that $\nabla \cdot \vec{r} = 3$ (Eq. (8.20)) and $\nabla(r^2) = 2r\nabla r = 2\vec{r}$.
- ★Q23. Why does the identity $\nabla^2 \vec{F} = \nabla(\nabla \cdot \vec{F}) - \nabla \times (\nabla \times \vec{F})$ make sense dimensionally, i.e. why is each side a vector field, not a scalar field?

Problems

- ***8.17 Prove $\nabla \cdot (f\vec{F}) = f(\nabla \cdot \vec{F}) + \vec{F} \cdot (\nabla f)$ for the specific fields $f = x^2 + y^2$ and $\vec{F} = (z, z, x + y)^T$: compute both sides directly and check they agree.
- **8.18 Prove the identity $\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B})$ using index notation (write $\nabla \cdot (\vec{A} \times \vec{B}) = \sum_i \frac{\partial}{\partial x_i} \sum_{j,k} \epsilon_{ijk} A_j B_k$, apply the product rule, and regroup).
- **8.19 Use Eq. (8.24) to simplify $\nabla \times (f\nabla f)$ (a curl of a field built from a single scalar field f). [Hint: Which of the two terms in the fifth identity of Eq. (8.24) vanishes, and why?]
- *8.20 Prove the identity $\nabla \times (\vec{A} \times \vec{B}) = \vec{A}(\nabla \cdot \vec{B}) - \vec{B}(\nabla \cdot \vec{A}) + (\vec{B} \cdot \nabla)\vec{A} - (\vec{A} \cdot \nabla)\vec{B}$ using thm:eps-delta.
- 🔧8.21 (drill: which identity to reach for) For each target expression, name which line of Eq. (8.24) you would use first: (a) $\nabla \times (r^2 \vec{F})$, (b) $\nabla \cdot (\nabla \times (\nabla f))$ (careful, this one is a trick), (c) $\nabla(f^2)$.

8.6 More Topics are Waiting for You

We have completed the overview of the **derivative** operations in vector calculus. You will use them in *electromagnetism*, but there you will also need **integral** operations. We postpone the discussions to the course *Mathematics and Codings on Physics* (物理數學與數值方法), which you will take in the next semester.

There, you will learn integrals about a scalar field $f(\vec{r})$, such as

$$\text{line integral } \int f(\vec{r})ds, \quad \text{surface integral } \iint f(\vec{r})dA, \quad \text{volume integral } \iiint f(\vec{r})dV,$$

and define integrals about a vector field $\vec{F}(\vec{r})$:

$$\text{line integral } \int \vec{F}(\vec{r}) \cdot d\vec{s}, \quad \text{surface integral } \iint \vec{F}(\vec{r}) \cdot d\vec{A}.$$

These integrals are used to describe the **Maxwell equations** in electromagnetism,

$$\begin{aligned} \iint_{\partial V} \vec{E}(\vec{r}, t) \cdot d\vec{A} &= \iiint_V \frac{\rho(\vec{r}, t)}{\epsilon_0} dV, & \int_{\partial S} \vec{E}(\vec{r}, t) \cdot d\vec{s} &= \iint_S \left(-\frac{\partial \vec{B}(\vec{r}, t)}{\partial t} \right) \cdot d\vec{A}, \\ \iint \vec{B}(\vec{r}, t) \cdot d\vec{A} &= 0, & \int_{\partial S} \vec{B}(\vec{r}, t) \cdot d\vec{s} &= \iint_S \mu_0 \left[\vec{J}(\vec{r}, t) + \epsilon_0 \frac{\partial \vec{E}(\vec{r}, t)}{\partial t} \right] \cdot d\vec{A}, \end{aligned} \quad (8.25)$$

or more precisely,

$$\begin{aligned} \vec{\nabla} \cdot \vec{E}(\vec{r}, t) &= \frac{\rho(\vec{r}, t)}{\epsilon_0}, & \vec{\nabla} \times \vec{E}(\vec{r}, t) &= -\frac{\partial \vec{B}(\vec{r}, t)}{\partial t}, \\ \vec{\nabla} \cdot \vec{B}(\vec{r}, t) &= 0, & \vec{\nabla} \times \vec{B}(\vec{r}, t) &= \mu_0 \vec{J}(\vec{r}, t) + \mu_0 \epsilon_0 \frac{\partial \vec{E}(\vec{r}, t)}{\partial t}. \end{aligned} \quad (8.26)$$

You will learn these equations are related by the **Gauss's theorem** and **Stokes' theorem**,

$$\iint_{\partial V} \vec{F} \cdot d\vec{A} = \iiint_V (\vec{\nabla} \cdot \vec{F}) dV, \quad \int_{\partial S} \vec{F} \cdot d\vec{s} = \iint_S (\vec{\nabla} \times \vec{F}) \cdot d\vec{A}. \quad (8.27)$$



You will also need the **cylindrical coordinates** (r, θ, z) and the **spherical coordinates** (r, θ, φ) to analyze the Maxwell equations. They are related to the **Cartesian coordinates** (x, y, z) by

$$\begin{aligned} \text{cylindrical: } (x, y, z) &= (r \cos \theta, r \sin \theta, \zeta), \\ \text{spherical: } (x, y, z) &= (r \sin \theta \cos \varphi, r \sin \theta \sin \varphi, r \cos \theta), \end{aligned} \quad (8.28)$$

and then, as the biggest challenge, you will convert the gradient, divergence, curl, and Laplacian, as well as dS and dV , into these coordinates.

Remark: Usually the cylindrical coordinate is written by (r, θ, z) because $z = \zeta$ is unchanged, but here we use ζ (zeta) to avoid confusion.

This document does not go into these advanced topics, but just in case you are motivated to prepare for the next course, several exercises are left below.

Problems

***8.22** Consider the cylindrical coordinates (r, θ, ζ) .

(1) The 3×3 matrix $J = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} & \frac{\partial x}{\partial \zeta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} & \frac{\partial y}{\partial \zeta} \\ \frac{\partial z}{\partial r} & \frac{\partial z}{\partial \theta} & \frac{\partial z}{\partial \zeta} \end{pmatrix}$, which we often write by $\frac{\partial(x, y, z)}{\partial(r, \theta, \zeta)}$, is called the

Jacobian matrix of the coordinate change $(r, \theta, \zeta) \rightarrow (x, y, z)$ and its determinant $\det J$ is called **Jacobian**. Compute J and confirm that $\det J = r$.

Be careful: Be aware that we here regard (r, θ, ζ) as the “old” coordinate and (x, y, z) as the “new” coordinate. So, x , y , and z are functions of (r, θ, ζ) , and thus J is a function of (r, θ, ζ) .

(2) Express (r, θ, ζ) in terms of (x, y, z) . This gives the inverse transformation, $(x, y, z) \rightarrow (r, \theta, \zeta)$. [Hint: Your answer will contain $\arctan(x/y)$, but what is “arctan”?]

(3) Let us $J' = \begin{pmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} & \frac{\partial r}{\partial z} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} & \frac{\partial \theta}{\partial z} \\ \frac{\partial \zeta}{\partial x} & \frac{\partial \zeta}{\partial y} & \frac{\partial \zeta}{\partial z} \end{pmatrix}$, i.e., the Jacobian matrix for the inverse transformation

$(x, y, z) \rightarrow (r, \theta, \zeta)$. Calculate J' and $\det J'$.

(4) Confirm $JJ' = I_3$. It means J' is the inverse matrix of J , and thus $\det J' = 1/\det J$.

***8.23** Do the same calculation for the spherical coordinates (r, θ, φ) .

(1) Calculate $J = \frac{\partial(x, y, z)}{\partial(r, \theta, \varphi)}$ and confirm that $\det J = r^2 \sin \theta$.

(2) Express (r, θ, φ) in terms of (x, y, z) .

(3) Calculate $J' = \frac{\partial(r, \theta, \varphi)}{\partial(x, y, z)}$ and $\det J'$. Confirm $JJ' = I_3$ by the direct calculation.

Advanced note: The **Jacobian** $\det J$ is used to transform integrals. With its **absolute value** $|\det J|$,

$$\iiint f(x, y, z) dx dy dz = \iiint f(x, y, z) \left| \det \frac{\partial(x, y, z)}{\partial(r, \theta, \varphi)} \right| dr d\theta d\varphi \quad \text{etc.} \quad (8.29)$$

(note the absolute-value symbol!), where $f(x, y, z)$ in the second expression is written in terms of (r, θ, φ) .

Problems

***8.24** We here compute the derivative operators in the cylindrical coordinates (r, θ, ζ) . Let us begin with the gradient. It should be

$$\vec{\nabla} f = \vec{e}_x \partial_x f + \vec{e}_y \partial_y f + \vec{e}_z \partial_z f = \vec{e}_r \bullet + \vec{e}_\theta \blacktriangle + \vec{e}_\zeta \blacksquare \quad (8.30)$$

and we want to find the unknown expressions $\bullet$, $\blacktriangle$, and $\blacksquare$.

- (1) Express $\partial_x f$, $\partial_y f$, and $\partial_z f$ in terms of $\partial_r f$, $\partial_\theta f$, and $\partial_\zeta f$.

Fail safe note: As x is a function of (r, θ, ζ) , the chain rule $\frac{\partial}{\partial x} = \frac{\partial r}{\partial x} \frac{\partial}{\partial r} + \frac{\partial \theta}{\partial x} \frac{\partial}{\partial \theta} + \frac{\partial \zeta}{\partial x} \frac{\partial}{\partial \zeta}$ serves.

- (2) The difficulty is that the meaning of $\vec{e}_r$, $\vec{e}_\theta$, and $\vec{e}_\zeta$. They should be the unit vectors forming an orthonormal basis and $\vec{e}_r$ should points in the positive r -direction. However, the positive r -direction depends on the position (x, y, z) , so $\vec{e}_r$ (and thus $\vec{e}_\theta$) depends on the position, as shown in Figure 1.

For point $P(x, y, z)$ or (r, θ, ζ) , the unit vectors are given by

$$\vec{e}_r = \vec{e}_x \cos \theta + \vec{e}_y \sin \theta, \quad \vec{e}_\theta = -\vec{e}_x \sin \theta + \vec{e}_y \cos \theta, \quad \vec{e}_\zeta = \vec{e}_z. \quad (8.31)$$

Express them in the matrix form. Confirm this is consistent with Figure 1. Express $\vec{e}_r$ and $\vec{e}_\theta$ without using r and θ . Confirm $(\vec{e}_r, \vec{e}_\theta, \vec{e}_\zeta)$ form an orthonormal basis and obey the right-hand rule.

Advanced note: In general, $\vec{e}_X$ has the same direction as $\frac{\partial \vec{r}}{\partial X}$. Namely, we can obtain the basis vectors by normalizing

$$\frac{\partial \vec{r}}{\partial r} = \frac{\partial}{\partial r} \begin{pmatrix} r \cos \theta \\ r \sin \theta \\ \zeta \end{pmatrix} = \begin{pmatrix} \cos \theta \\ \sin \theta \\ 0 \end{pmatrix}, \quad \frac{\partial \vec{r}}{\partial \theta} = \begin{pmatrix} -r \sin \theta \\ r \cos \theta \\ 0 \end{pmatrix}, \quad \text{and} \quad \frac{\partial \vec{r}}{\partial \zeta} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}. \quad (8.32)$$

This method is useful for the spherical coordinates.

- (3) Express $\vec{e}_x$, $\vec{e}_y$, and $\vec{e}_z$ only with θ , $\vec{e}_r$, $\vec{e}_\theta$, and $\vec{e}_\zeta$.
- (4) Substitute the above expressions into $\vec{\nabla} f = \vec{e}_x \partial_x f + \vec{e}_y \partial_y f + \vec{e}_z \partial_z f$ and reach

$$\vec{\nabla} f = \vec{e}_r \frac{\partial f}{\partial r} + \frac{\vec{e}_\theta}{r} \frac{\partial f}{\partial \theta} + \vec{e}_\zeta \frac{\partial f}{\partial \zeta}. \quad (8.33)$$

***8.25** Let us consider divergence in cylindrical coordinates. As divergence is for a vector field $\vec{F} = F_x \vec{e}_x + F_y \vec{e}_y + F_z \vec{e}_z = F_r \vec{e}_r + F_\theta \vec{e}_\theta + F_\zeta \vec{e}_\zeta$, we should use F_r , F_θ , and F_ζ instead of F_x , F_y , and F_z . So, what we need to calculate is

$$\vec{\nabla} \cdot \vec{F} = (\vec{e}_r \partial_r + (\vec{e}_\theta / r) \partial_\theta + \vec{e}_\zeta \partial_\zeta) \cdot (F_r \vec{e}_r + F_\theta \vec{e}_\theta + F_\zeta \vec{e}_\zeta). \quad (8.34)$$

- (1) Calculate $\partial_r \vec{e}_r$, $\partial_r \vec{e}_\theta$, $\partial_r \vec{e}_\zeta$, $\partial_\theta \vec{e}_r$, $\partial_\theta \vec{e}_\theta$, $\partial_\theta \vec{e}_\zeta$, $\partial_\zeta \vec{e}_r$, $\partial_\zeta \vec{e}_\theta$, and $\partial_\zeta \vec{e}_\zeta$.
- (2) Confirm $\vec{\nabla} \cdot \vec{F} = \frac{1}{r} \partial_r (r F_r) + \frac{1}{r} \partial_\theta F_\theta + \partial_\zeta F_\zeta$
- (3) Find $\vec{\nabla} \times \vec{F}$ and Δf . Compare your answer with online resources.

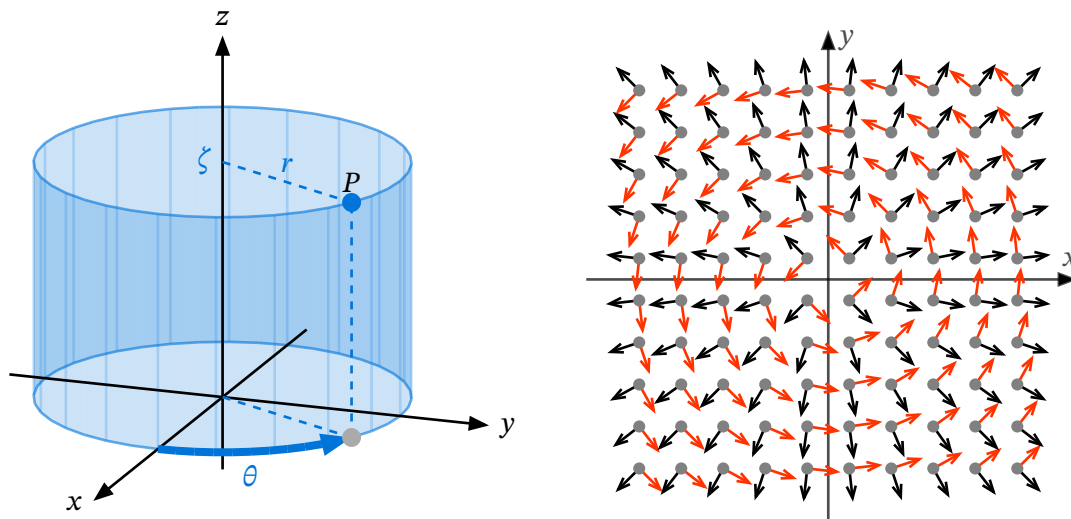


Figure 1: (left) The cylindrical coordinates, where the position of point P is described by (r, θ, ζ) . (right) The basis vector, $(\vec{e}_r, \vec{e}_\theta, \vec{e}_\zeta)$, of the cylindrical coordinates. Since the direction of $\vec{e}_r$ and $\vec{e}_\theta$ depends on the position, they are drawn for each point, described by a gray dot; $\vec{e}_r$ are drawn in black and $\vec{e}_\theta$ are drawn in red. Meanwhile, $\vec{e}_\zeta$ is a constant vector $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. Notice that $(\vec{e}_r, \vec{e}_\theta, \vec{e}_\zeta)$ forms an orthonormal basis and obeys the **right-hand rule**.